MASTER TRAINER SLIDES · WSQ

AWS Certified Cloud Practitioner Training

AWS Cloud Practitioner · CLF-C02

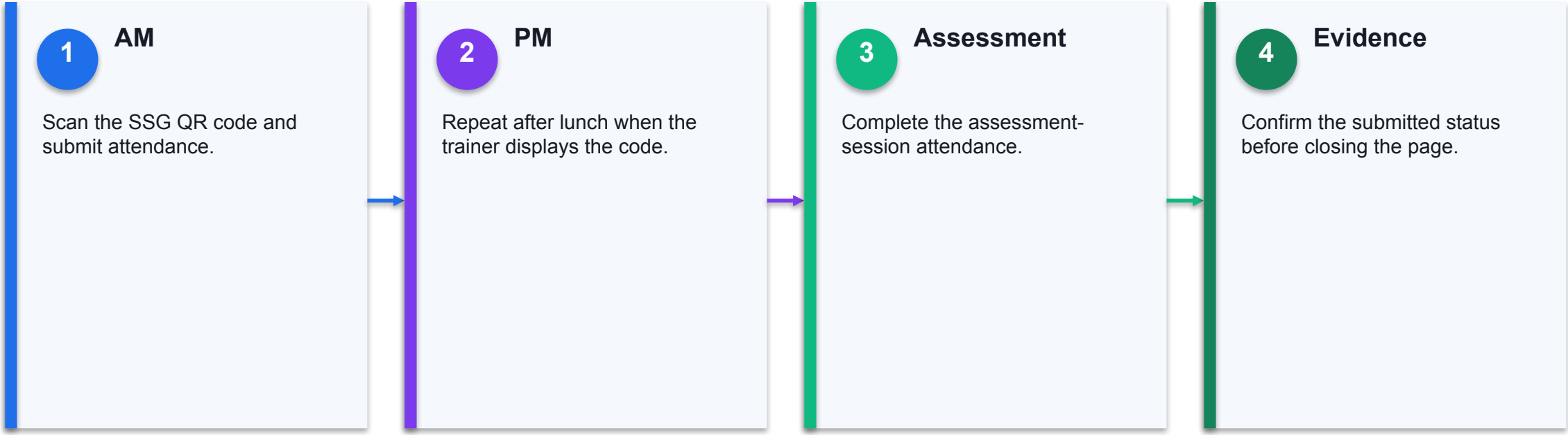
WSQ Course Code: TGS-2023039183
Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v15 · 25 August 2026



COURSE ADMINISTRATION

Digital Attendance (Mandatory)



MECHANISM

Read left to right: input, control, state change, observable evidence.

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About the Trainer



Appointed Trainer

General trainer profile · confirm before delivery

QUALIFICATIONS AWS certification or equivalent cloud-platform competence

EXPERIENCE Cloud architecture, security, operations and cost governance

PREPARATION Review v15 materials, lab controls and assessment boundaries

CONTACT Use the approved class communication channel

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About the Trainer



Dr. Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE Principal Trainer and courseware lead

FOCUS Cloud, AI, automation and software engineering

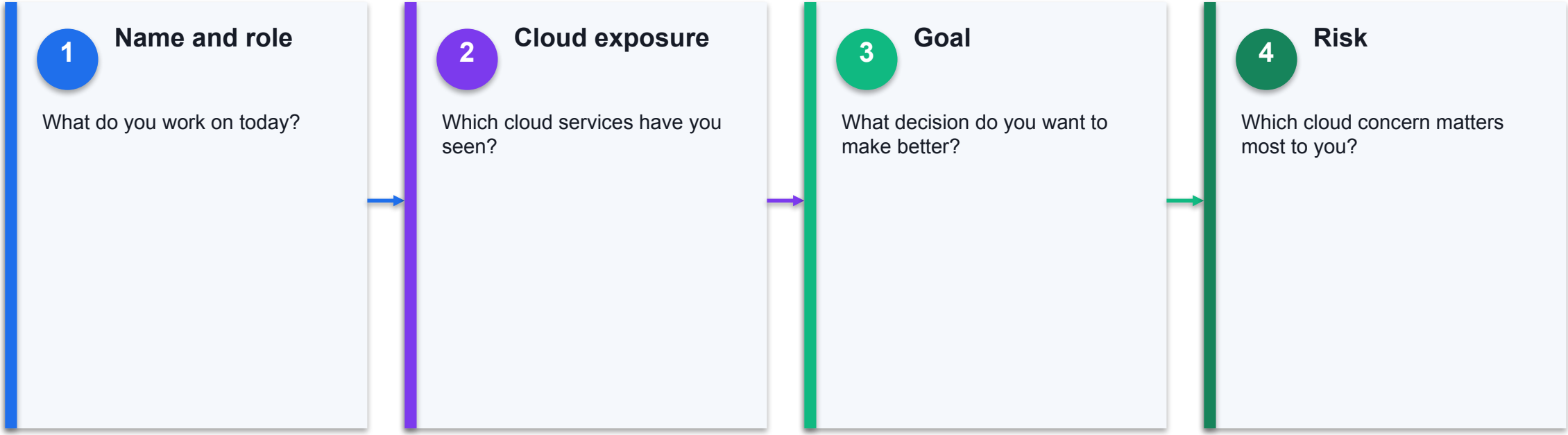
DELIVERY Mechanism-led explanation with practical evidence

CONTACT enquiry@tertiaryinfotech.com · 6100 0613

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ICE-BREAKER

Learner introductions



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Ground rules

1 Participate

Ask, test and explain your evidence.

2 Protect access

Never share credentials or capture secrets.

3 Control cost

Create only approved resources and clean up.

4 Respect time

Return from breaks and complete attendance.

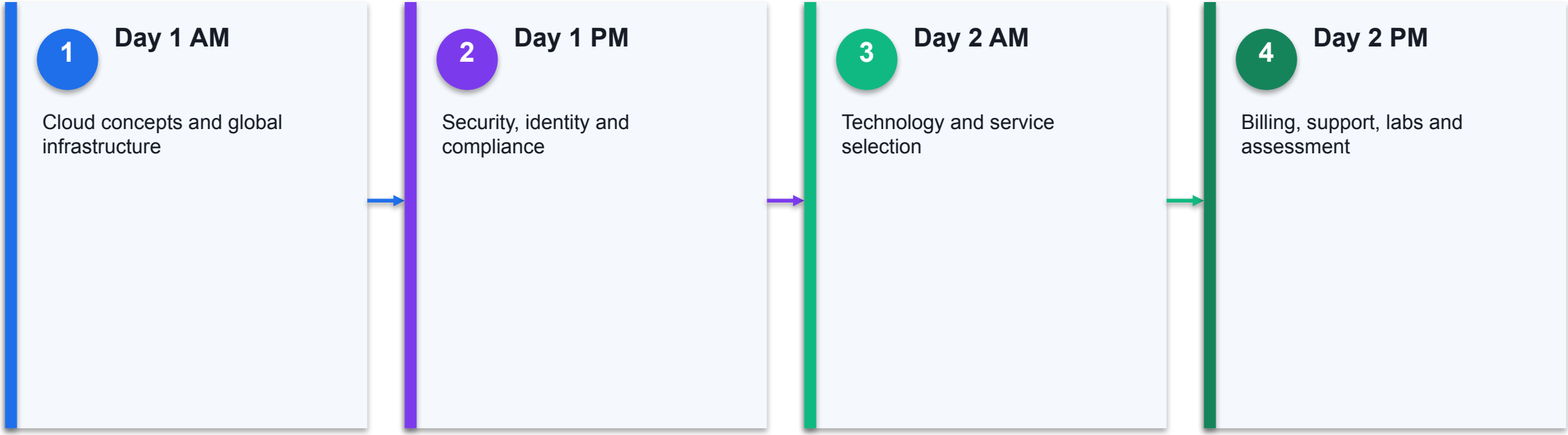
MECHANISM

Read left to right: input, control, state change, observable evidence.

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LESSON PLAN

Two-day course map

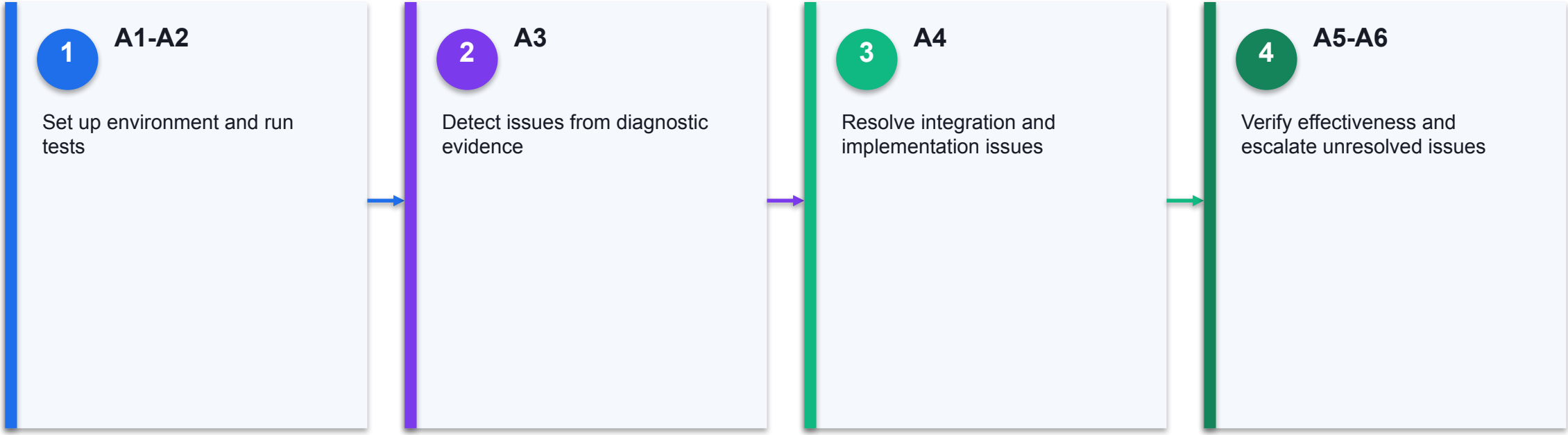


MECHANISM

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TSC ability pathway



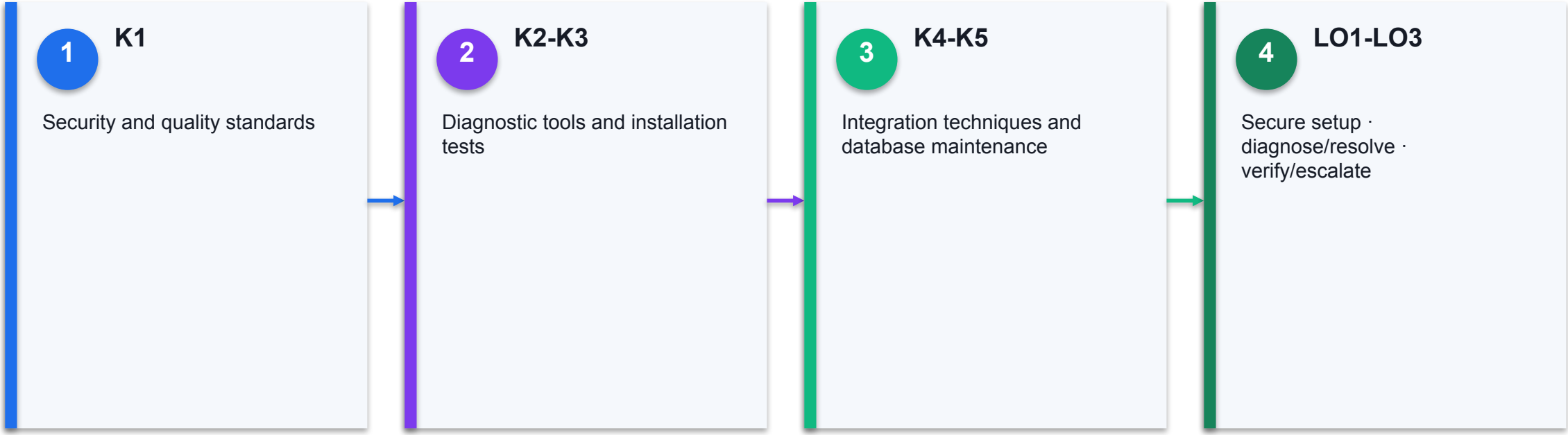
MECHANISM

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TSC ALIGNMENT

Knowledge and learning outcomes



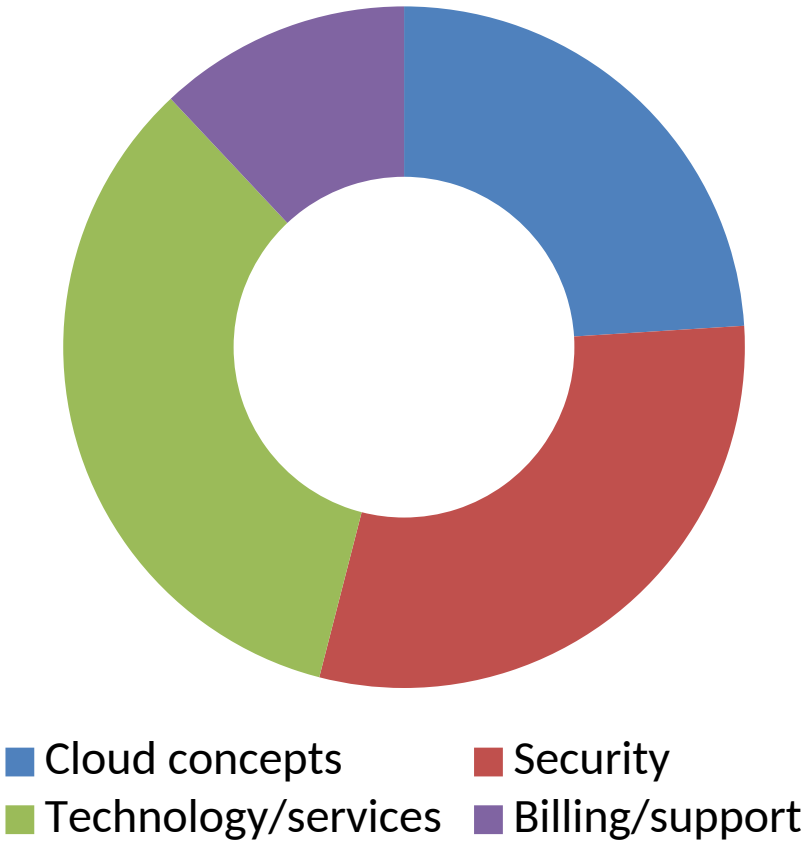
MECHANISM

Read left to right: input, control, state change, observable evidence.

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EXAM BLUEPRINT

CLF-C02 exam domain weighting



WHAT THE DATA SHOWS

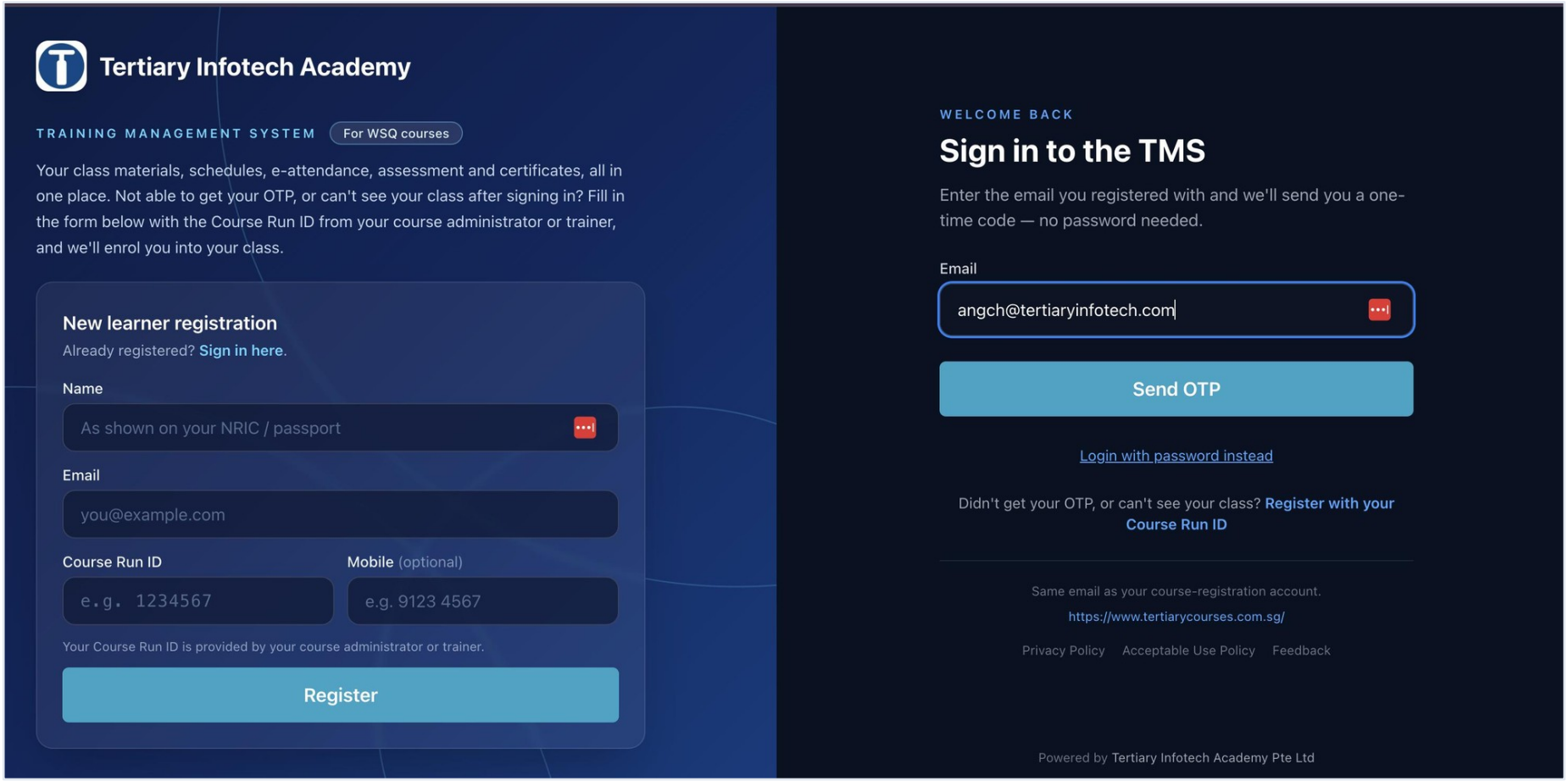
Technology and services carries the largest single share, but security plus cloud concepts together account for more than half of scored content.

Editable PowerPoint chart

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COURSE PORTAL

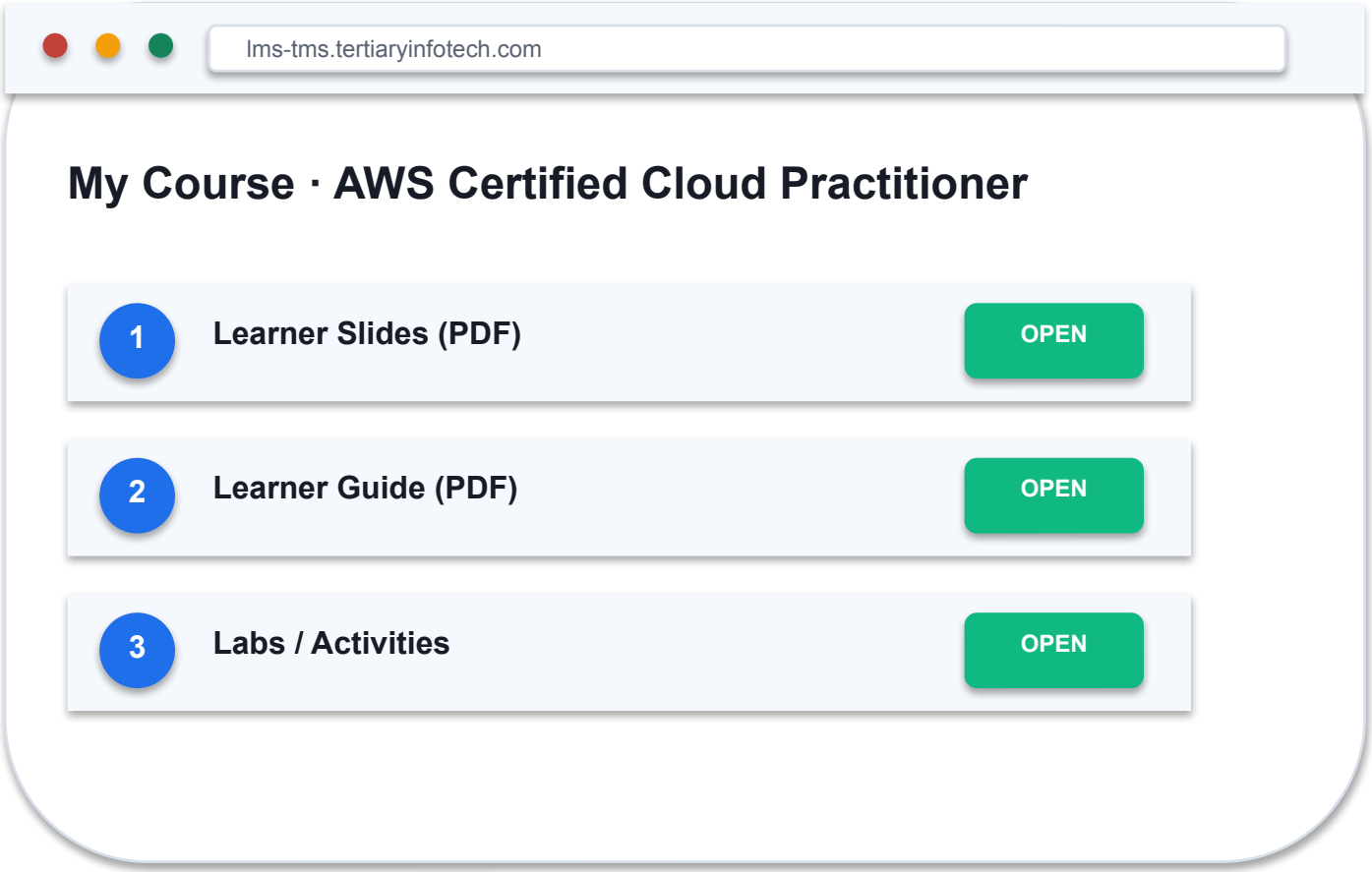
Login to LMS and Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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Download the Course Materials



1 **Sign in**
Use the registered learner email.

2 **Open course**
Select the current class instance.

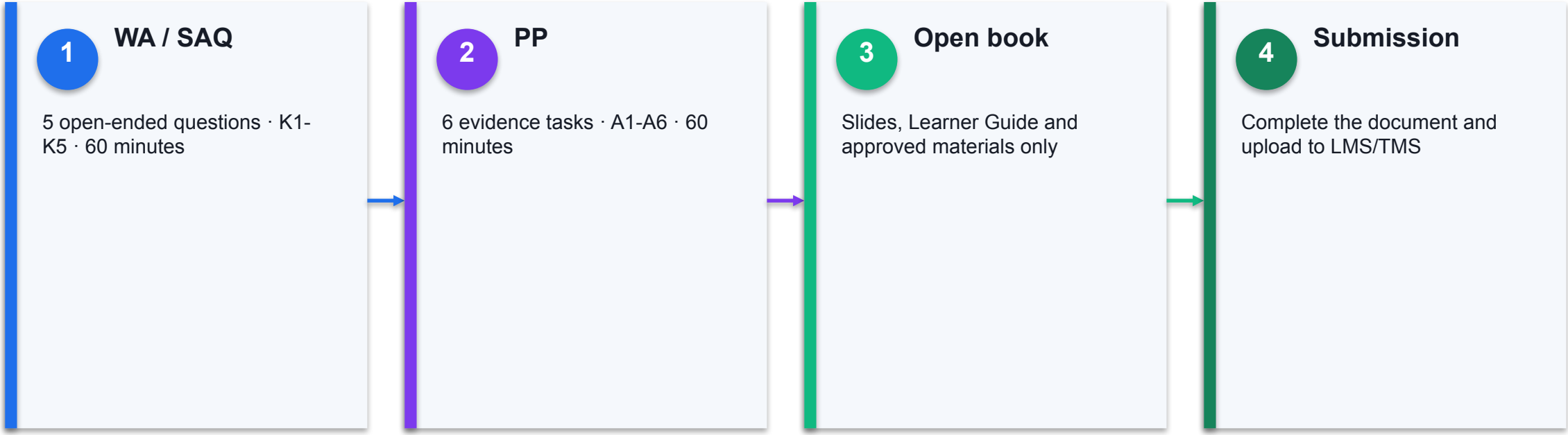
3 **Download**
Open PDF/LG/Labs; save locally.

4 **Verify**
Check filename and current version.

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ASSESSMENT

Assessment format



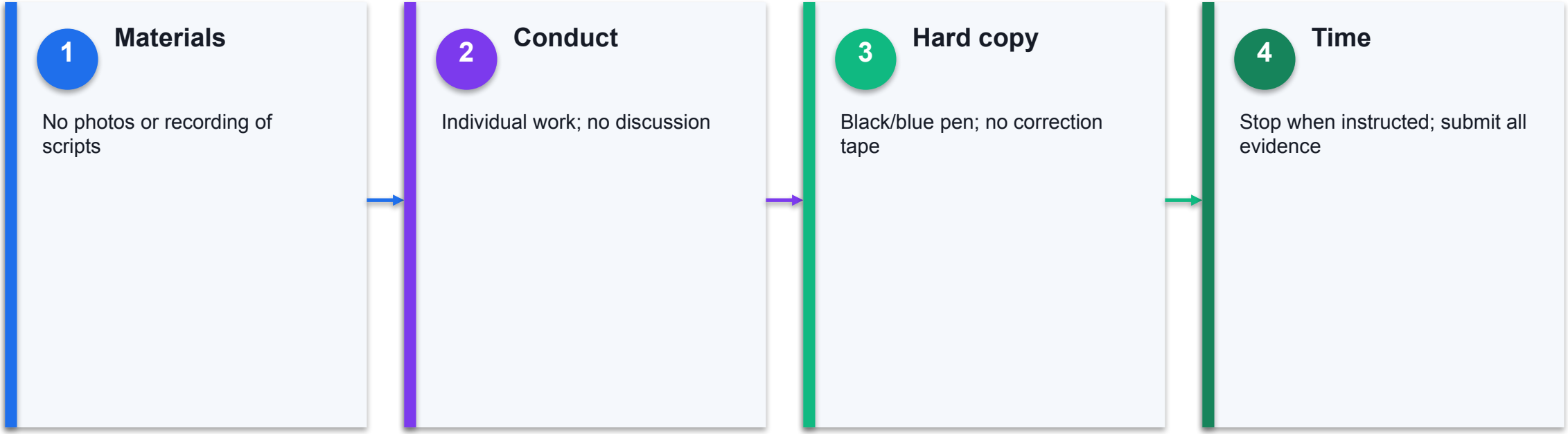
MECHANISM

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ASSESSMENT BRIEFING

Briefing before assessment

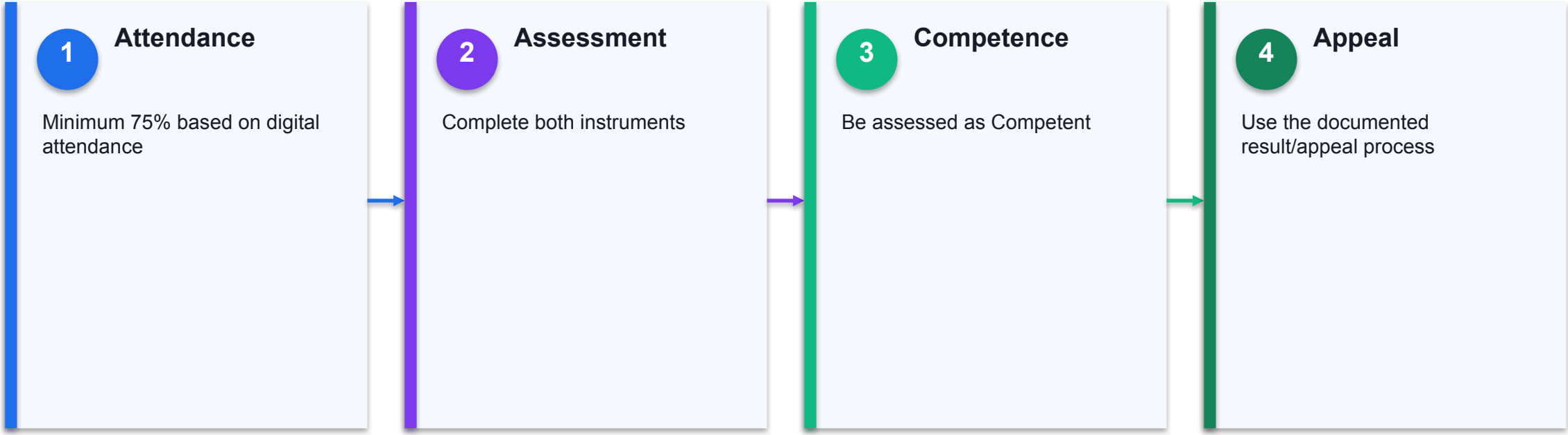


MECHANISM

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Funding and completion criteria



MECHANISM

Read left to right: input, control, state change, observable evidence.

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01

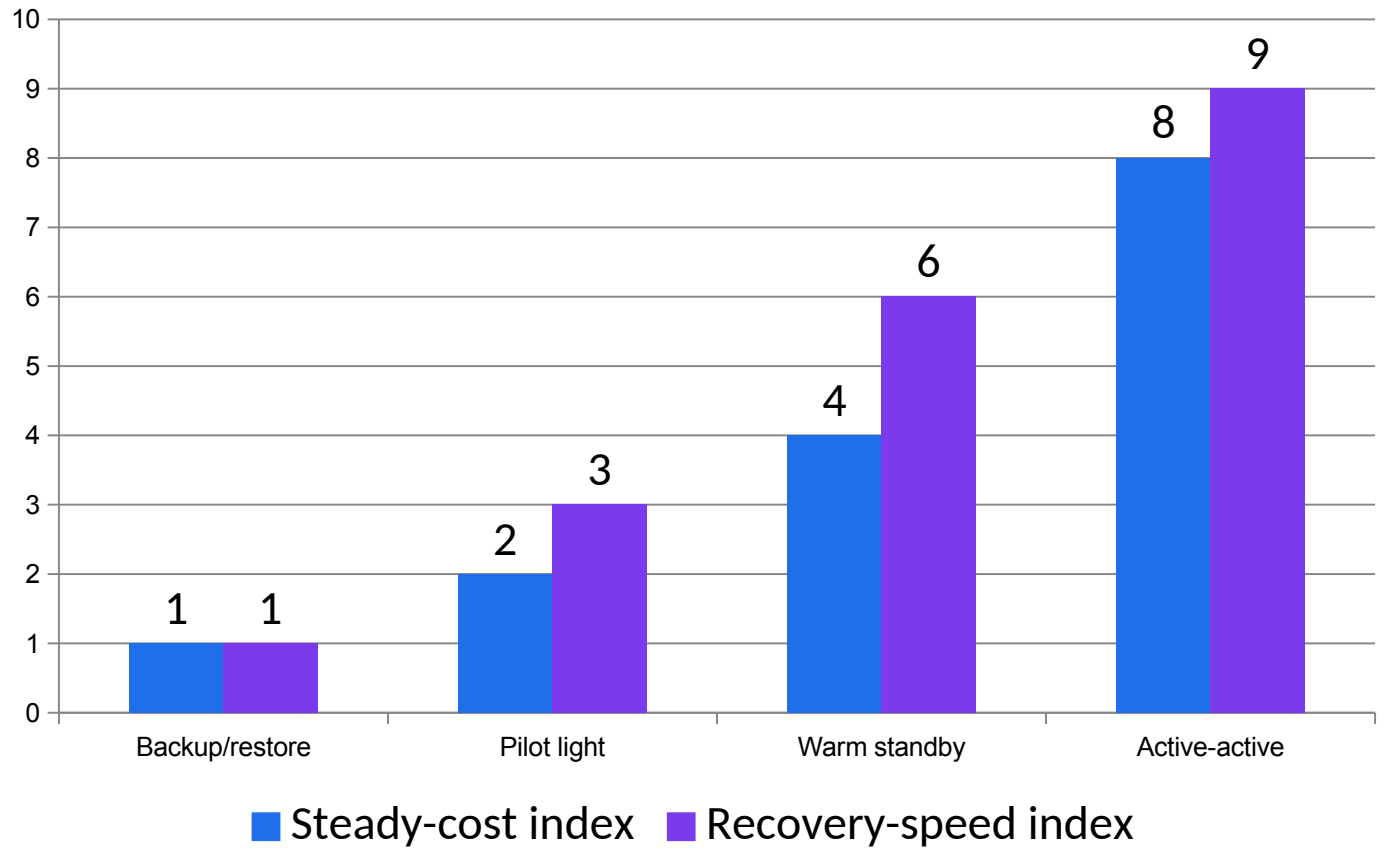
DOMAIN 1 · EXAM WEIGHT 24%

Cloud Concepts

Cloud value · global infrastructure · architecture · migration and transformation

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Illustrative recovery trade-off



WHAT THE DATA SHOWS

More standing capacity can shorten recovery, but only a tested design proves RPO/RTO.

Editable PowerPoint chart

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Cloud operating model: request to evidence



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Cloud operating model: request to evidence · technical contract

1	INPUT / INTENT **Request**	A team declares capacity, region, security and service-level needs.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Provision**	APIs create metered resources from a service catalogue.	RETAIN **Approved configuration**
3	STATE CHANGE **Operate**	Metrics, logs, policies and automation maintain desired state.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Evidence**	Usage, configuration and audit records prove outcome and cost.	RETAIN **Audit, metric or test record**

CONTROL RISK Treating cloud as rented servers preserves slow manual processes and weak cost visibility.

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Cloud operating model: request to evidence · decision and control

1 On premises

USE WHEN

Stable utilisation and owned facilities

TRADE-OFF

Capital commitment and long lead time

2 Cloud

USE WHEN

Variable demand and rapid experiments

TRADE-OFF

Governance must keep pace with speed

3 Hybrid

USE WHEN

Regulated or latency-bound estates

TRADE-OFF

Integration and operating complexity

FAILURE MODE

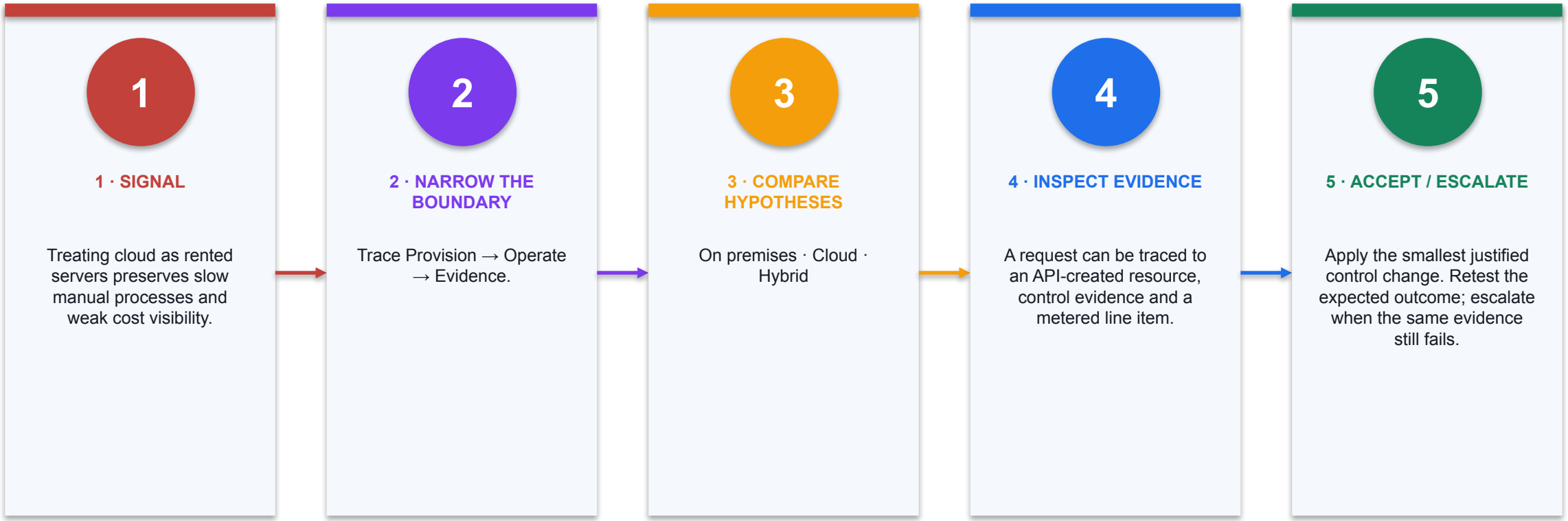
Treating cloud as rented servers preserves slow manual processes and weak cost visibility.

VERIFICATION EVIDENCE

A request can be traced to an API-created resource, control evidence and a metered line item.

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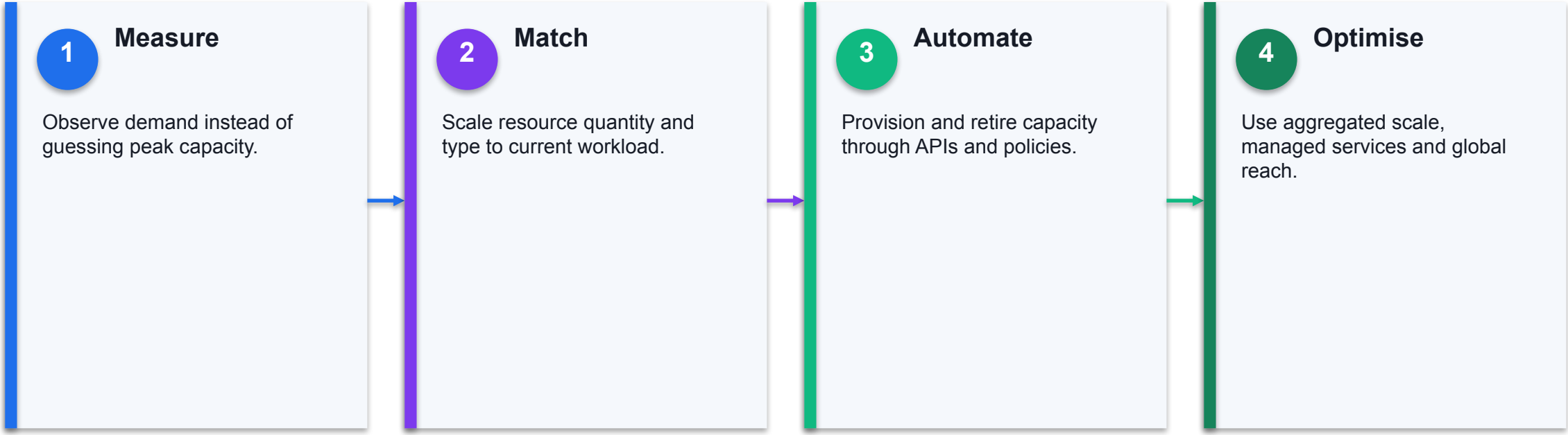
Cloud operating model: request to evidence · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Six cloud advantages as operating levers



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Six cloud advantages as operating levers · technical contract

1	INPUT / INTENT **Measure**	Observe demand instead of guessing peak capacity.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Match**	Scale resource quantity and type to current workload.	RETAIN **Approved configuration**
3	STATE CHANGE **Automate**	Provision and retire capacity through APIs and policies.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Optimise**	Use aggregated scale, managed services and global reach.	RETAIN **Audit, metric or test record**

CONTROL RISK Elastic infrastructure without termination rules can scale cost as quickly as performance.

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Six cloud advantages as operating levers · decision and control

1 CAPEX to OPEX

USE WHEN

Uncertain growth or short-lived workloads

TRADE-OFF

Variable bills need budgets

2 Economies of scale

USE WHEN

Commodity infrastructure

TRADE-OFF

Service choice still matters

3 Global in minutes

USE WHEN

Distributed users

TRADE-OFF

Data residency can constrain region

FAILURE MODE

Elastic infrastructure without termination rules can scale cost as quickly as performance.

VERIFICATION EVIDENCE

Utilisation, deployment lead time and unit cost improve against an agreed baseline.

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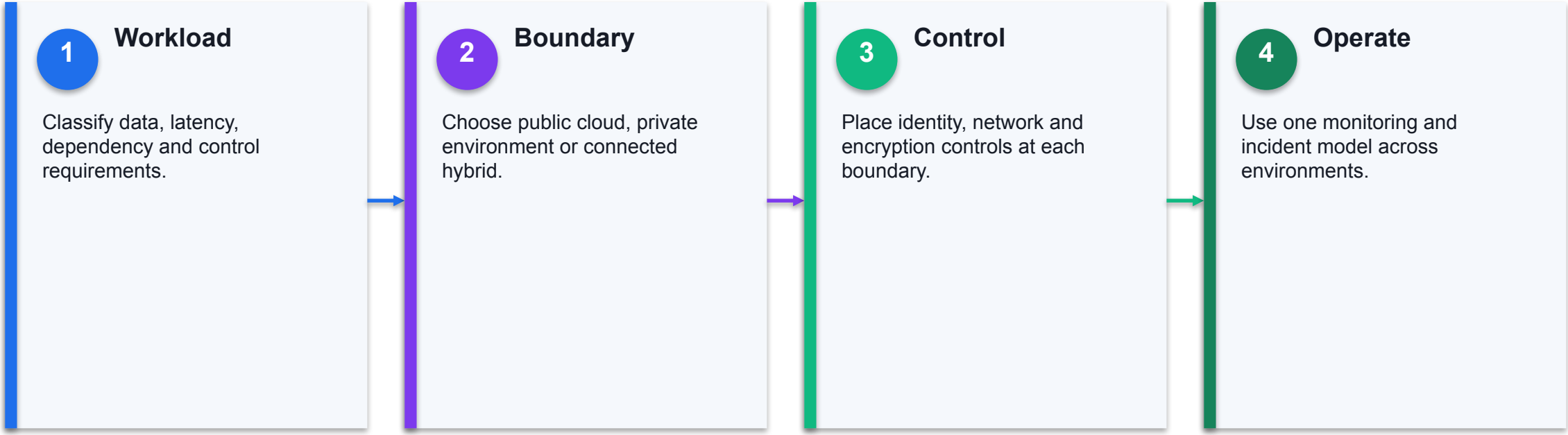
Six cloud advantages as operating levers · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Deployment models and trust boundaries



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Deployment models and trust boundaries · technical contract

1	INPUT / INTENT **Workload**	Classify data, latency, dependency and control requirements.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Boundary**	Choose public cloud, private environment or connected hybrid.	RETAIN **Approved configuration**
3	STATE CHANGE **Control**	Place identity, network and encryption controls at each boundary.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Operate**	Use one monitoring and incident model across environments.	RETAIN **Audit, metric or test record**

CONTROL RISK A hybrid design without explicit ownership leaves gaps between on-premises and AWS teams.

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Deployment models and trust boundaries · decision and control

1 Public cloud

USE WHEN

Elastic digital workloads

TRADE-OFF

Provider controls the physical layer

2 Private cloud

USE WHEN

Dedicated control requirements

TRADE-OFF

Organisation carries capacity burden

3 Hybrid

USE WHEN

Migration and regulated integration

TRADE-OFF

Connectivity becomes critical path

FAILURE MODE

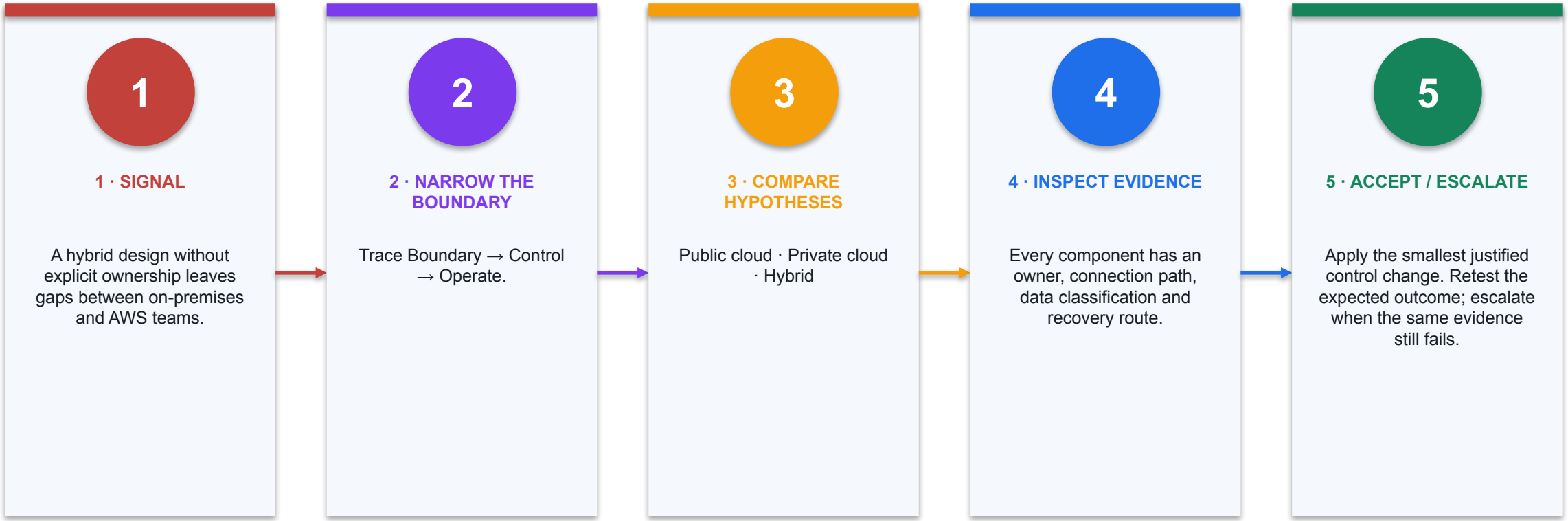
A hybrid design without explicit ownership leaves gaps between on-premises and AWS teams.

VERIFICATION EVIDENCE

Every component has an owner, connection path, data classification and recovery route.

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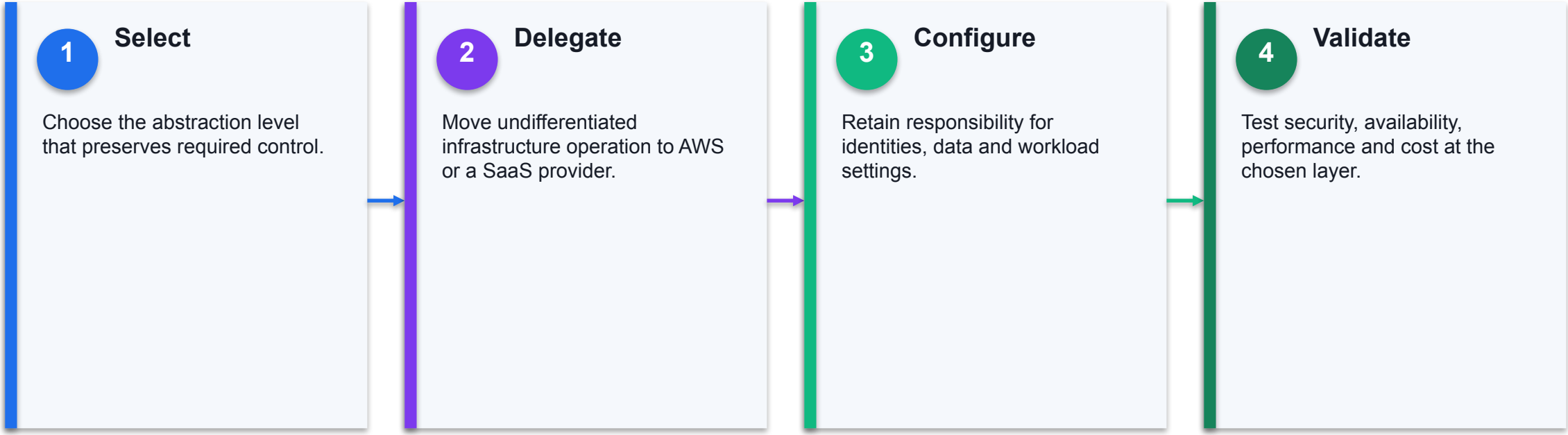
Deployment models and trust boundaries · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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IaaS, PaaS, serverless and SaaS responsibility



MECHANISM

Read left to right: input, control, state change, observable evidence.

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IaaS, PaaS, serverless and SaaS responsibility · technical contract

1	INPUT / INTENT **Select**	Choose the abstraction level that preserves required control.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Delegate**	Move undifferentiated infrastructure operation to AWS or a SaaS provider.	RETAIN **Approved configuration**
3	STATE CHANGE **Configure**	Retain responsibility for identities, data and workload settings.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Validate**	Test security, availability, performance and cost at the chosen layer.	RETAIN **Audit, metric or test record**

CONTROL RISK Choosing a managed service does not transfer responsibility for access, data or configuration.

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IaaS, PaaS, serverless and SaaS responsibility · decision and control

1 IaaS

USE WHEN

Custom OS or legacy runtime

TRADE-OFF

Patch and scale more layers

2 Managed/serverless

USE WHEN

Focus on application and data

TRADE-OFF

Accept service constraints

3 SaaS

USE WHEN

Consume a complete capability

TRADE-OFF

Least infrastructure control

FAILURE MODE

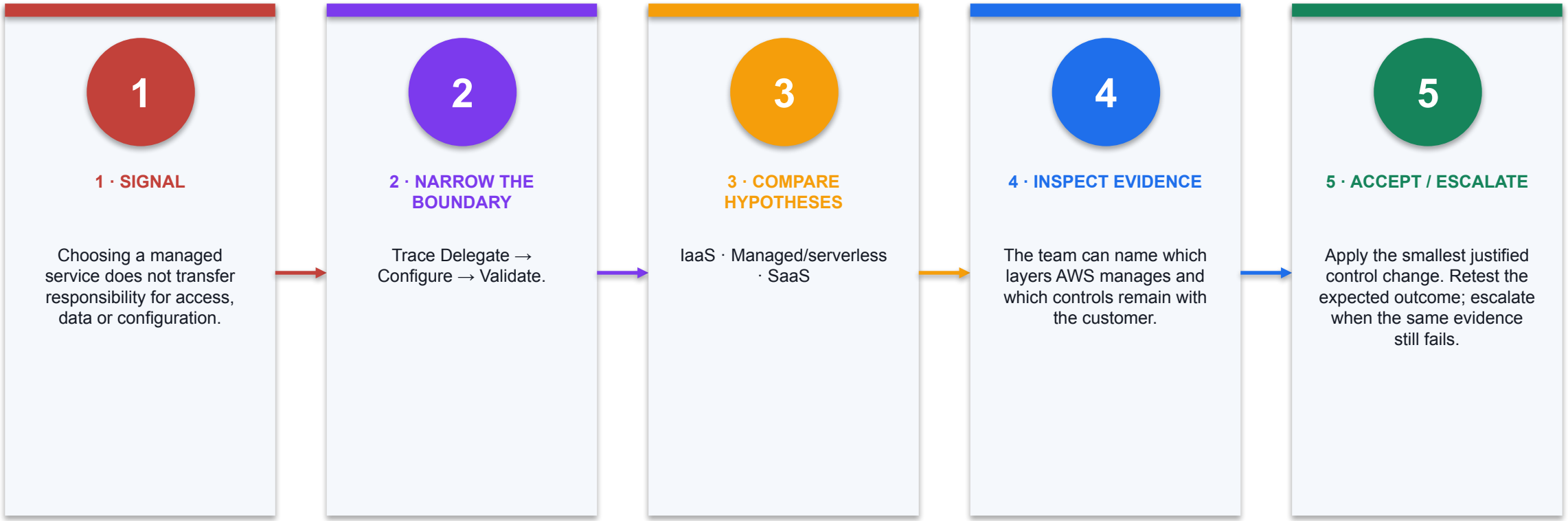
Choosing a managed service does not transfer responsibility for access, data or configuration.

VERIFICATION EVIDENCE

The team can name which layers AWS manages and which controls remain with the customer.

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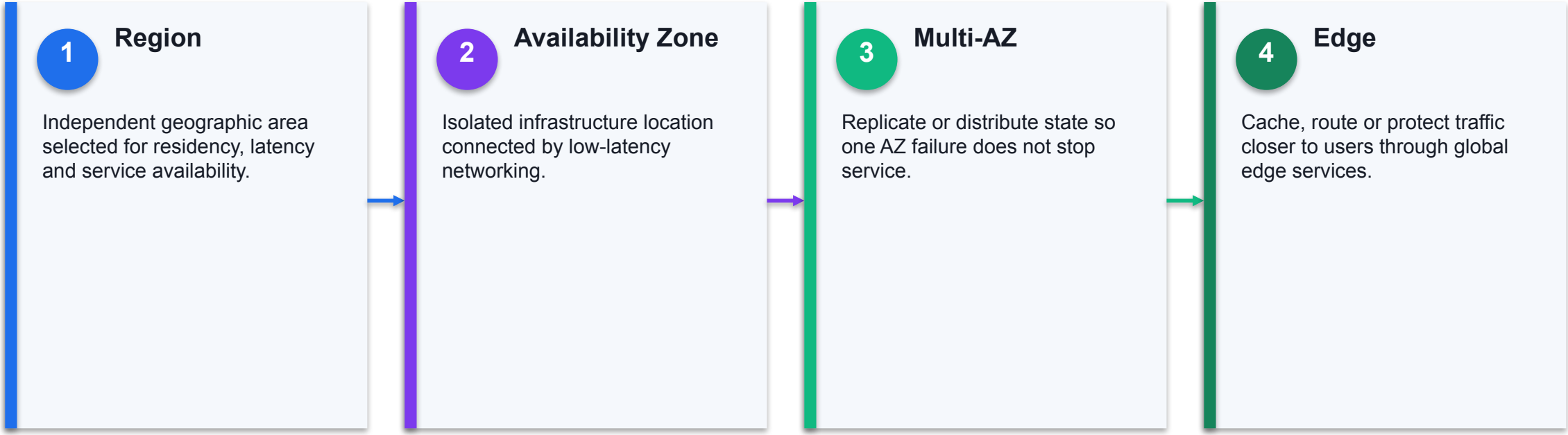
IaaS, PaaS, serverless and SaaS responsibility · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Global infrastructure: Region to edge



MECHANISM

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Global infrastructure: Region to edge · technical contract

1	INPUT / INTENT **Region**	Independent geographic area selected for residency, latency and service availability.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Availability Zone**	Isolated infrastructure location connected by low-latency networking.	RETAIN **Approved configuration**
3	STATE CHANGE **Multi-AZ**	Replicate or distribute state so one AZ failure does not stop service.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Edge**	Cache, route or protect traffic closer to users through global edge services.	RETAIN **Audit, metric or test record**

CONTROL RISK A resource being in a Region does not automatically make it resilient across AZs.

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Global infrastructure: Region to edge · decision and control

1 Single AZ

USE WHEN

Development or non-critical state

TRADE-OFF

One facility fault can interrupt service

2 Multi-AZ

USE WHEN

Production availability

TRADE-OFF

Higher cost and replication design

3 Multi-Region

USE WHEN

Regional disaster or global users

TRADE-OFF

Data consistency and failover complexity

FAILURE MODE

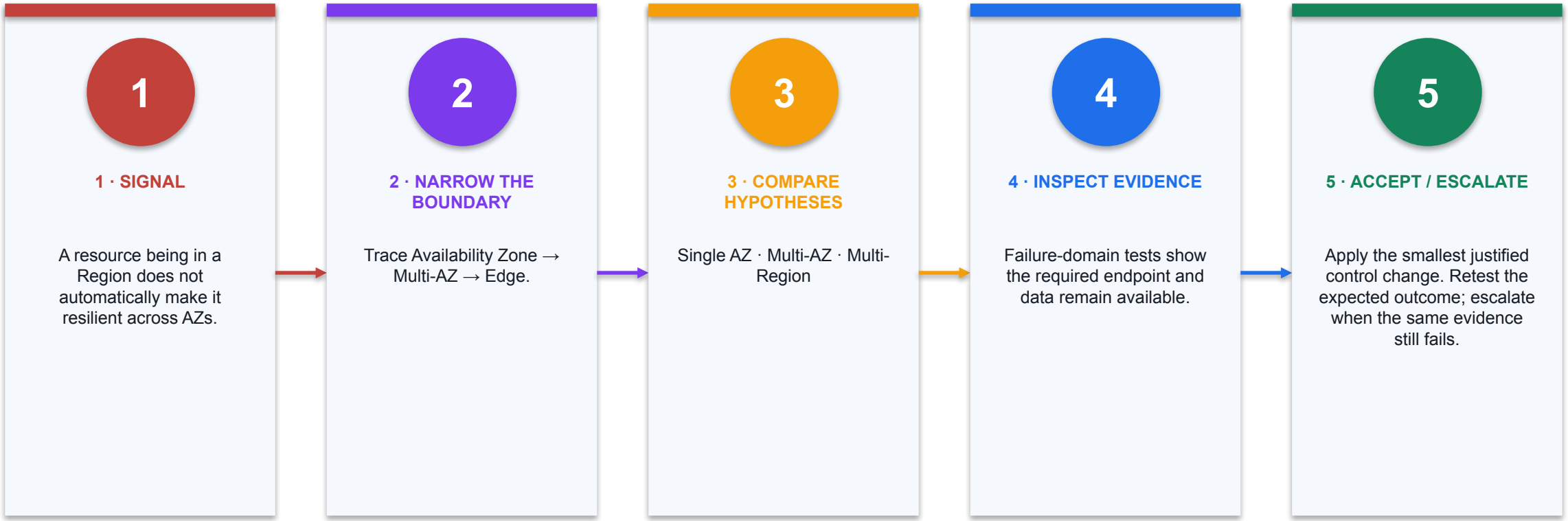
A resource being in a Region does not automatically make it resilient across AZs.

VERIFICATION EVIDENCE

Failure-domain tests show the required endpoint and data remain available.

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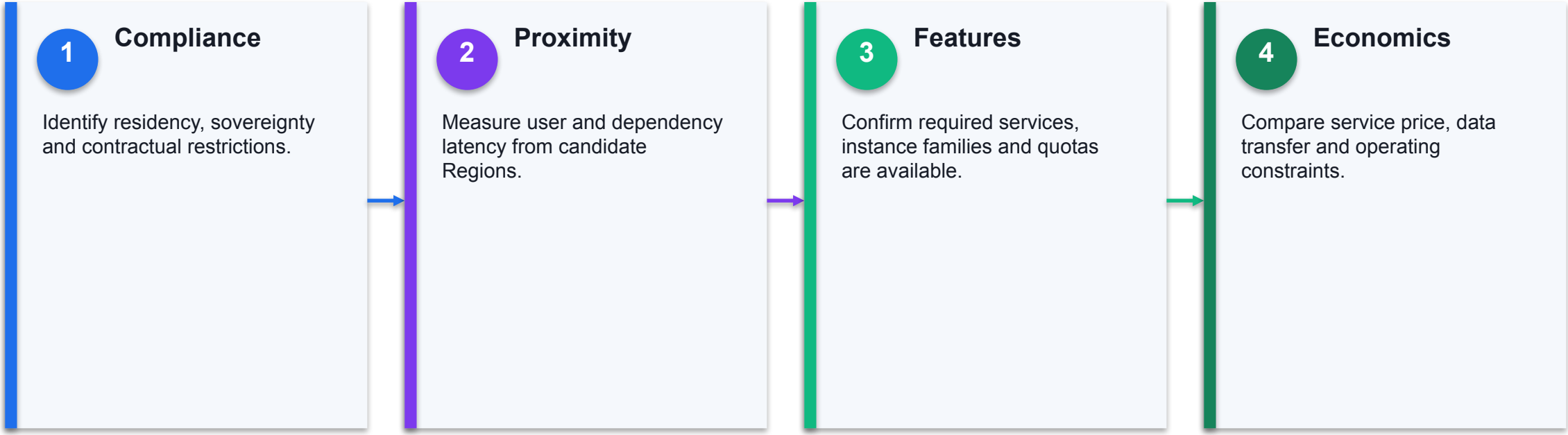
Global infrastructure: Region to edge · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Region selection is a four-constraint decision



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Region selection is a four-constraint decision · technical contract

1	INPUT / INTENT **Compliance**	Identify residency, sovereignty and contractual restrictions.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Proximity**	Measure user and dependency latency from candidate Regions.	RETAIN **Approved configuration**
3	STATE CHANGE **Features**	Confirm required services, instance families and quotas are available.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Economics**	Compare service price, data transfer and operating constraints.	RETAIN **Audit, metric or test record**

CONTROL RISK Selecting a Region from price alone can violate residency or increase network transfer cost.

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Region selection is a four-constraint decision · decision and control

1 Compliance first

USE WHEN

Regulated data

TRADE-OFF

May limit the candidate list

2 Latency first

USE WHEN

Interactive applications

TRADE-OFF

Edge services may alter the decision

3 Cost first

USE WHEN

Batch or portable workloads

TRADE-OFF

Do not ignore resilience and transfer

FAILURE MODE

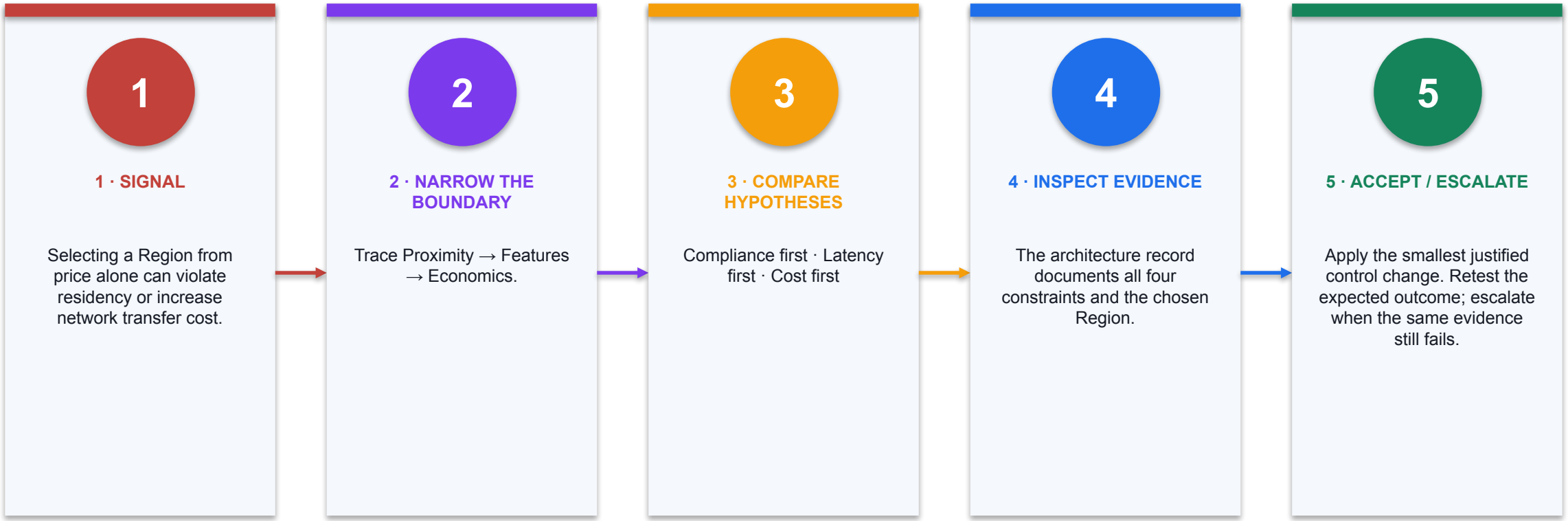
Selecting a Region from price alone can violate residency or increase network transfer cost.

VERIFICATION EVIDENCE

The architecture record documents all four constraints and the chosen Region.

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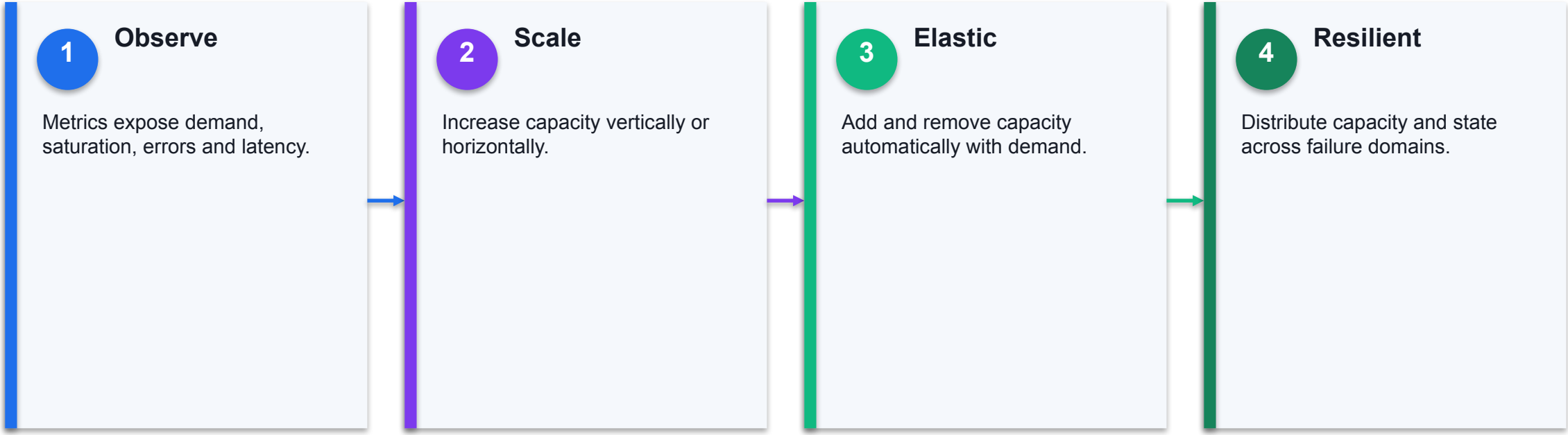
Region selection is a four-constraint decision · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Scalability, elasticity and availability are different



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Scalability, elasticity and availability are different · technical contract

1	INPUT / INTENT **Observe**	Metrics expose demand, saturation, errors and latency.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Scale**	Increase capacity vertically or horizontally.	RETAIN **Approved configuration**
3	STATE CHANGE **Elastic**	Add and remove capacity automatically with demand.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Resilient**	Distribute capacity and state across failure domains.	RETAIN **Audit, metric or test record**

CONTROL RISK Auto Scaling in one AZ improves capacity but does not remove the AZ as a single failure domain.

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Scalability, elasticity and availability are different · decision and control

1 Vertical scale

USE WHEN

Fast lift for a single node

TRADE-OFF

Has a ceiling and restart risk

2 Horizontal scale

USE WHEN

Stateless or partitionable workloads

TRADE-OFF

Needs load distribution

3 Multi-AZ

USE WHEN

Availability requirement

TRADE-OFF

Does not itself handle traffic growth

FAILURE MODE

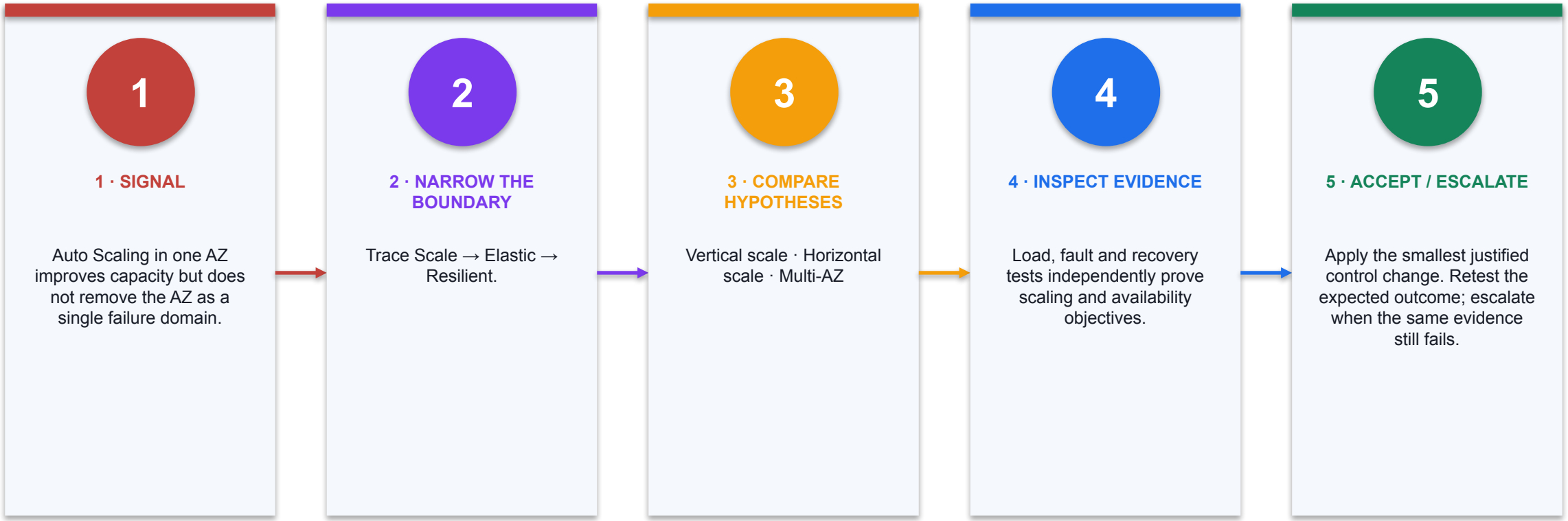
Auto Scaling in one AZ improves capacity but does not remove the AZ as a single failure domain.

VERIFICATION EVIDENCE

Load, fault and recovery tests independently prove scaling and availability objectives.

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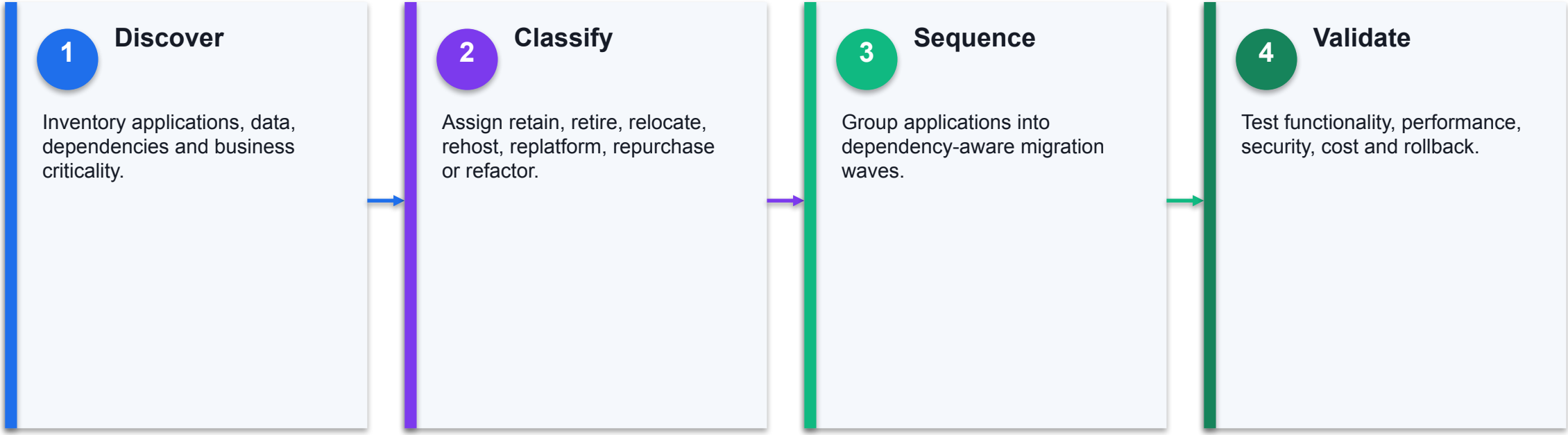
Scalability, elasticity and availability are different · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Migration 7 Rs: portfolio to wave plan



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Migration 7 Rs: portfolio to wave plan · technical contract

1	INPUT / INTENT **Discover**	Inventory applications, data, dependencies and business criticality.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Classify**	Assign retain, retire, relocate, rehost, replatform, repurchase or refactor.	RETAIN **Approved configuration**
3	STATE CHANGE **Sequence**	Group applications into dependency-aware migration waves.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Validate**	Test functionality, performance, security, cost and rollback.	RETAIN **Audit, metric or test record**

CONTROL RISK Migrating a database before its dependent application or network path creates avoidable outage.

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Migration 7 Rs: portfolio to wave plan · decision and control

1 Rehost

USE WHEN

Speed and minimal change

TRADE-OFF

Carries legacy inefficiency

2 Replatform

USE WHEN

Moderate managed-service benefit

TRADE-OFF

Needs targeted change and testing

3 Refactor

USE WHEN

Maximum cloud-native value

TRADE-OFF

Highest time and delivery risk

FAILURE MODE

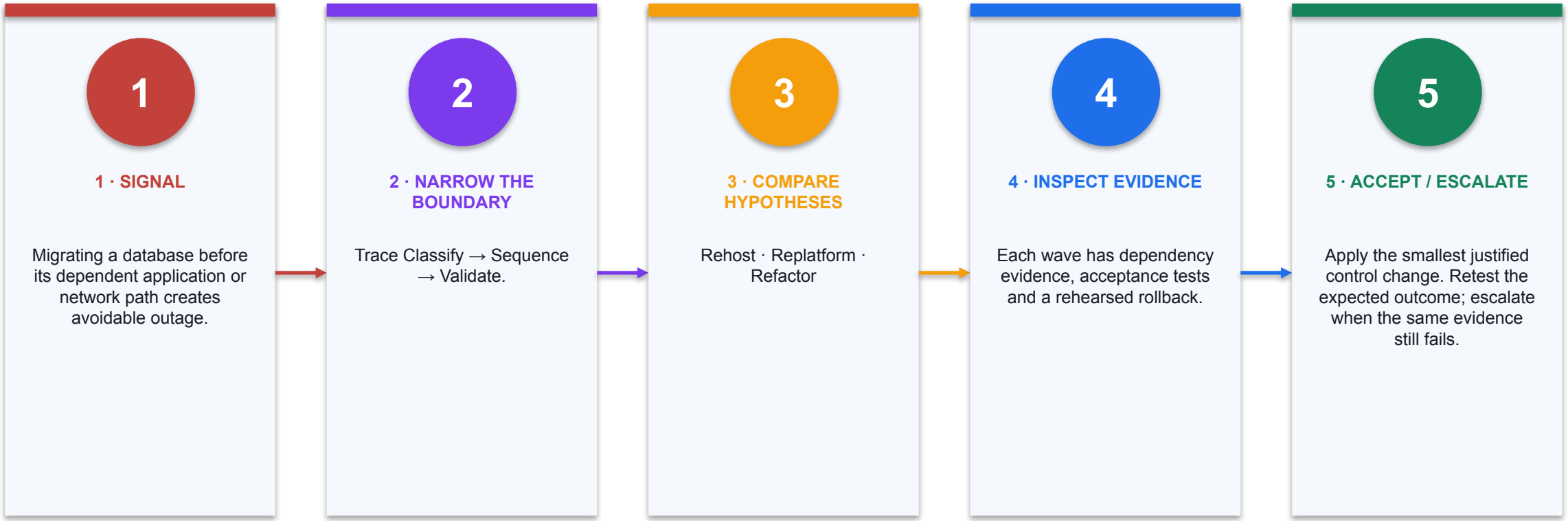
Migrating a database before its dependent application or network path creates avoidable outage.

VERIFICATION EVIDENCE

Each wave has dependency evidence, acceptance tests and a rehearsed rollback.

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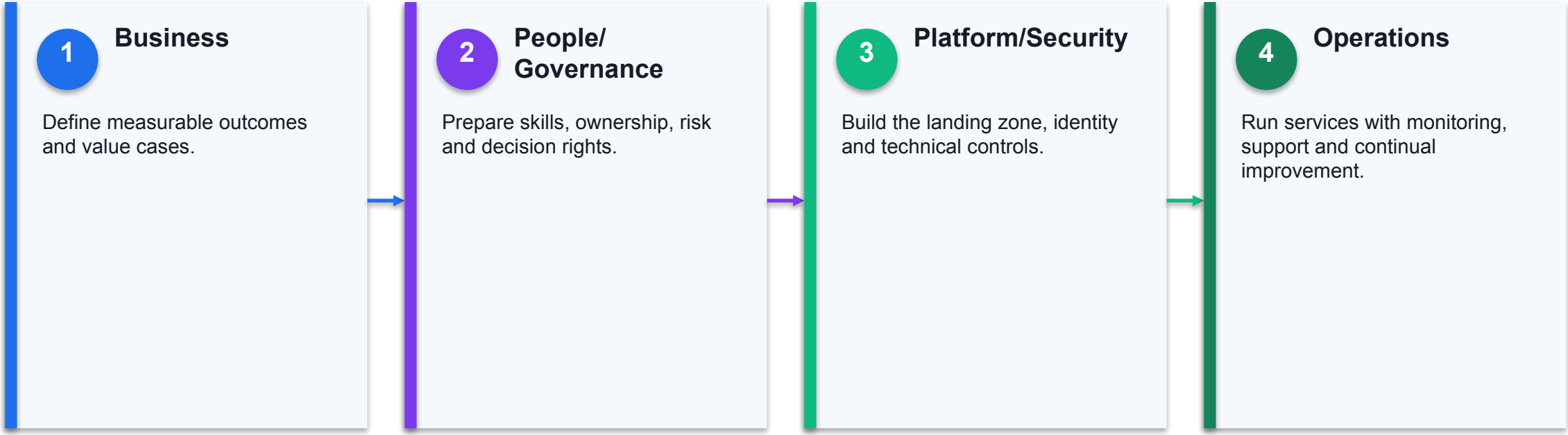
Migration 7 Rs: portfolio to wave plan · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Cloud Adoption Framework perspectives



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Cloud Adoption Framework perspectives · technical contract

1	INPUT / INTENT **Business**	Define measurable outcomes and value cases.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **People/Governance**	Prepare skills, ownership, risk and decision rights.	RETAIN **Approved configuration**
3	STATE CHANGE **Platform/Security**	Build the landing zone, identity and technical controls.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Operations**	Run services with monitoring, support and continual improvement.	RETAIN **Audit, metric or test record**

CONTROL RISK A platform launch without people, governance and operations creates an unmanaged cloud estate.

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Cloud Adoption Framework perspectives · decision and control

1 Envision

USE WHEN

Agree transformation outcomes

TRADE-OFF

Avoid technology-only goals

2 Align

USE WHEN

Identify capability gaps

TRADE-OFF

Requires cross-functional ownership

3 Launch/Scale

USE WHEN

Deliver pilots then expand

TRADE-OFF

Evidence must drive repeatability

FAILURE MODE

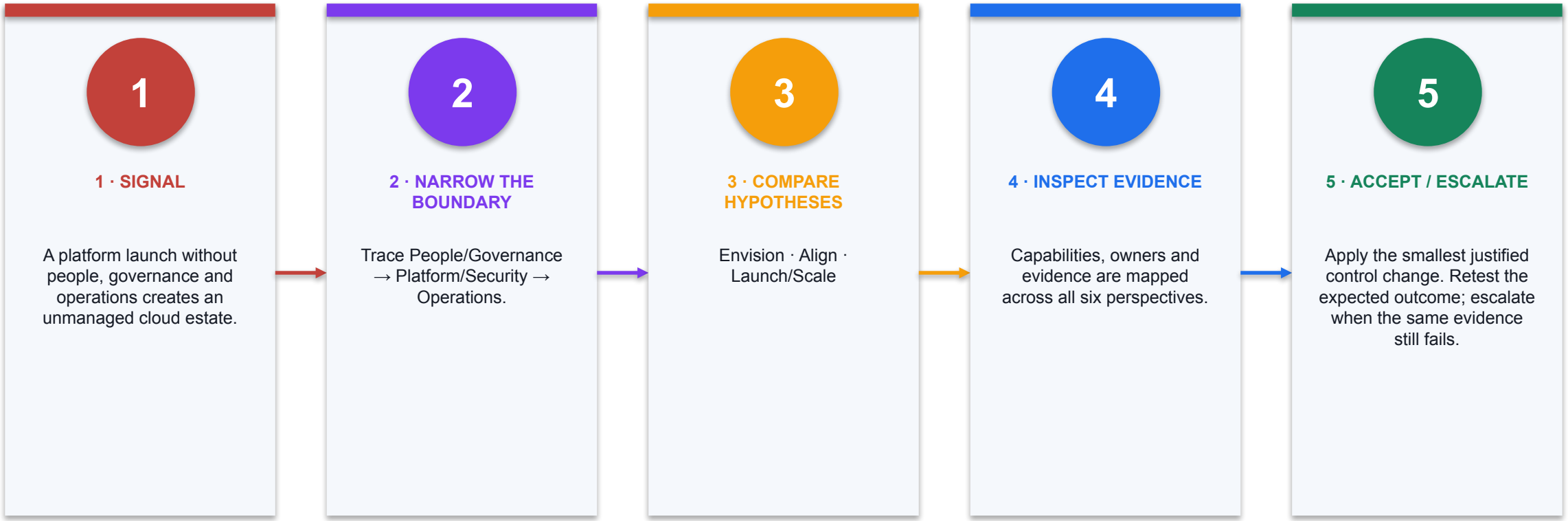
A platform launch without people, governance and operations creates an unmanaged cloud estate.

VERIFICATION EVIDENCE

Capabilities, owners and evidence are mapped across all six perspectives.

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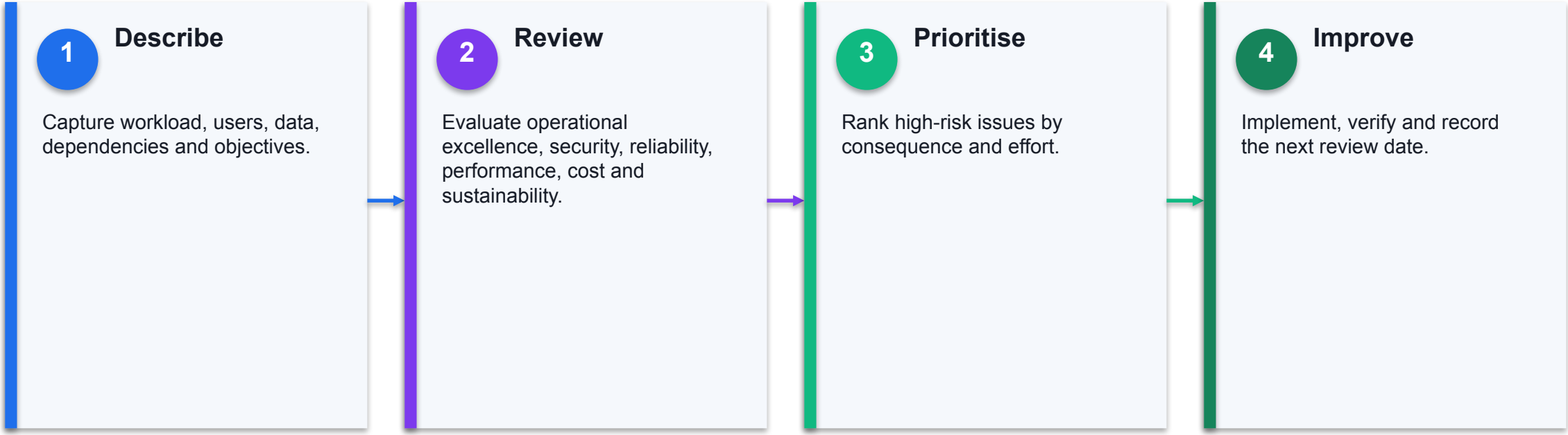
Cloud Adoption Framework perspectives · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Well-Architected: six-pillar review loop



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Well-Architected: six-pillar review loop · technical contract

1	INPUT / INTENT **Describe**	Capture workload, users, data, dependencies and objectives.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Review**	Evaluate operational excellence, security, reliability, performance, cost and sustainability.	RETAIN **Approved configuration**
3	STATE CHANGE **Prioritise**	Rank high-risk issues by consequence and effort.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Improve**	Implement, verify and record the next review date.	RETAIN **Audit, metric or test record**

CONTROL RISK Treating pillars as a one-time checklist leaves new risks undiscovered as the workload changes.

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Well-Architected: six-pillar review loop · decision and control

1 Operational

USE WHEN

Run and improve the workload

TRADE-OFF

Operations as code and observability

2 Risk

USE WHEN

Protect and recover

TRADE-OFF

Security and reliability controls

3 Efficiency

USE WHEN

Use resources well

TRADE-OFF

Performance, cost and sustainability

FAILURE MODE

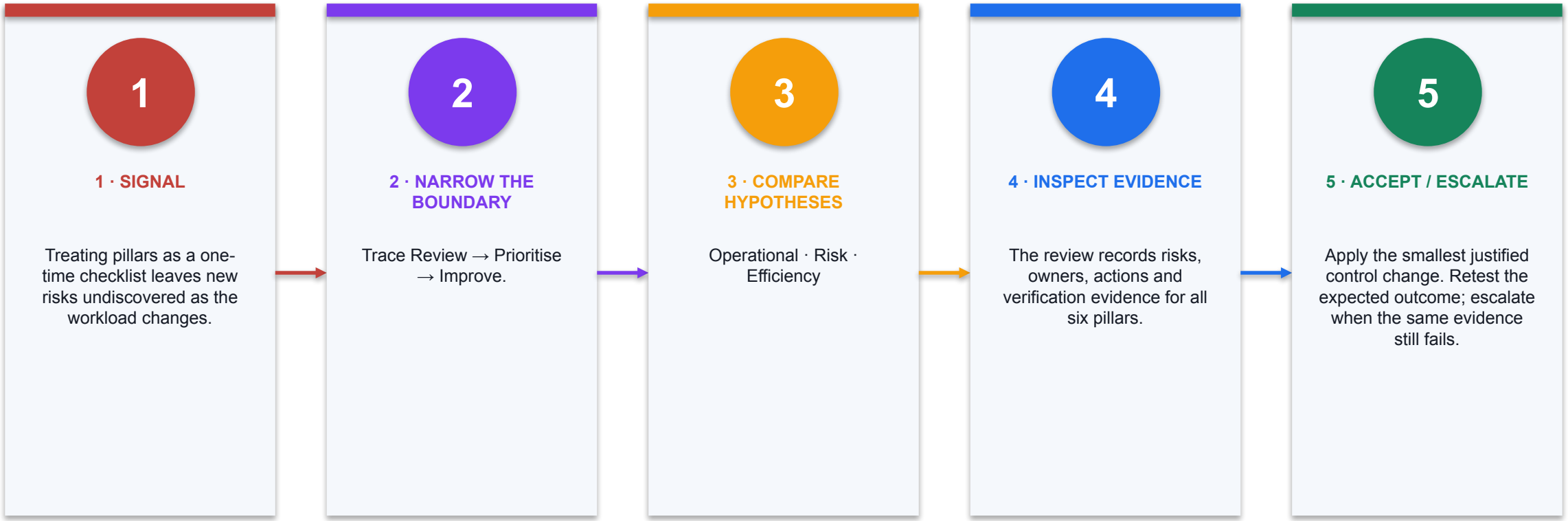
Treating pillars as a one-time checklist leaves new risks undiscovered as the workload changes.

VERIFICATION EVIDENCE

The review records risks, owners, actions and verification evidence for all six pillars.

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Well-Architected: six-pillar review loop · failure-to-evidence case



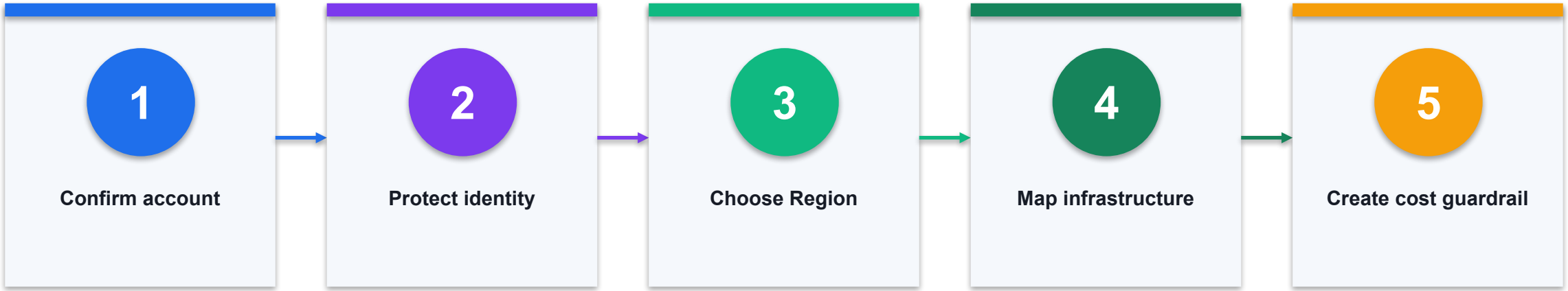
Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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DOMAIN 1 · LAB 1 · 40 MIN

Secure Account, Region and Cost Baseline

A project team needs a controlled AWS starting point before any workload resources are created.



YOU'LL PRODUCE **Account safety checklist, Region decision record, global-infrastructure sketch and budget evidence.**

Tools: AWS Management Console, IAM, Billing and Cost Management, AWS Budgets · Detailed procedure: Learner Guide, Lab 1

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Secure Account, Region and Cost Baseline · verification gate

1

PURPOSE

LO1 / A1 - establish a safe AWS learning environment and region decision.

2

ARTIFACT SET

Account safety checklist, Region decision record, global-infrastructure sketch and budget evidence.

3

ACCEPTANCE TEST

No credentials are captured; root use is avoided; the Region decision covers all four constraints; a budget threshold and recipient are visible.

EVIDENCE BOUNDARY **AWS Management Console, IAM, Billing and Cost Management, AWS Budgets**

If an acceptance condition fails, retain the timestamped evidence and escalate with the tested boundary and observed result.

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RECAP · VERIFY BEFORE MOVING ON

Domain 1 evidence recap

1 Operating model

A request can be traced to an API-created resource, control evidence and a metered line item.

2 Cloud advantages

Utilisation, deployment lead time and unit cost improve against an agreed baseline.

3 Deployment models

Every component has an owner, connection path, data classification and recovery route.

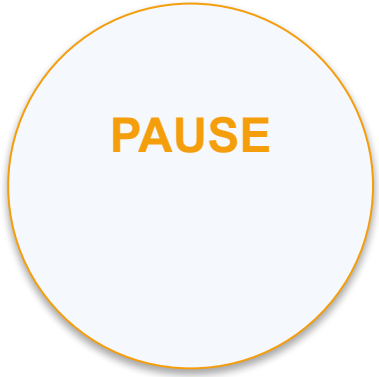
4 Service abstraction

The team can name which layers AWS manages and which controls remain with the customer.

MECHANISM

Read left to right: input, control, state change, observable evidence.

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Lunch Break

60 minutes

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02

DOMAIN 2 · EXAM WEIGHT 30%

Security and Compliance

Shared responsibility · identity · data protection · detection · governance

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GENERATED VISUAL · ANNOTATED

Shared responsibility and resilient boundaries



Provider boundary

Physical facilities, hardware and managed-service infrastructure.

Customer boundary

Identity, data, workload and configuration controls.

Fault isolation

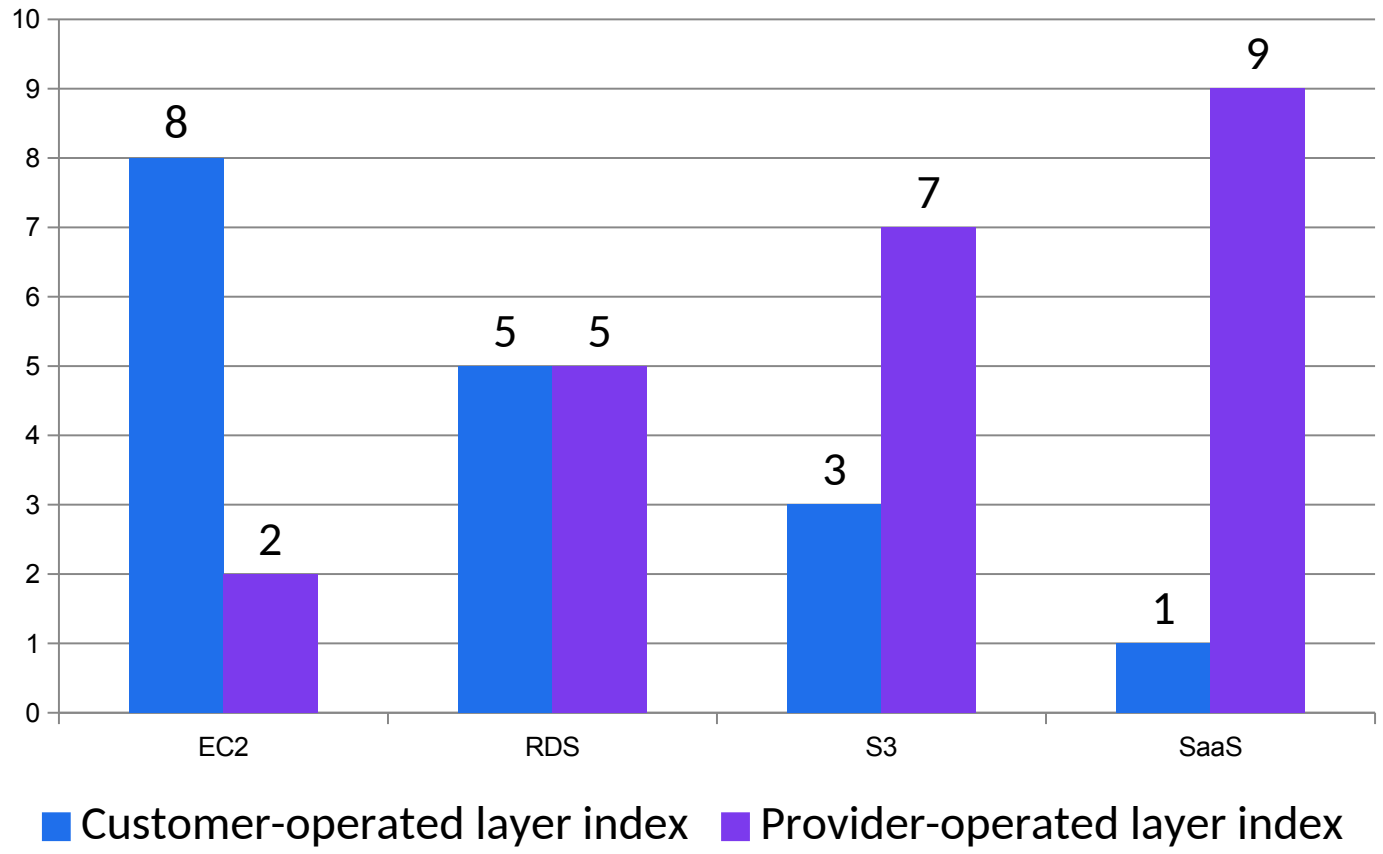
Multi-AZ design contains one infrastructure failure.

Evidence

Policies, logs and tests prove the customer controls.

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Responsibility shifts with abstraction



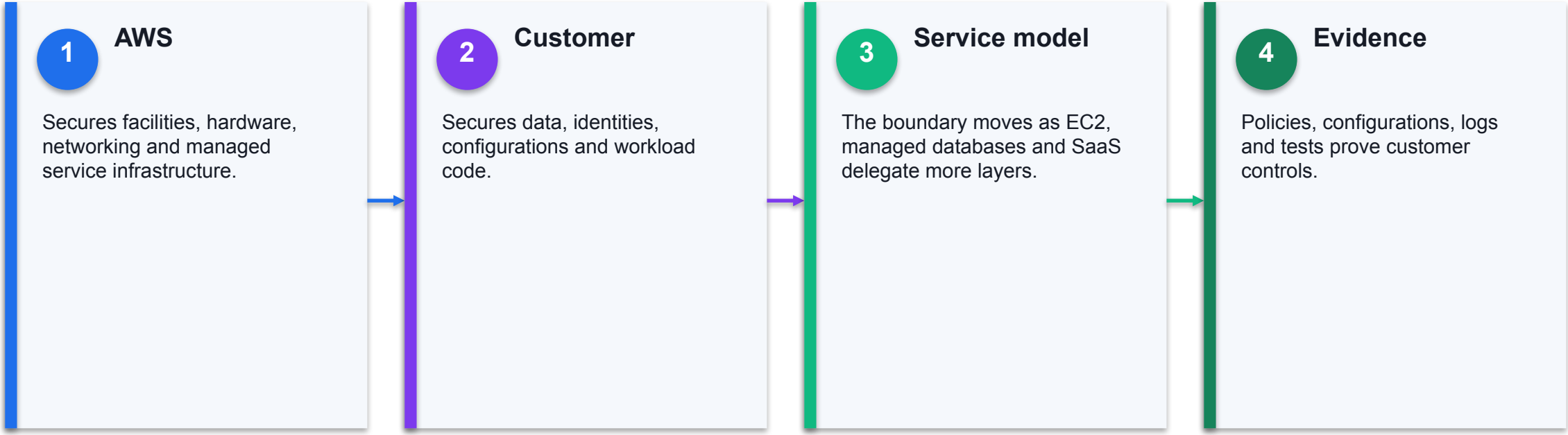
WHAT THE DATA SHOWS

Managed services move more infrastructure operation to the provider; customer responsibility for identities, data and configuration remains.

Editable PowerPoint chart

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Shared responsibility changes by service



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Shared responsibility changes by service · technical contract

1	INPUT / INTENT **AWS**	Secures facilities, hardware, networking and managed service infrastructure.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Customer**	Secures data, identities, configurations and workload code.	RETAIN **Approved configuration**
3	STATE CHANGE **Service model**	The boundary moves as EC2, managed databases and SaaS delegate more layers.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Evidence**	Policies, configurations, logs and tests prove customer controls.	RETAIN **Audit, metric or test record**

CONTROL RISK Assuming AWS configures least privilege or data classification for the customer causes exposure.

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Shared responsibility changes by service · decision and control

1 EC2

USE WHEN

Customer needs OS control

TRADE-OFF

Customer patches guest OS

2 RDS

USE WHEN

Managed database engine

TRADE-OFF

Customer controls accounts and data

3 S3

USE WHEN

Managed object storage

TRADE-OFF

Customer controls access and data lifecycle

FAILURE MODE

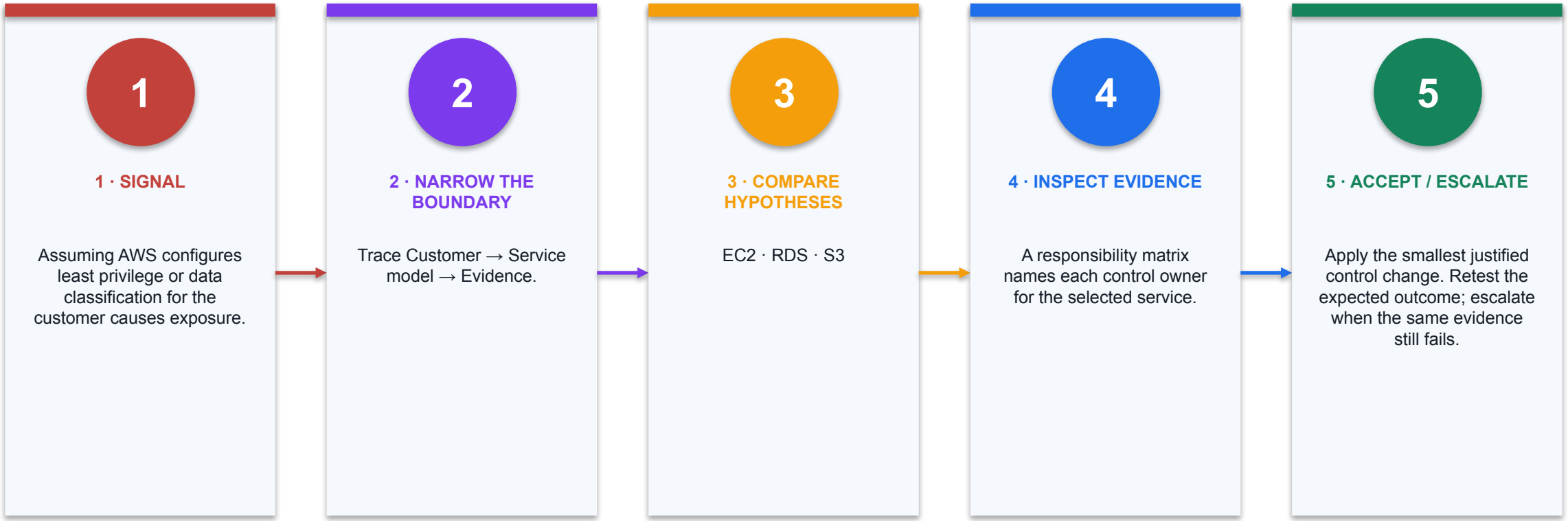
Assuming AWS configures least privilege or data classification for the customer causes exposure.

VERIFICATION EVIDENCE

A responsibility matrix names each control owner for the selected service.

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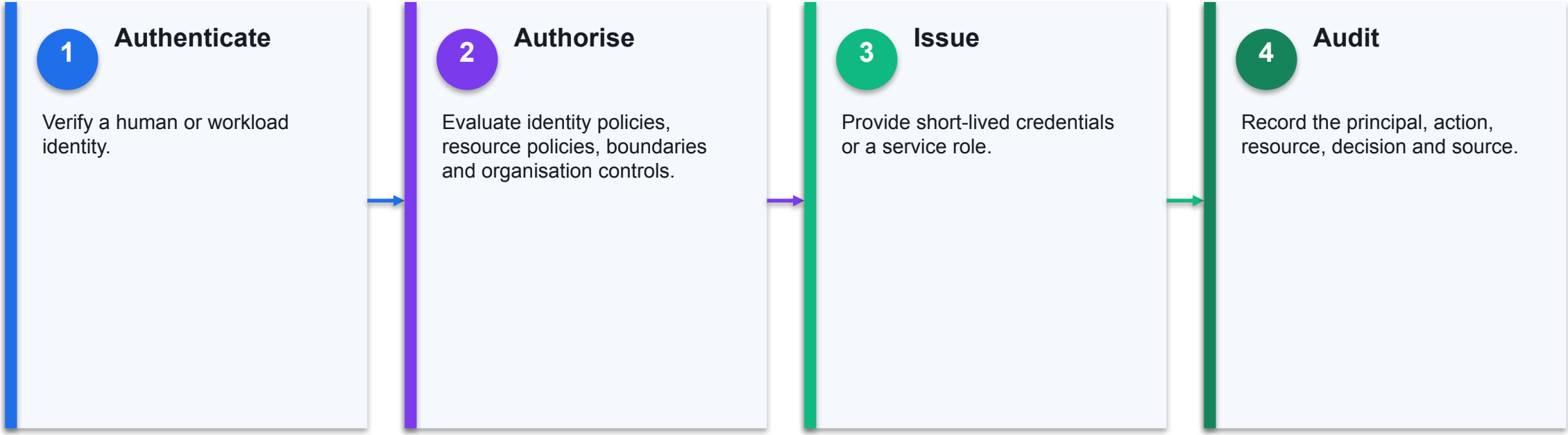
Shared responsibility changes by service · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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IAM identities: human, workload and federation



MECHANISM

Read left to right: input, control, state change, observable evidence.

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IAM identities: human, workload and federation · technical contract

1	INPUT / INTENT **Authenticate**	Verify a human or workload identity.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Authorise**	Evaluate identity policies, resource policies, boundaries and organisation controls.	RETAIN **Approved configuration**
3	STATE CHANGE **Issue**	Provide short-lived credentials or a service role.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Audit**	Record the principal, action, resource, decision and source.	RETAIN **Audit, metric or test record**

CONTROL RISK Embedding long-lived access keys in code bypasses rotation and increases blast radius.

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IAM identities: human, workload and federation · decision and control

1 IAM user

USE WHEN

Exceptional long-lived human/API identity

TRADE-OFF

Credential rotation burden

2 IAM role

USE WHEN

AWS service or workload access

TRADE-OFF

Requires correct trust policy

3 Federation/Identity Center

USE WHEN

Workforce single sign-on

TRADE-OFF

Depends on identity-provider governance

FAILURE MODE

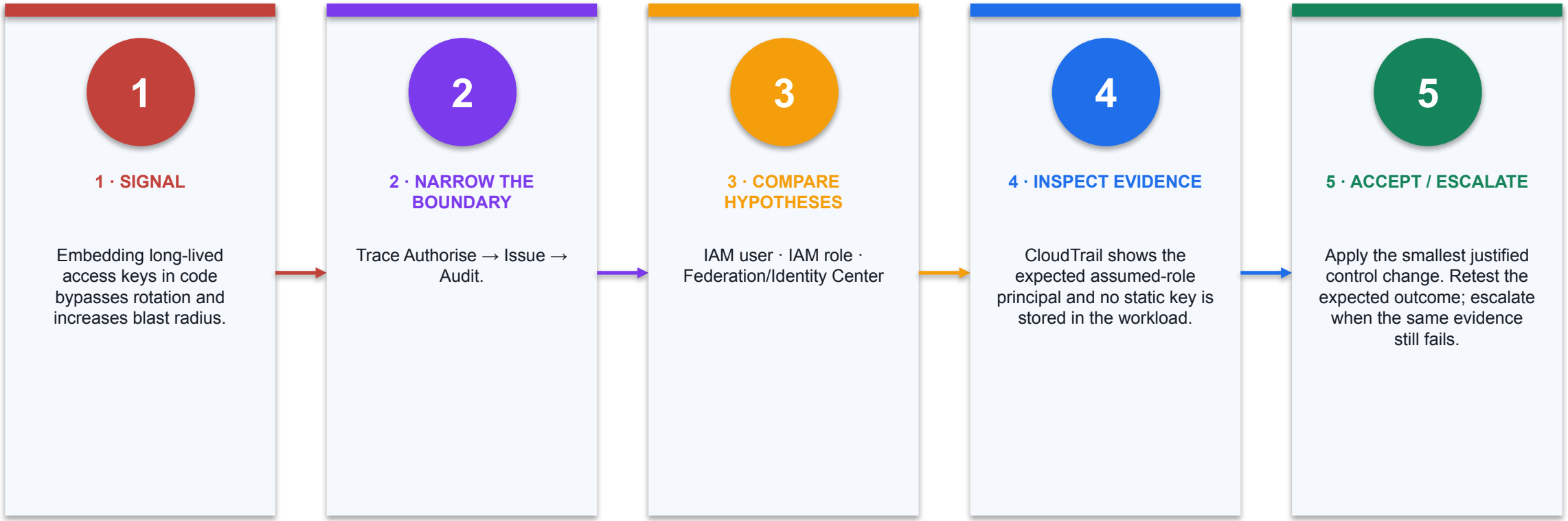
Embedding long-lived access keys in code bypasses rotation and increases blast radius.

VERIFICATION EVIDENCE

CloudTrail shows the expected assumed-role principal and no static key is stored in the workload.

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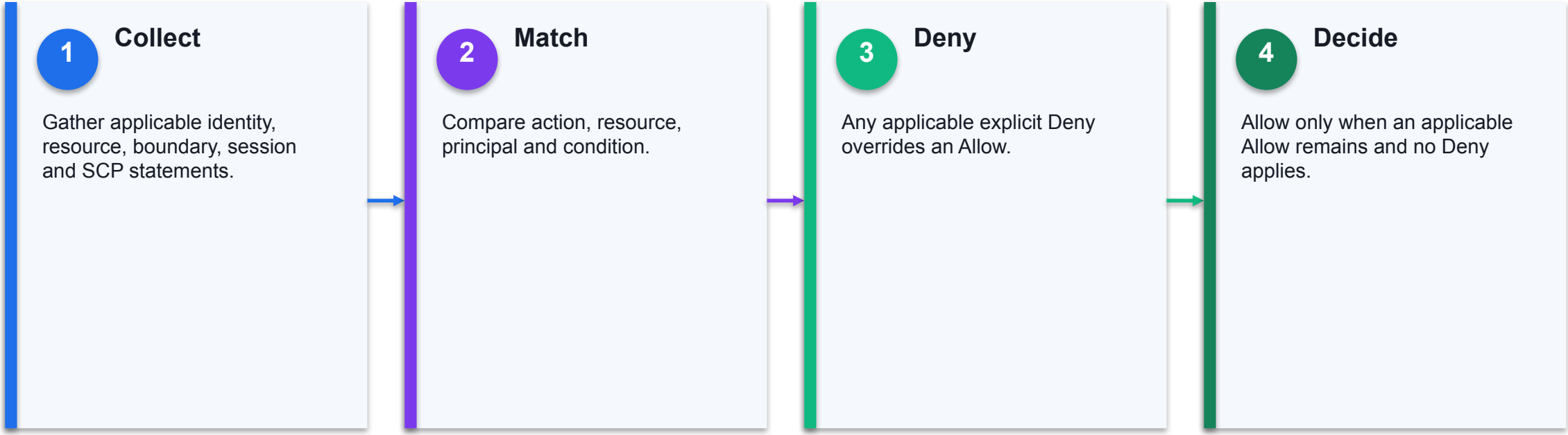
IAM identities: human, workload and federation · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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IAM policy evaluation is an explicit decision engine



MECHANISM

Read left to right: input, control, state change, observable evidence.

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IAM policy evaluation is an explicit decision engine · technical contract

1	INPUT / INTENT **Collect**	Gather applicable identity, resource, boundary, session and SCP statements.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Match**	Compare action, resource, principal and condition.	RETAIN **Approved configuration**
3	STATE CHANGE **Deny**	Any applicable explicit Deny overrides an Allow.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Decide**	Allow only when an applicable Allow remains and no Deny applies.	RETAIN **Audit, metric or test record**

CONTROL RISK An Allow in one policy cannot override an explicit Deny in another applicable policy.

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IAM policy evaluation is an explicit decision engine · decision and control

1 Identity policy

USE WHEN

Permissions for a principal

TRADE-OFF

Moves with the identity

2 Resource policy

USE WHEN

Who can access one resource

TRADE-OFF

Principal is part of the policy

3 Permissions boundary

USE WHEN

Maximum delegated permissions

TRADE-OFF

Does not grant permissions alone

FAILURE MODE

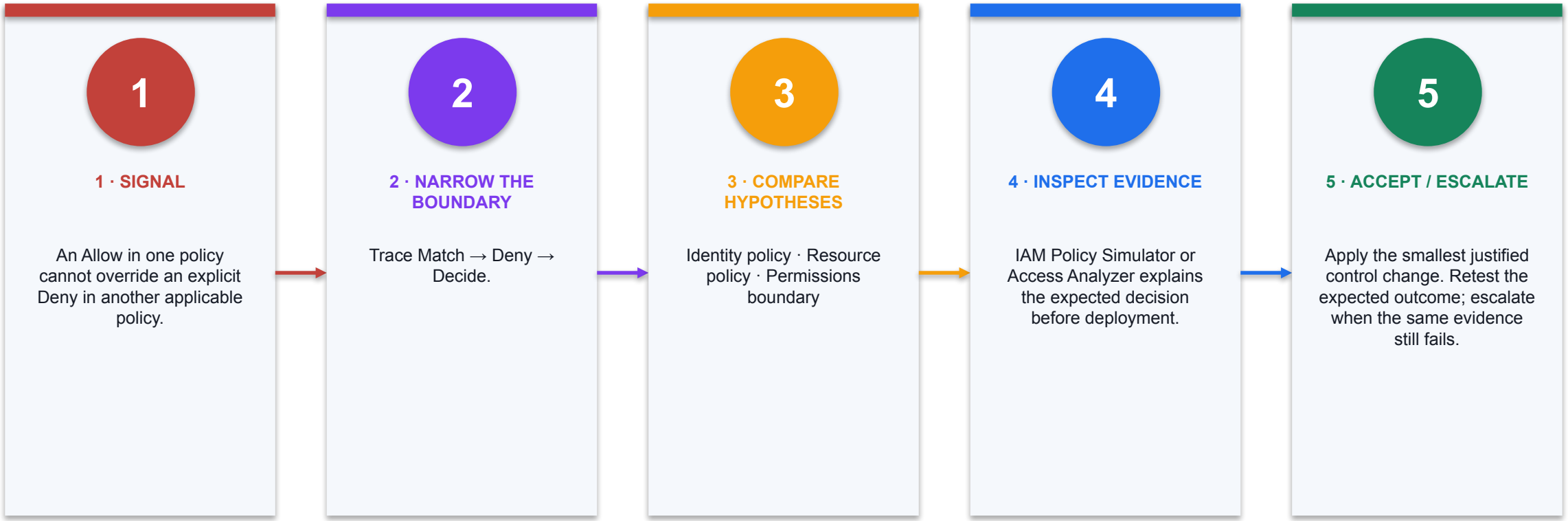
An Allow in one policy cannot override an explicit Deny in another applicable policy.

VERIFICATION EVIDENCE

IAM Policy Simulator or Access Analyzer explains the expected decision before deployment.

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IAM policy evaluation is an explicit decision engine · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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MFA, temporary credentials and least privilege



MECHANISM

Read left to right: input, control, state change, observable evidence.

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MFA, temporary credentials and least privilege · technical contract

1	INPUT / INTENT **Strong sign-in**	Require MFA and protect the root user.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Short sessions**	Use STS-issued temporary credentials instead of static keys.	RETAIN **Approved configuration**
3	STATE CHANGE **Narrow scope**	Limit actions, resources and conditions to the task.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Review**	Use access analysis and last-accessed data to remove excess.	RETAIN **Audit, metric or test record**

CONTROL RISK MFA on an administrator does not make an over-broad policy safe.

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MFA, temporary credentials and least privilege · decision and control

1 MFA

USE WHEN

Human sign-in and sensitive actions

TRADE-OFF

Does not replace authorisation

2 Role session

USE WHEN

Workloads and cross-account access

TRADE-OFF

Trust policy is critical

3 Access Analyzer

USE WHEN

Find external or unused access

TRADE-OFF

Findings require owner action

FAILURE MODE

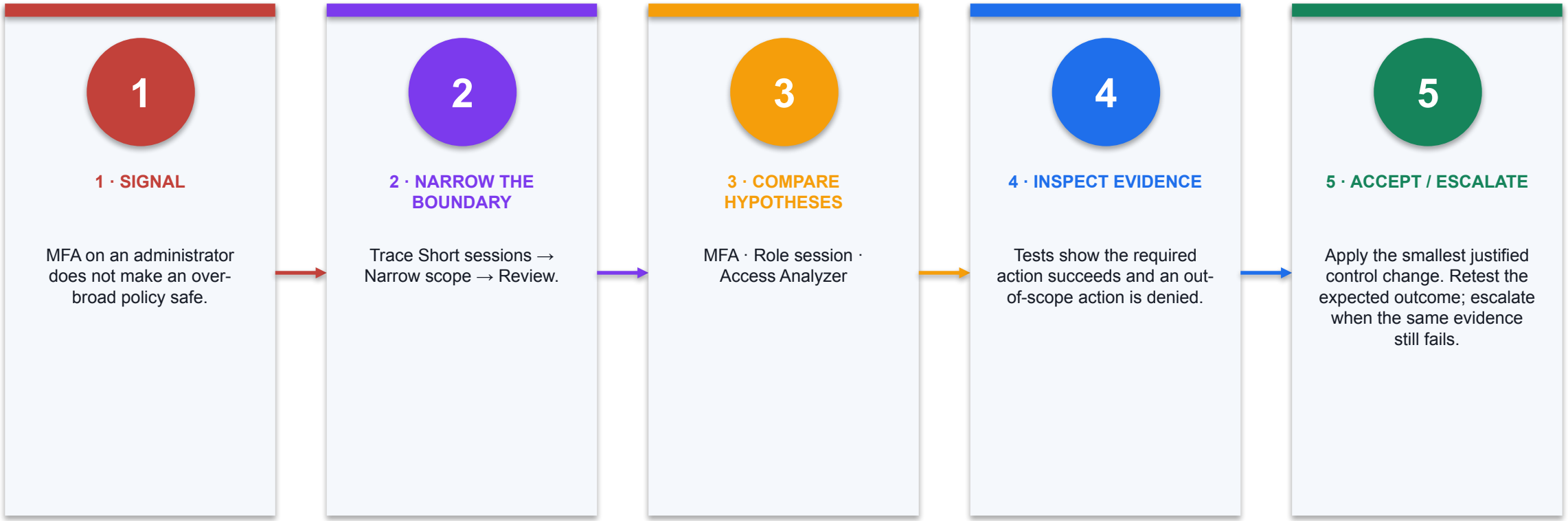
MFA on an administrator does not make an over-broad policy safe.

VERIFICATION EVIDENCE

Tests show the required action succeeds and an out-of-scope action is denied.

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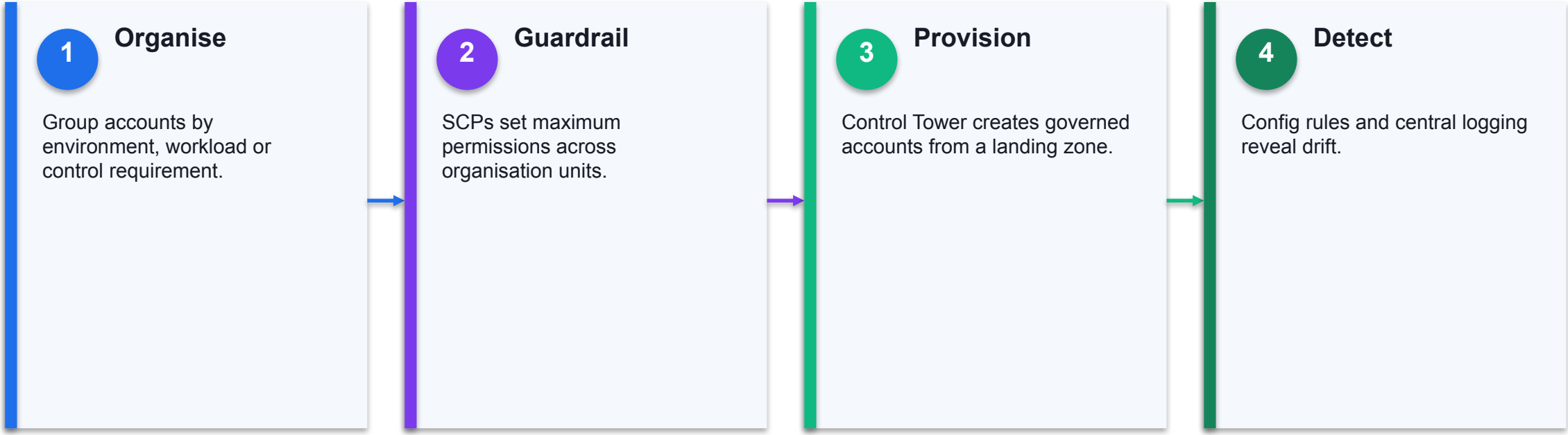
MFA, temporary credentials and least privilege · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Organisations, SCPs and Control Tower



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Organisations, SCPs and Control Tower · technical contract

1	INPUT / INTENT **Organise**	Group accounts by environment, workload or control requirement.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Guardrail**	SCPs set maximum permissions across organisation units.	RETAIN **Approved configuration**
3	STATE CHANGE **Provision**	Control Tower creates governed accounts from a landing zone.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Detect**	Config rules and central logging reveal drift.	RETAIN **Audit, metric or test record**

CONTROL RISK Attaching a restrictive SCP without testing can block required operations across many accounts.

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Organisations, SCPs and Control Tower · decision and control

1 AWS Organizations

USE WHEN

Multi-account billing and hierarchy

TRADE-OFF

Requires account design

2 SCP

USE WHEN

Prevent prohibited actions

TRADE-OFF

Does not grant permissions

3 Control Tower

USE WHEN

Opinionated landing-zone governance

TRADE-OFF

Customisation must respect controls

FAILURE MODE

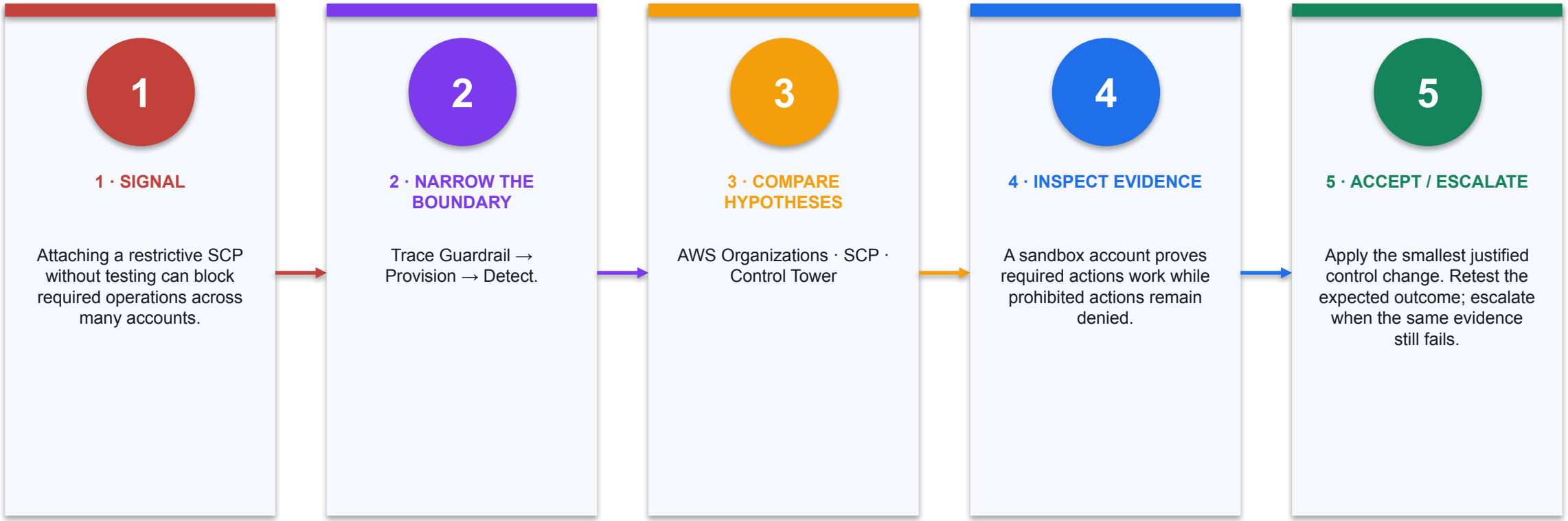
Attaching a restrictive SCP without testing can block required operations across many accounts.

VERIFICATION EVIDENCE

A sandbox account proves required actions work while prohibited actions remain denied.

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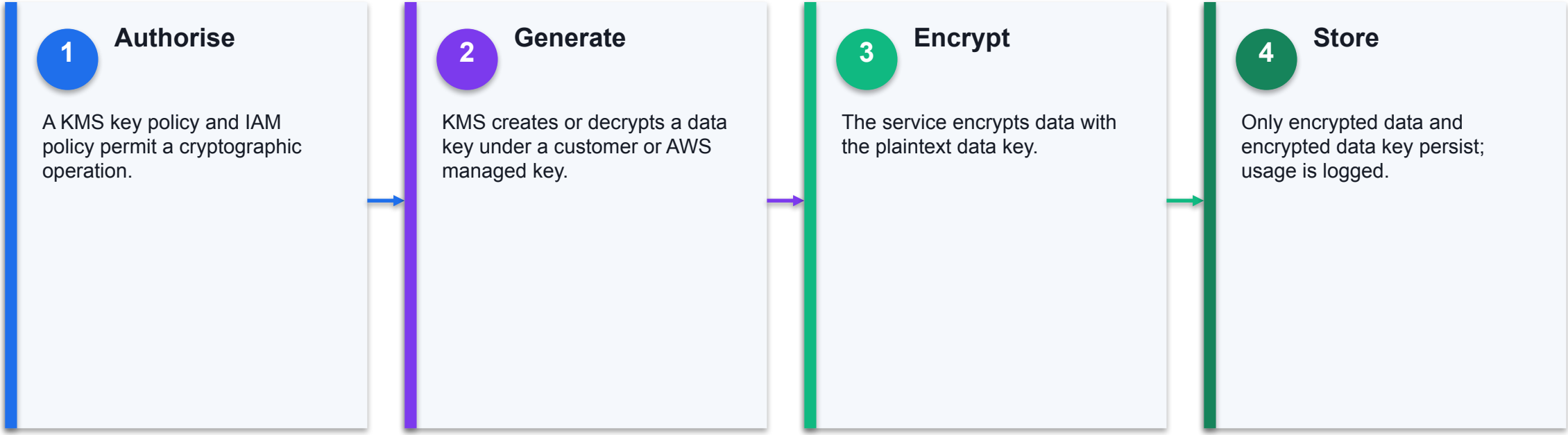
Organisations, SCPs and Control Tower · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Encryption: data keys, KMS and CloudHSM



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Encryption: data keys, KMS and CloudHSM · technical contract

1	INPUT / INTENT **Authorise**	A KMS key policy and IAM policy permit a cryptographic operation.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Generate**	KMS creates or decrypts a data key under a customer or AWS managed key.	RETAIN **Approved configuration**
3	STATE CHANGE **Encrypt**	The service encrypts data with the plaintext data key.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Store**	Only encrypted data and encrypted data key persist; usage is logged.	RETAIN **Audit, metric or test record**

CONTROL RISK Encrypting data at rest while exposing the KMS key through a broad policy defeats the control.

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Encryption: data keys, KMS and CloudHSM · decision and control

1 AWS owned key

USE WHEN

Default service protection

TRADE-OFF

Least customer visibility

2 Customer managed KMS key

USE WHEN

Control rotation, policy and audit

TRADE-OFF

Management responsibility and cost

3 CloudHSM

USE WHEN

Dedicated HSM control

TRADE-OFF

Higher complexity and operating burden

FAILURE MODE

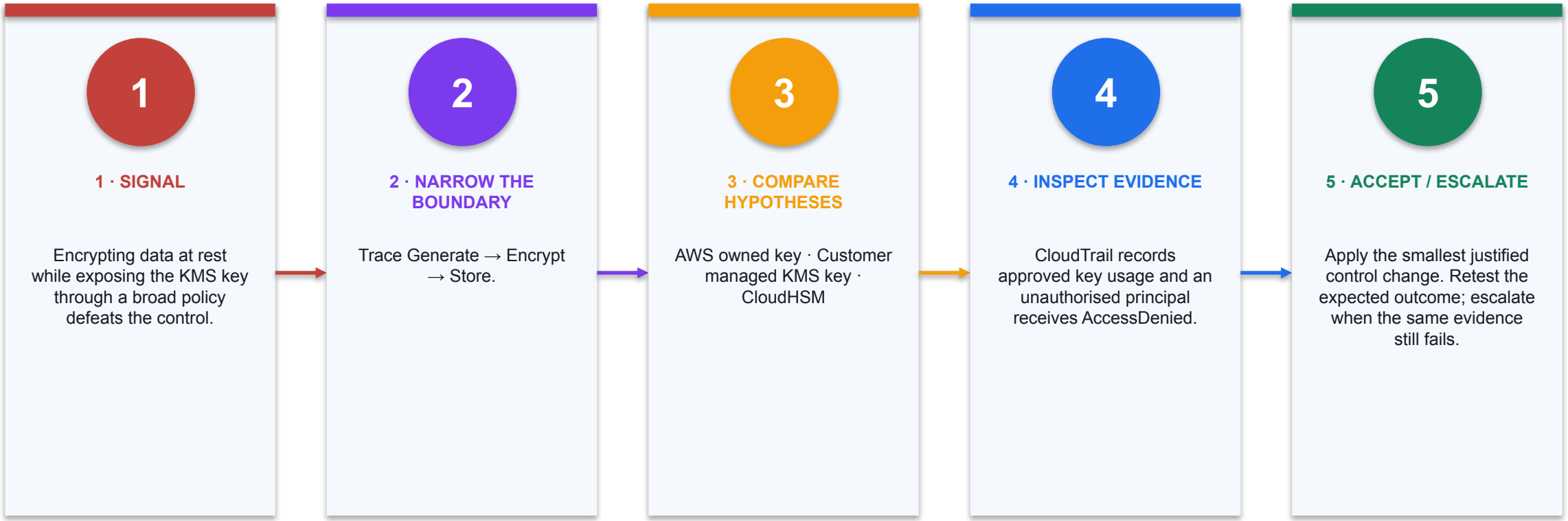
Encrypting data at rest while exposing the KMS key through a broad policy defeats the control.

VERIFICATION EVIDENCE

CloudTrail records approved key usage and an unauthorised principal receives AccessDenied.

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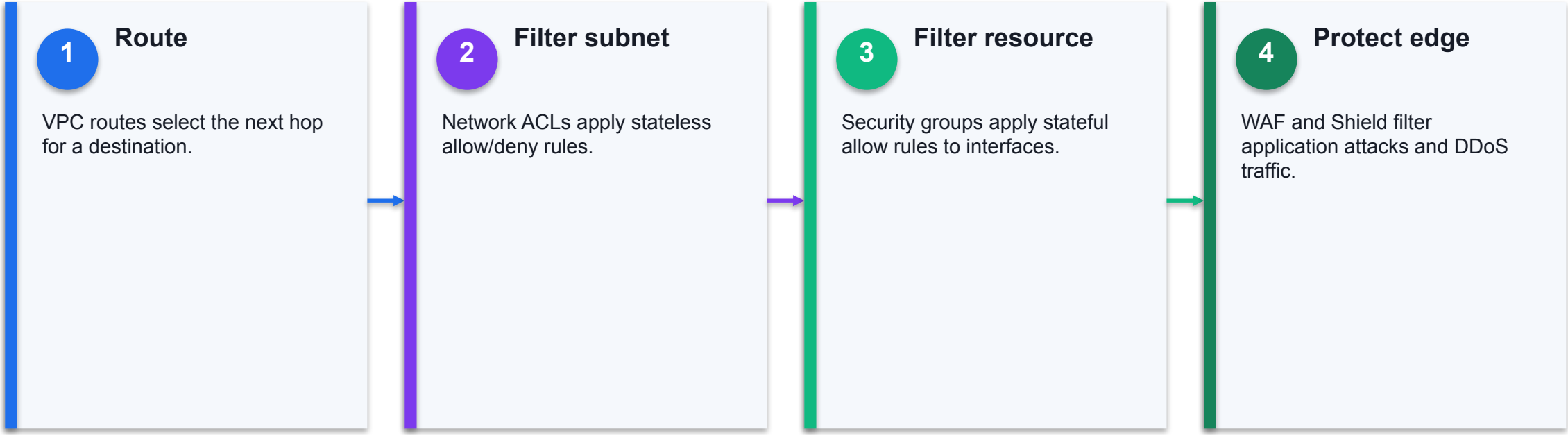
Encryption: data keys, KMS and CloudHSM · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Network security controls operate at different layers



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Network security controls operate at different layers · technical contract

1	INPUT / INTENT **Route**	VPC routes select the next hop for a destination.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Filter subnet**	Network ACLs apply stateless allow/deny rules.	RETAIN **Approved configuration**
3	STATE CHANGE **Filter resource**	Security groups apply stateful allow rules to interfaces.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Protect edge**	WAF and Shield filter application attacks and DDoS traffic.	RETAIN **Audit, metric or test record**

CONTROL RISK Opening 0.0.0.0/0 on an administrative port bypasses network segmentation intent.

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Network security controls operate at different layers · decision and control

1 Security group

USE WHEN

Instance/ENI traffic

TRADE-OFF

Stateful and allow-only

2 Network ACL

USE WHEN

Subnet boundary

TRADE-OFF

Stateless and ordered allow/deny

3 AWS WAF

USE WHEN

HTTP/S application requests

TRADE-OFF

Requires rule tuning

FAILURE MODE

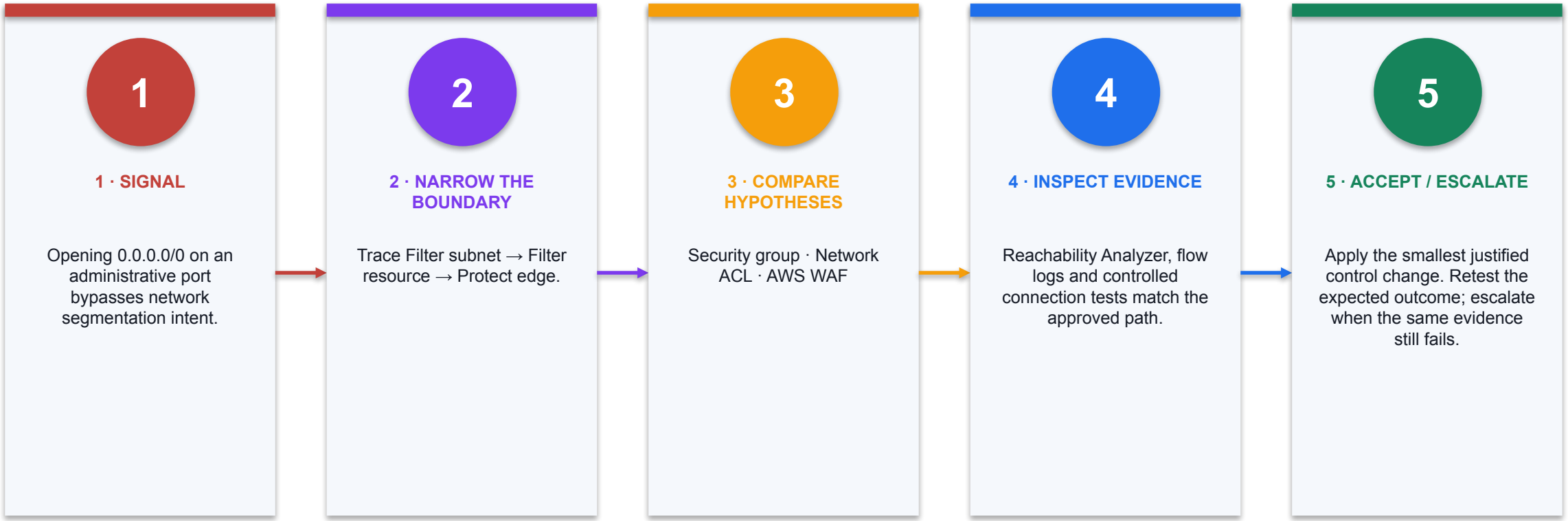
Opening 0.0.0.0/0 on an administrative port bypasses network segmentation intent.

VERIFICATION EVIDENCE

Reachability Analyzer, flow logs and controlled connection tests match the approved path.

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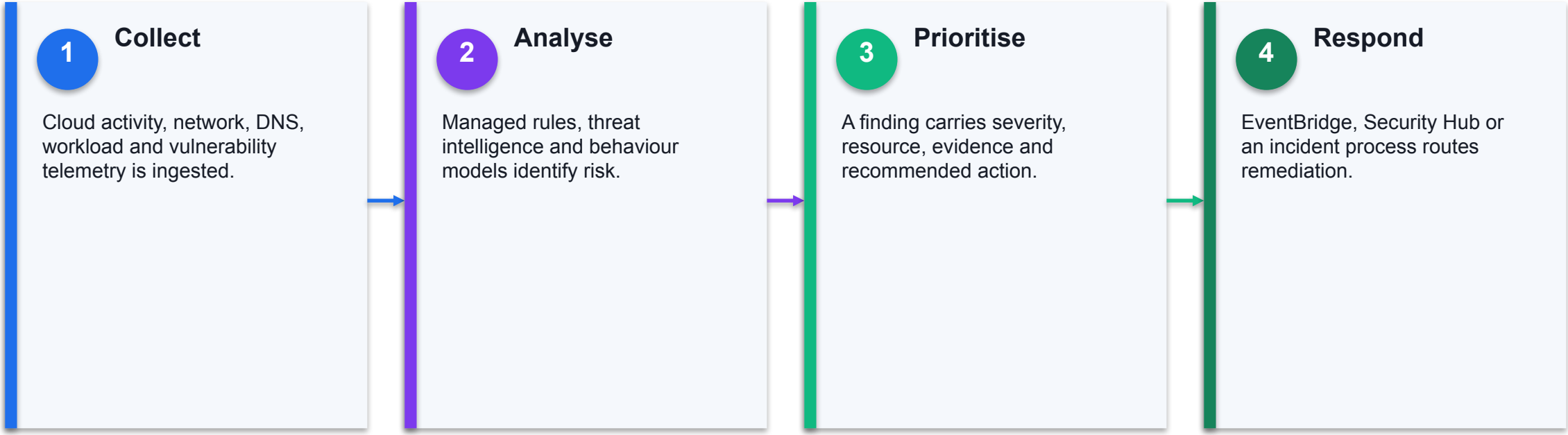
Network security controls operate at different layers · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Detection services: signal to finding



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Detection services: signal to finding · technical contract

1	INPUT / INTENT **Collect**	Cloud activity, network, DNS, workload and vulnerability telemetry is ingested.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Analyse**	Managed rules, threat intelligence and behaviour models identify risk.	RETAIN **Approved configuration**
3	STATE CHANGE **Prioritise**	A finding carries severity, resource, evidence and recommended action.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Respond**	EventBridge, Security Hub or an incident process routes remediation.	RETAIN **Audit, metric or test record**

CONTROL RISK Enabling a service without routing or owning findings produces detection without response.

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Detection services: signal to finding · decision and control

1 GuardDuty

USE WHEN

Threat detection from activity and telemetry

TRADE-OFF

Finding is not automatic containment

2 Inspector

USE WHEN

Workload and image vulnerabilities

TRADE-OFF

Coverage depends on supported resources

3 Macie

USE WHEN

Sensitive data discovery in S3

TRADE-OFF

Discovery scope and cost need control

FAILURE MODE

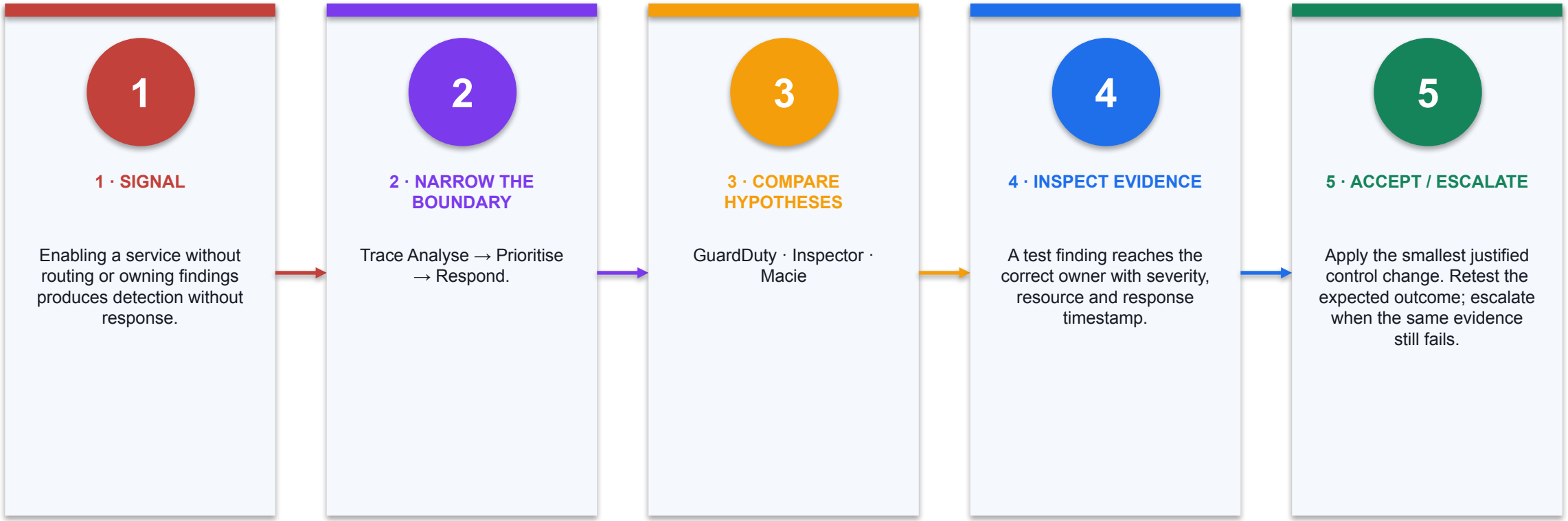
Enabling a service without routing or owning findings produces detection without response.

VERIFICATION EVIDENCE

A test finding reaches the correct owner with severity, resource and response timestamp.

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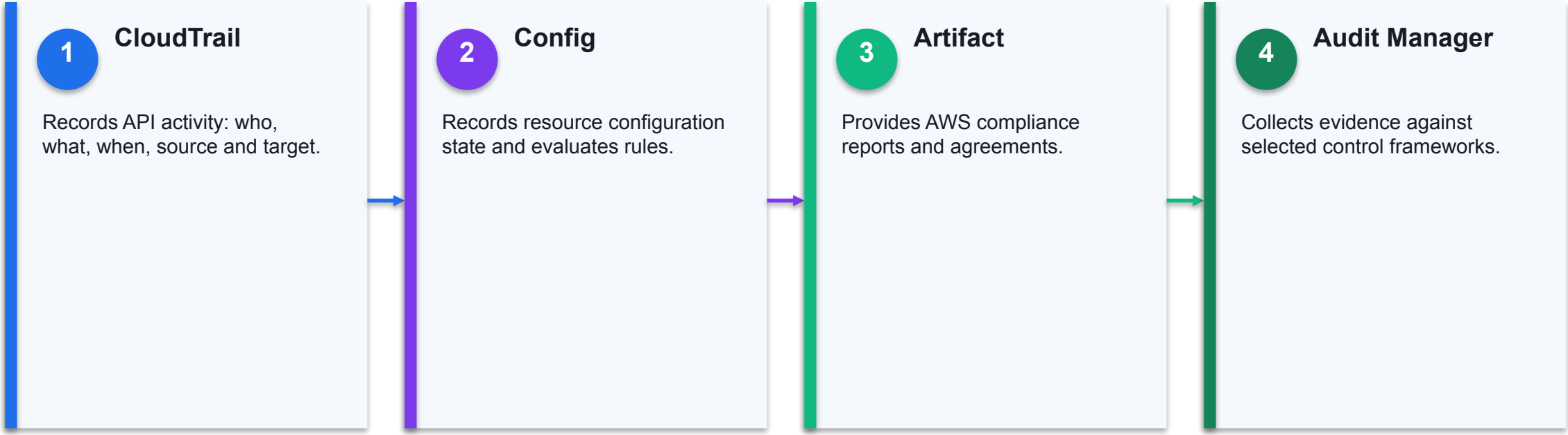
Detection services: signal to finding · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Audit, configuration and compliance evidence



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Audit, configuration and compliance evidence · technical contract

1	INPUT / INTENT **CloudTrail**	Records API activity: who, what, when, source and target.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Config**	Records resource configuration state and evaluates rules.	RETAIN **Approved configuration**
3	STATE CHANGE **Artifact**	Provides AWS compliance reports and agreements.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Audit Manager**	Collects evidence against selected control frameworks.	RETAIN **Audit, metric or test record**

CONTROL RISK CloudWatch metrics cannot substitute for an API audit trail or configuration history.

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Audit, configuration and compliance evidence · decision and control

1 Event history

USE WHEN

Recent investigation

TRADE-OFF

Limited history and scope

2 Trail/Lake

USE WHEN

Durable organisation-wide audit

TRADE-OFF

Storage and query design required

3 Config rule

USE WHEN

Detect configuration drift

TRADE-OFF

Remediation is a separate action

FAILURE MODE

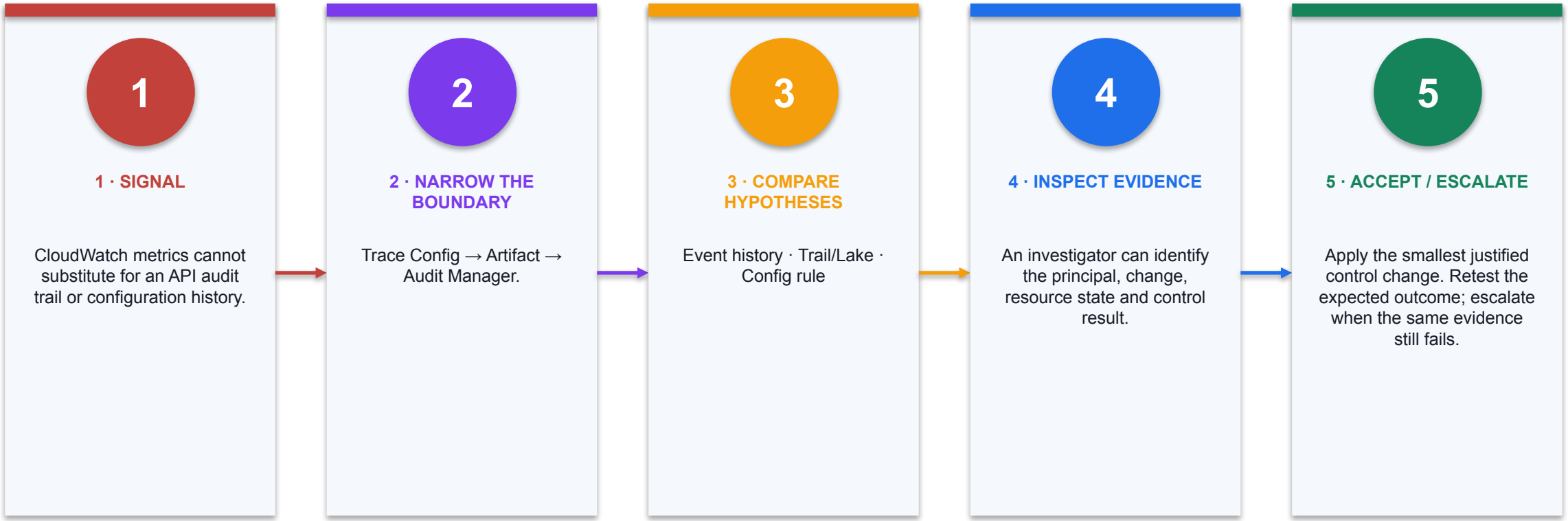
CloudWatch metrics cannot substitute for an API audit trail or configuration history.

VERIFICATION EVIDENCE

An investigator can identify the principal, change, resource state and control result.

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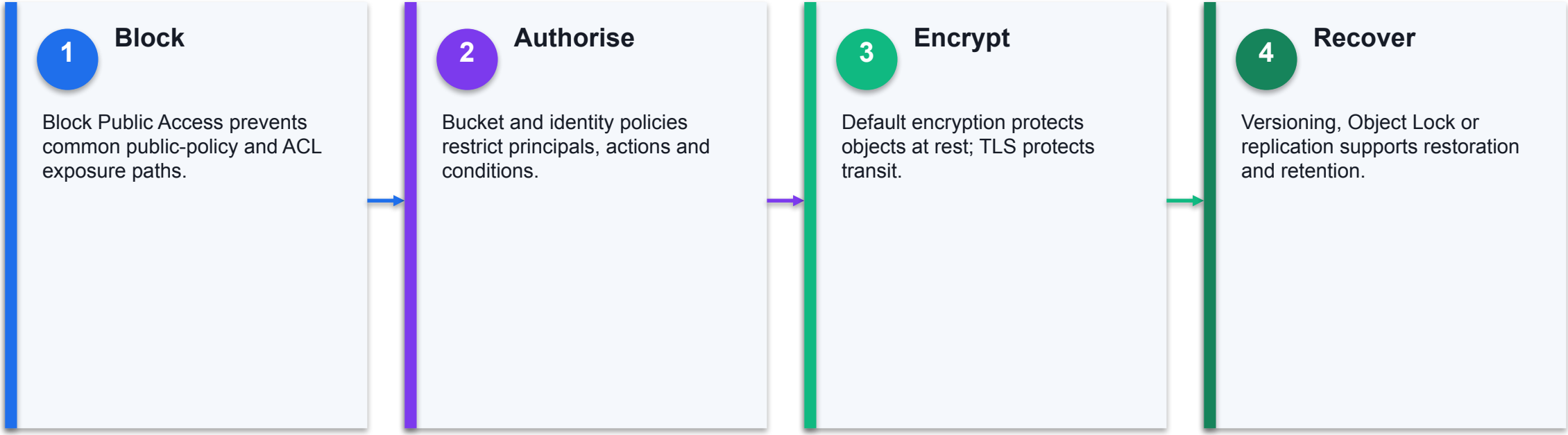
Audit, configuration and compliance evidence · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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S3 data protection is a layered control



MECHANISM

Read left to right: input, control, state change, observable evidence.

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S3 data protection is a layered control · technical contract

1	INPUT / INTENT **Block**	Block Public Access prevents common public-policy and ACL exposure paths.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Authorise**	Bucket and identity policies restrict principals, actions and conditions.	RETAIN **Approved configuration**
3	STATE CHANGE **Encrypt**	Default encryption protects objects at rest; TLS protects transit.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Recover**	Versioning, Object Lock or replication supports restoration and retention.	RETAIN **Audit, metric or test record**

CONTROL RISK Encryption does not prevent an authorised-but-over-broad principal from reading data.

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S3 data protection is a layered control · decision and control

1 Versioning

USE WHEN

Recover overwrite or deletion

TRADE-OFF

Adds storage versions and cost

2 Object Lock

USE WHEN

WORM retention requirement

TRADE-OFF

Requires governance/compliance design

3 Replication

USE WHEN

Separate failure domain or account

TRADE-OFF

Permissions and transfer cost

FAILURE MODE

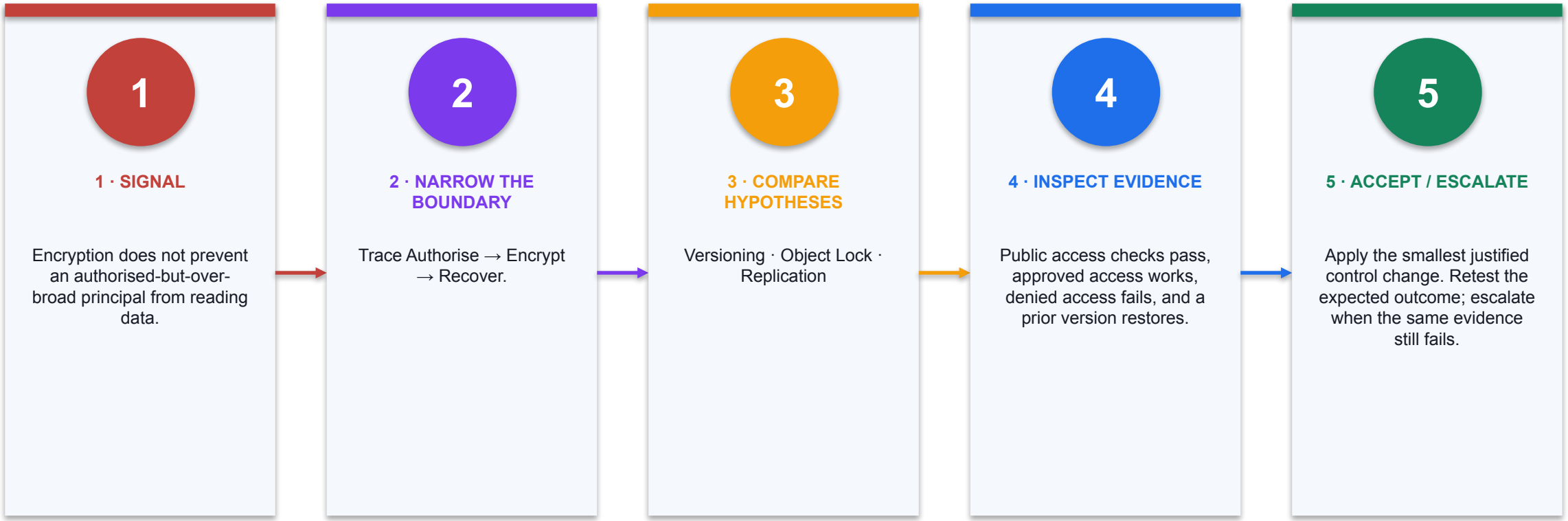
Encryption does not prevent an authorised-but-over-broad principal from reading data.

VERIFICATION EVIDENCE

Public access checks pass, approved access works, denied access fails, and a prior version restores.

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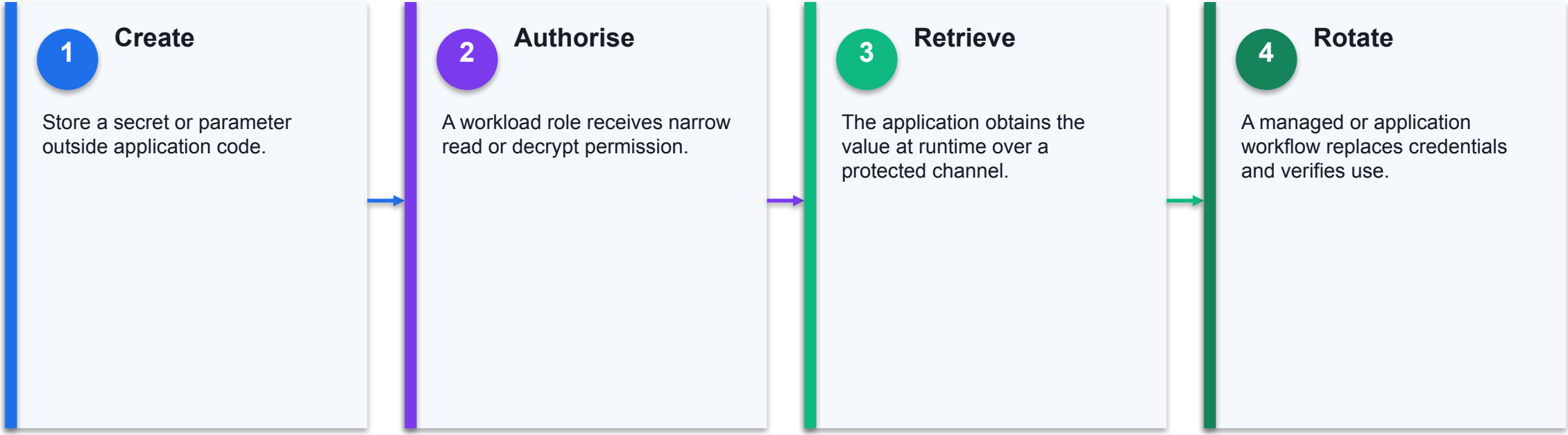
S3 data protection is a layered control · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Secrets, credentials and parameter storage



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Secrets, credentials and parameter storage · technical contract

1	INPUT / INTENT **Create**	Store a secret or parameter outside application code.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Authorise**	A workload role receives narrow read or decrypt permission.	RETAIN **Approved configuration**
3	STATE CHANGE **Retrieve**	The application obtains the value at runtime over a protected channel.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Rotate**	A managed or application workflow replaces credentials and verifies use.	RETAIN **Audit, metric or test record**

CONTROL RISK A secret copied into a container image remains exposed after the source value rotates.

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Secrets, credentials and parameter storage · decision and control

1 Secrets Manager

USE WHEN

Credentials needing rotation

TRADE-OFF

Per-secret and API cost

2 Parameter Store

USE WHEN

Configuration and secure strings

TRADE-OFF

Different feature and throughput tiers

3 Environment variable

USE WHEN

Non-secret runtime configuration

TRADE-OFF

Can leak through diagnostics

FAILURE MODE

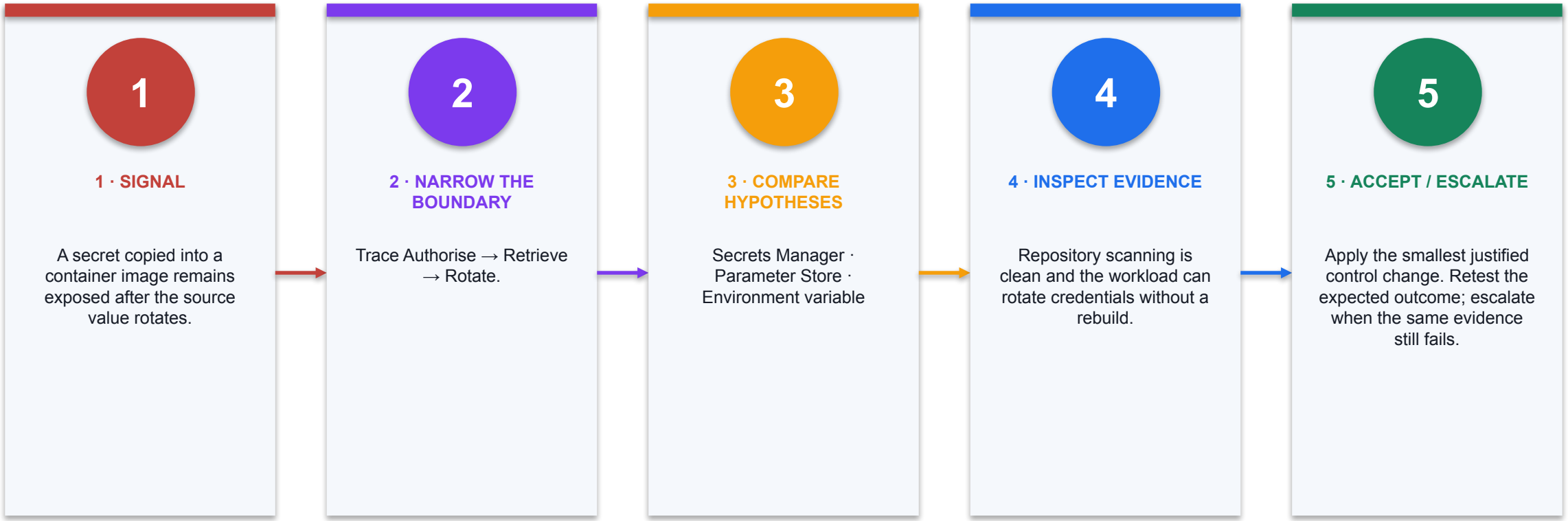
A secret copied into a container image remains exposed after the source value rotates.

VERIFICATION EVIDENCE

Repository scanning is clean and the workload can rotate credentials without a rebuild.

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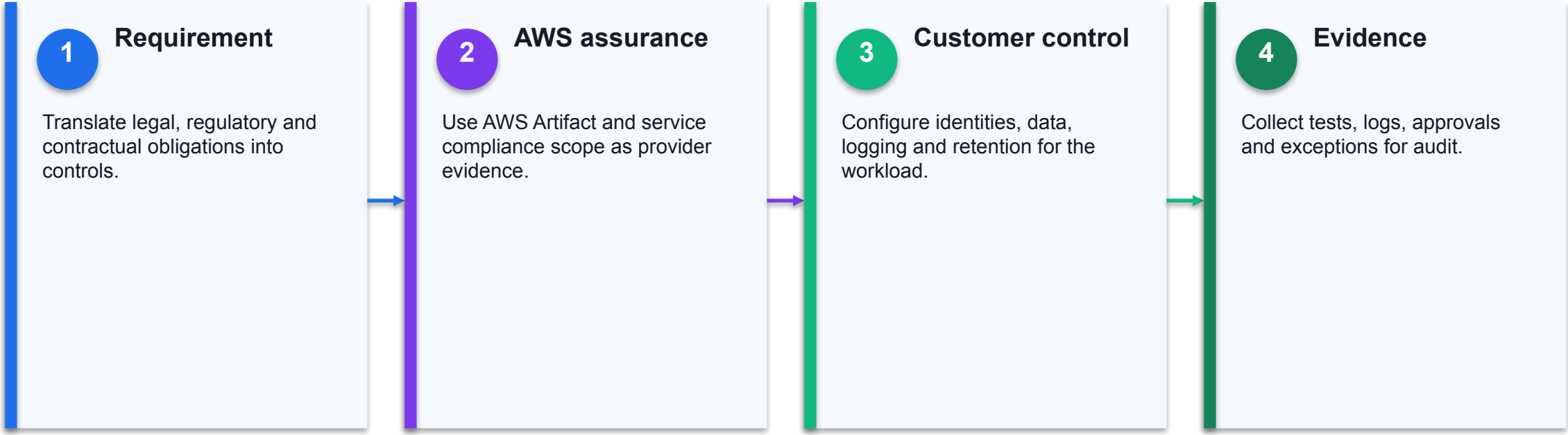
Secrets, credentials and parameter storage · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Compliance programmes do not secure a workload



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Compliance programmes do not secure a workload · technical contract

1	INPUT / INTENT **Requirement**	Translate legal, regulatory and contractual obligations into controls.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **AWS assurance**	Use AWS Artifact and service compliance scope as provider evidence.	RETAIN **Approved configuration**
3	STATE CHANGE **Customer control**	Configure identities, data, logging and retention for the workload.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Evidence**	Collect tests, logs, approvals and exceptions for audit.	RETAIN **Audit, metric or test record**

CONTROL RISK Listing AWS certifications without proving customer-side controls creates a false compliance claim.

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Compliance programmes do not secure a workload · decision and control

1 AWS Artifact

USE WHEN

Provider reports and agreements

TRADE-OFF

Not workload configuration evidence

2 Audit Manager

USE WHEN

Automated evidence collection

TRADE-OFF

Framework tailoring still required

3 Config conformance pack

USE WHEN

Configuration control set

TRADE-OFF

Does not cover every process control

FAILURE MODE

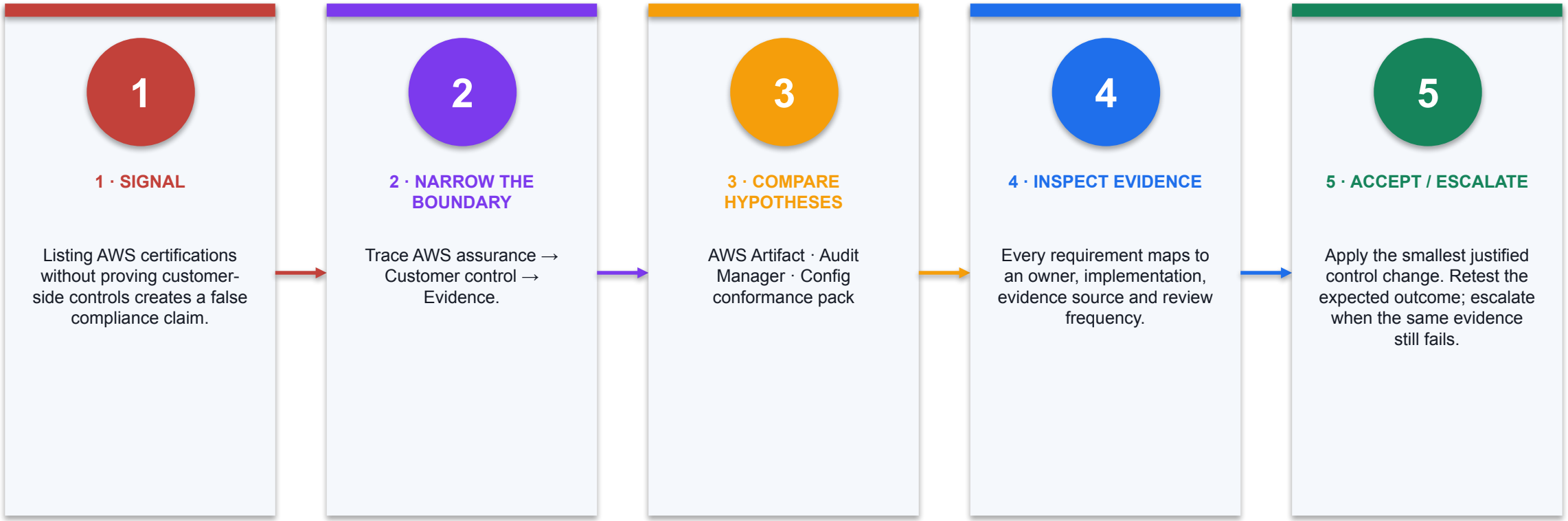
Listing AWS certifications without proving customer-side controls creates a false compliance claim.

VERIFICATION EVIDENCE

Every requirement maps to an owner, implementation, evidence source and review frequency.

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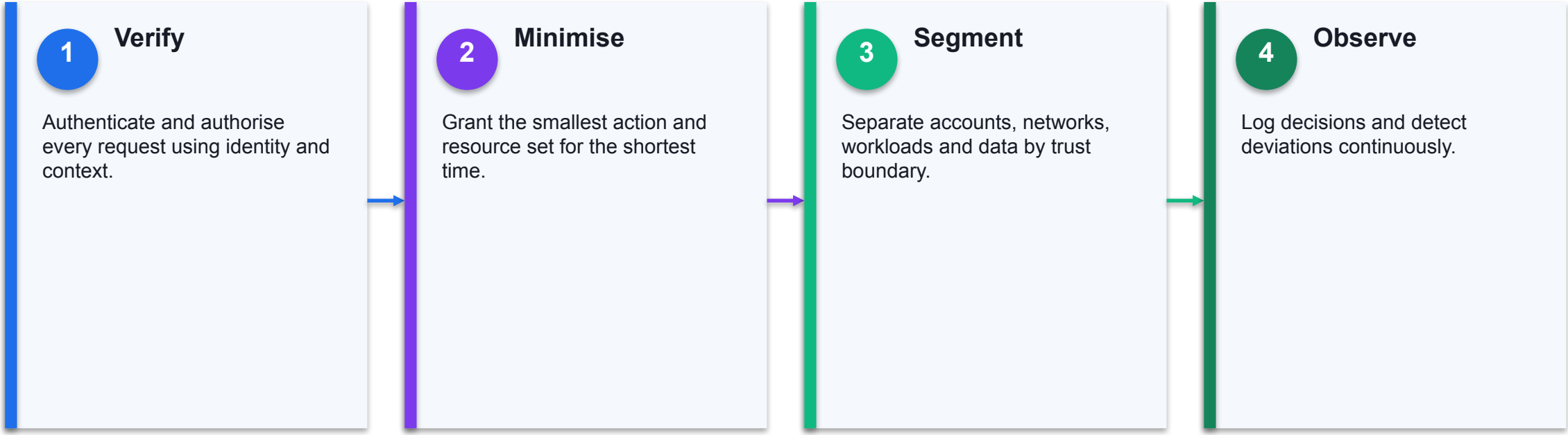
Compliance programmes do not secure a workload · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Zero trust and least privilege reduce blast radius



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Zero trust and least privilege reduce blast radius · technical contract

1	INPUT / INTENT **Verify**	Authenticate and authorise every request using identity and context.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Minimise**	Grant the smallest action and resource set for the shortest time.	RETAIN **Approved configuration**
3	STATE CHANGE **Segment**	Separate accounts, networks, workloads and data by trust boundary.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Observe**	Log decisions and detect deviations continuously.	RETAIN **Audit, metric or test record**

CONTROL RISK A flat network with administrator credentials makes one compromise an estate-wide event.

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Zero trust and least privilege reduce blast radius · decision and control

1 Account boundary

USE WHEN

Strong workload/environment isolation

TRADE-OFF

More cross-account design

2 Network segment

USE WHEN

Limit traffic paths

TRADE-OFF

Identity controls remain necessary

3 Resource policy

USE WHEN

Constrain direct access

TRADE-OFF

Policy complexity can hide errors

FAILURE MODE

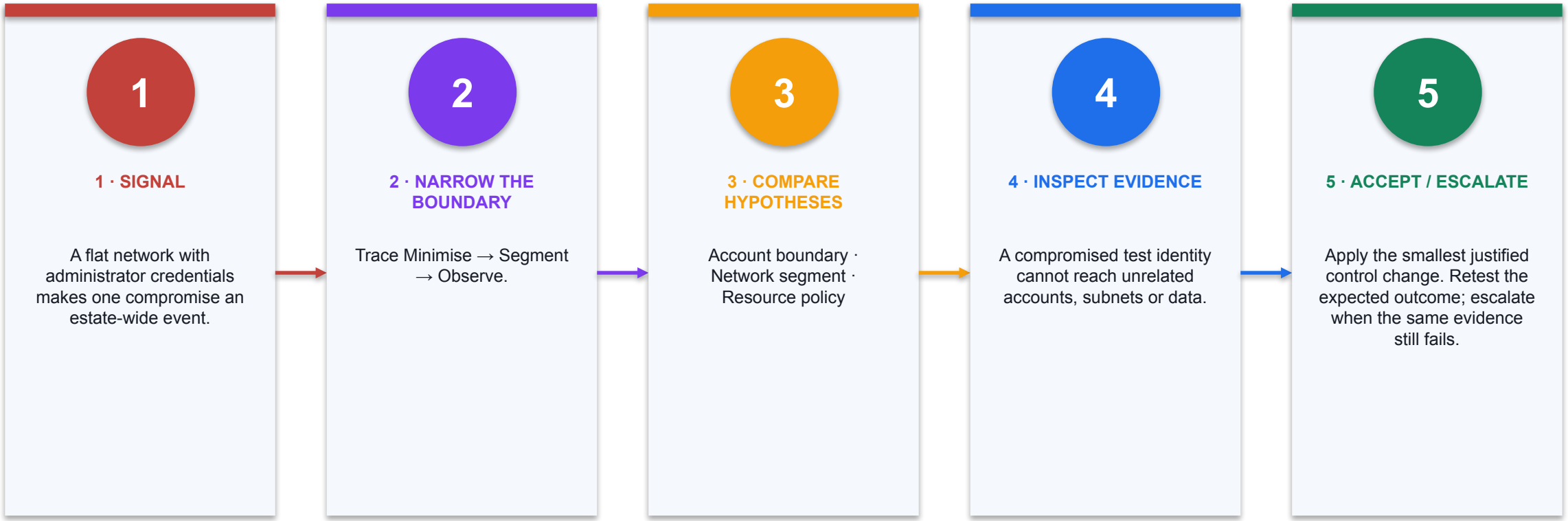
A flat network with administrator credentials makes one compromise an estate-wide event.

VERIFICATION EVIDENCE

A compromised test identity cannot reach unrelated accounts, subnets or data.

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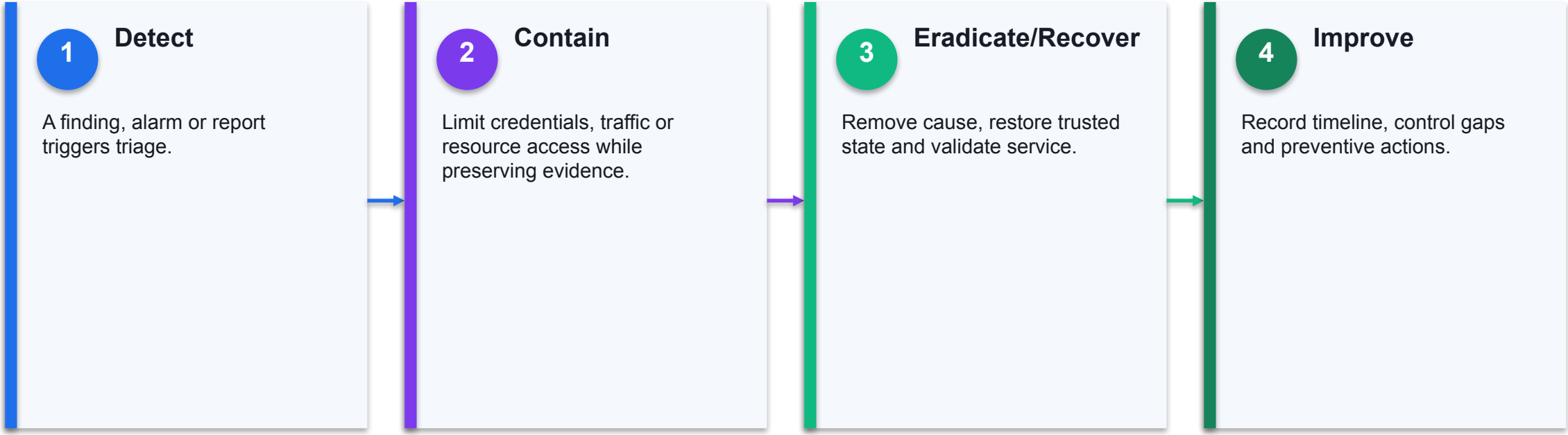
Zero trust and least privilege reduce blast radius · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Incident response: detect to improve



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Incident response: detect to improve · technical contract

1	INPUT / INTENT **Detect**	A finding, alarm or report triggers triage.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Contain**	Limit credentials, traffic or resource access while preserving evidence.	RETAIN **Approved configuration**
3	STATE CHANGE **Eradicate/Recover**	Remove cause, restore trusted state and validate service.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Improve**	Record timeline, control gaps and preventive actions.	RETAIN **Audit, metric or test record**

CONTROL RISK Deleting a compromised resource before preserving evidence can destroy the incident timeline.

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Incident response: detect to improve · decision and control

1 Automation

USE WHEN

Repeatable low-risk containment

TRADE-OFF

Guard against false positives

2 Human approval

USE WHEN

High-impact isolation

TRADE-OFF

Adds response latency

3 Forensics account

USE WHEN

Preserve evidence

TRADE-OFF

Requires pre-built access and logging

FAILURE MODE

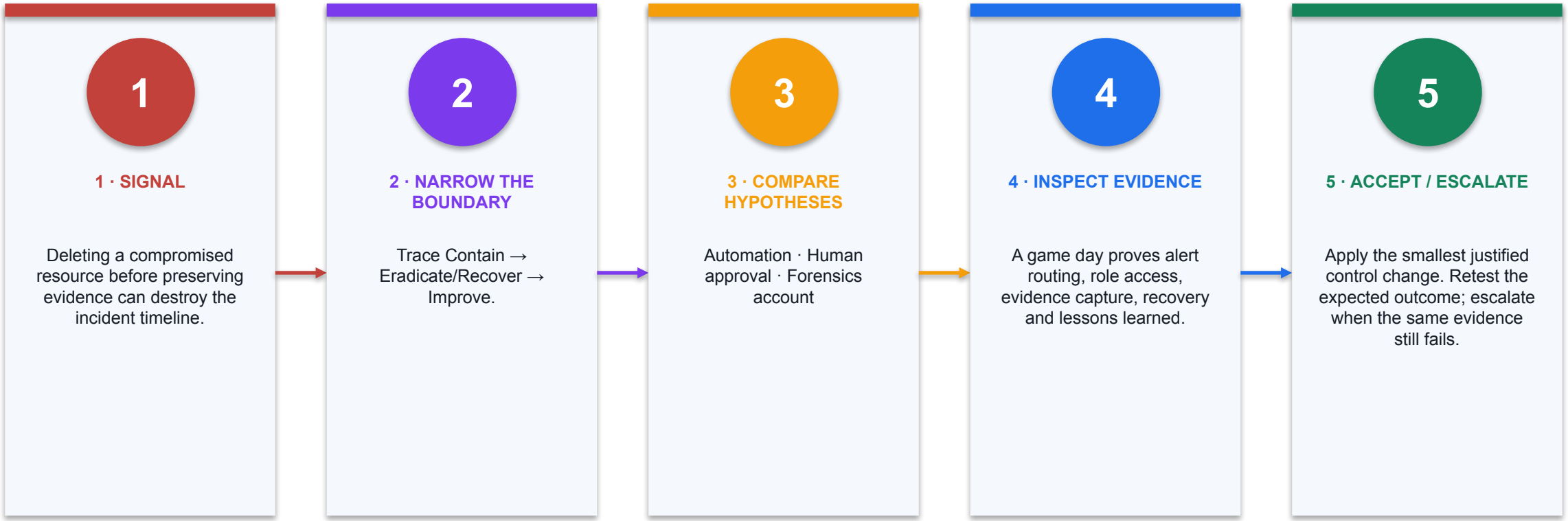
Deleting a compromised resource before preserving evidence can destroy the incident timeline.

VERIFICATION EVIDENCE

A game day proves alert routing, role access, evidence capture, recovery and lessons learned.

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Incident response: detect to improve · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Disaster recovery patterns trade RTO for cost



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Disaster recovery patterns trade RTO for cost · technical contract

1	INPUT / INTENT **Backup**	Protect data and configuration in a separate recovery path.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Prepare**	Pre-provision identity, networking and automation.	RETAIN **Approved configuration**
3	STATE CHANGE **Fail over**	Restore or activate capacity when recovery criteria are met.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Validate**	Test data integrity, endpoint switch, performance and return.	RETAIN **Audit, metric or test record**

CONTROL RISK A backup that has never been restored is not proven recovery capability.

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Disaster recovery patterns trade RTO for cost · decision and control

1 Backup/restore

USE WHEN

Longer RTO and lowest steady cost

TRADE-OFF

Capacity is created after event

2 Pilot light/warm standby

USE WHEN

Moderate RTO

TRADE-OFF

Some duplicate environment cost

3 Multi-site active

USE WHEN

Shortest RTO

TRADE-OFF

Highest cost and consistency complexity

FAILURE MODE

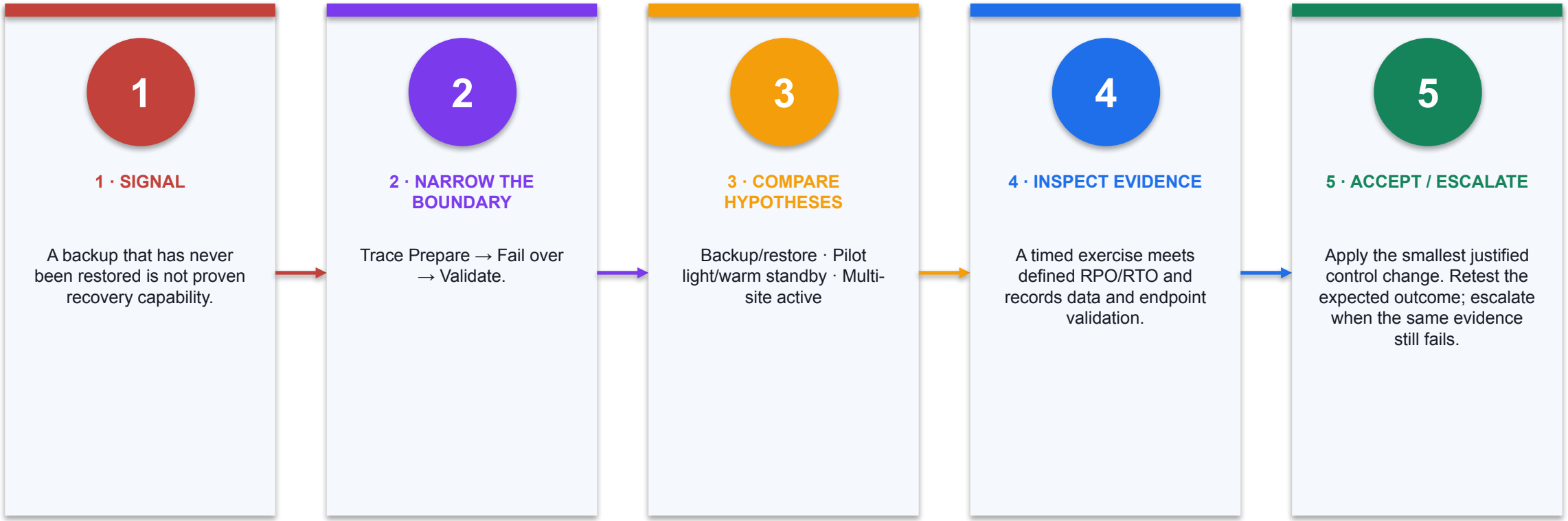
A backup that has never been restored is not proven recovery capability.

VERIFICATION EVIDENCE

A timed exercise meets defined RPO/RTO and records data and endpoint validation.

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Disaster recovery patterns trade RTO for cost · failure-to-evidence case



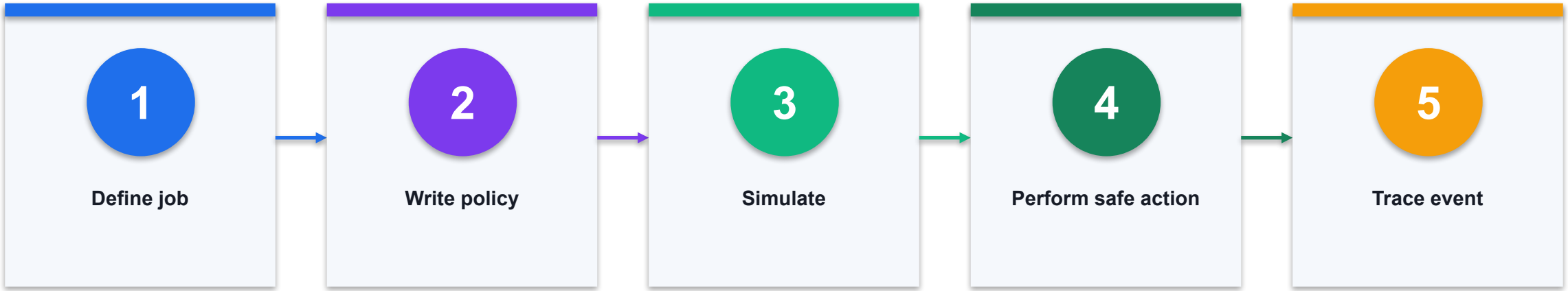
Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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DOMAIN 2 · LAB 2 · 55 MIN

IAM Policy and CloudTrail Evidence

A support analyst must inspect S3 metadata but must not delete objects or change bucket policy.



YOU'LL PRODUCE **Least-privilege policy excerpt, allow/deny test matrix and one CloudTrail event evidence record.**

Tools: IAM, IAM Policy Simulator or Access Analyzer, Amazon S3, AWS CloudTrail · Detailed procedure: Learner Guide, Lab 2

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IAM Policy and CloudTrail Evidence · verification gate

1

PURPOSE

LO1 / K1 / A1 - configure and test least-privilege access with an audit trail.

2

ARTIFACT SET

Least-privilege policy excerpt, allow/deny test matrix and one CloudTrail event evidence record.

3

ACCEPTANCE TEST

Required reads are allowed, prohibited writes are denied, and CloudTrail identifies the expected assumed principal and action.

EVIDENCE BOUNDARY **IAM, IAM Policy Simulator or Access Analyzer, Amazon S3, AWS CloudTrail**

If an acceptance condition fails, retain the timestamped evidence and escalate with the tested boundary and observed result.

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RECAP · VERIFY BEFORE MOVING ON

Domain 2 evidence recap

1 Shared responsibility

A responsibility matrix names each control owner for the selected service.

2 Identity types

CloudTrail shows the expected assumed-role principal and no static key is stored in the workload.

3 Policy evaluation

IAM Policy Simulator or Access Analyzer explains the expected decision before deployment.

4 Least privilege

Tests show the required action succeeds and an out-of-scope action is denied.

MECHANISM

Read left to right: input, control, state change, observable evidence.

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03

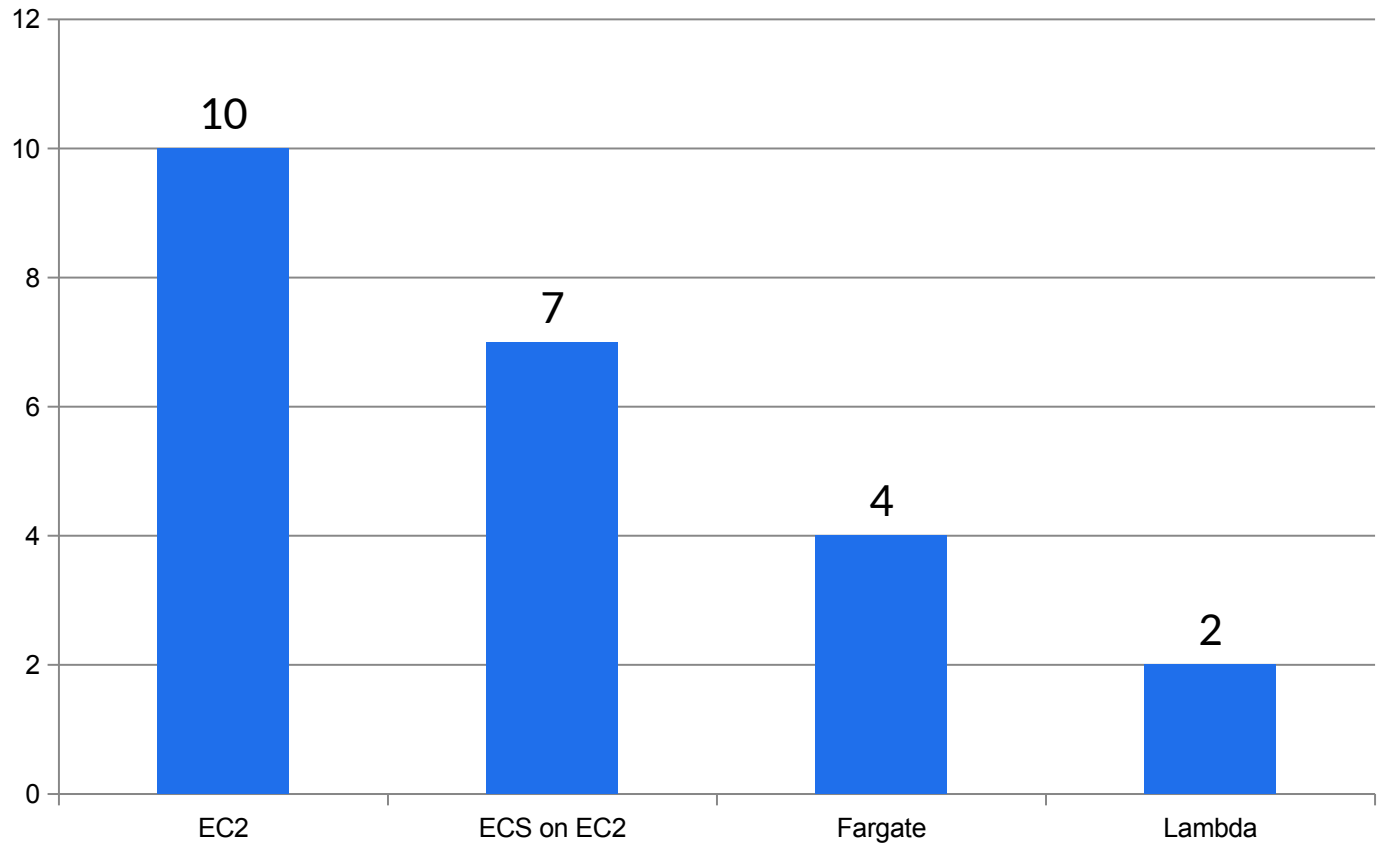
DOMAIN 3 · EXAM WEIGHT 34%

Cloud Technology and Services

Compute · storage · databases · networking · integration · operations · analytics

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Compute operations responsibility index



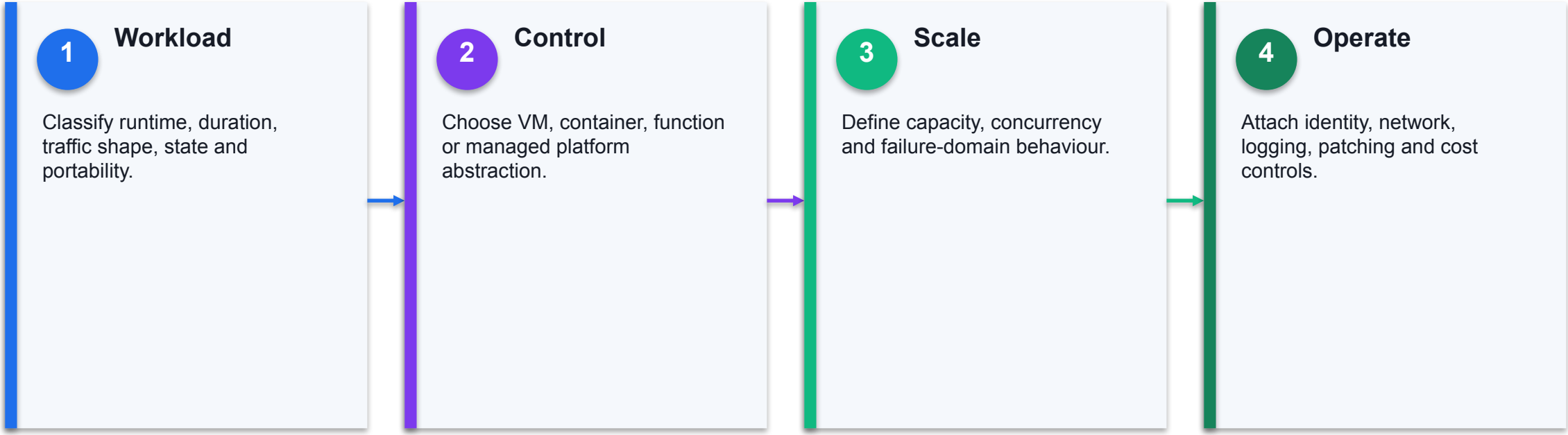
WHAT THE DATA SHOWS

Higher abstraction reduces server operations, while runtime, identity, data and application controls still require ownership.

Editable PowerPoint chart

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Compute selection: control, unit and duration



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Compute selection: control, unit and duration · technical contract

1	INPUT / INTENT **Workload**	Classify runtime, duration, traffic shape, state and portability.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Control**	Choose VM, container, function or managed platform abstraction.	RETAIN **Approved configuration**
3	STATE CHANGE **Scale**	Define capacity, concurrency and failure-domain behaviour.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Operate**	Attach identity, network, logging, patching and cost controls.	RETAIN **Audit, metric or test record**

CONTROL RISK Selecting compute by familiarity can create unnecessary operating work or hard service limits.

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Compute selection: control, unit and duration · decision and control

1 EC2

USE WHEN

OS control or long-running general compute

TRADE-OFF

Patch and capacity responsibility

2 Lambda

USE WHEN

Event-driven short execution

TRADE-OFF

Runtime and duration constraints

3 Fargate

USE WHEN

Containers without server management

TRADE-OFF

Task sizing and service cost

FAILURE MODE

Selecting compute by familiarity can create unnecessary operating work or hard service limits.

VERIFICATION EVIDENCE

A decision record ties workload attributes to the selected execution model.

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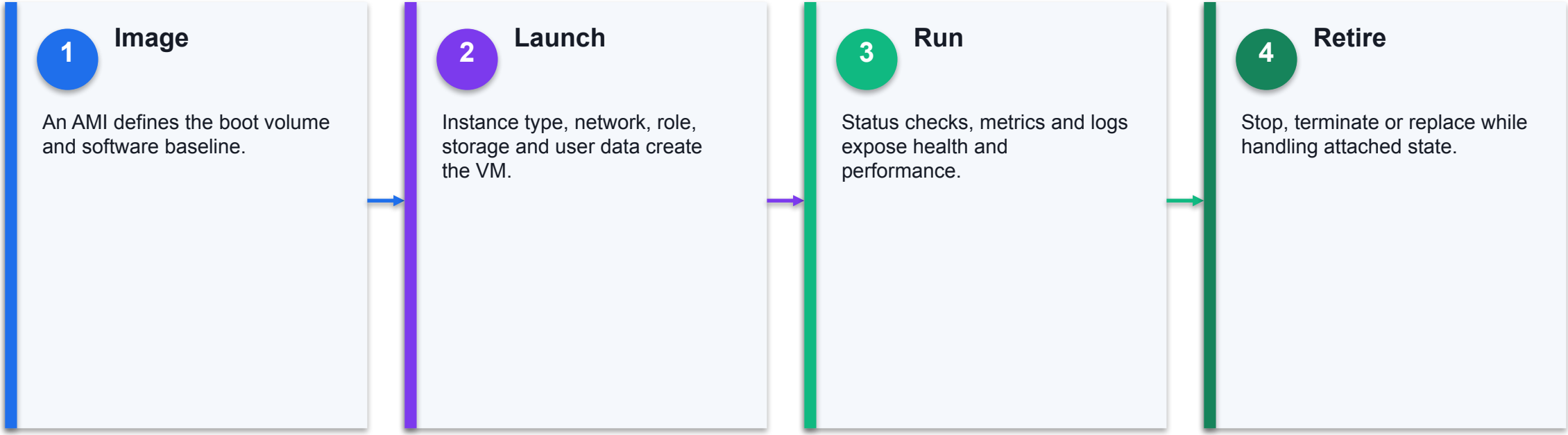
Compute selection: control, unit and duration · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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EC2 instance lifecycle and evidence



MECHANISM

Read left to right: input, control, state change, observable evidence.

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EC2 instance lifecycle and evidence · technical contract

1	INPUT / INTENT **Image**	An AMI defines the boot volume and software baseline.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Launch**	Instance type, network, role, storage and user data create the VM.	RETAIN **Approved configuration**
3	STATE CHANGE **Run**	Status checks, metrics and logs expose health and performance.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Retire**	Stop, terminate or replace while handling attached state.	RETAIN **Audit, metric or test record**

CONTROL RISK Passing the EC2 system check does not prove the application endpoint is healthy.

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EC2 instance lifecycle and evidence · decision and control

1 General purpose

USE WHEN

Balanced CPU and memory

TRADE-OFF

Not optimal for specialised load

2 Compute/memory/storage optimised

USE WHEN

Dominant resource requirement

TRADE-OFF

Higher unit price

3 Accelerated

USE WHEN

GPU or specialised hardware

TRADE-OFF

Availability and cost constraints

FAILURE MODE

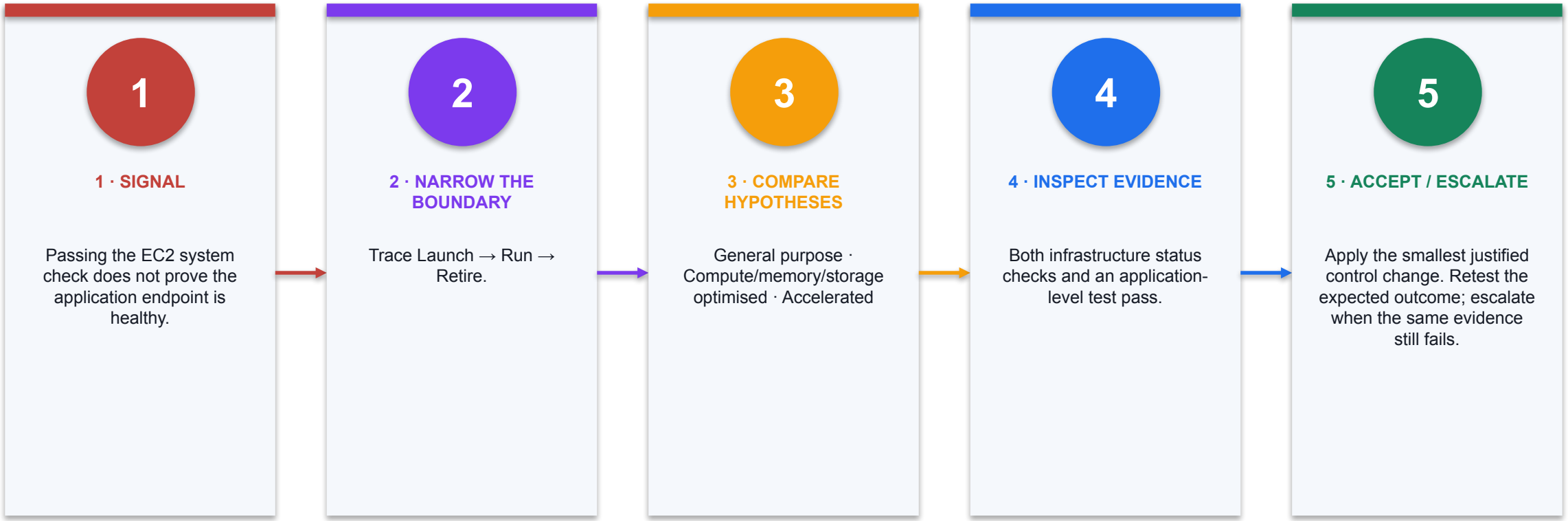
Passing the EC2 system check does not prove the application endpoint is healthy.

VERIFICATION EVIDENCE

Both infrastructure status checks and an application-level test pass.

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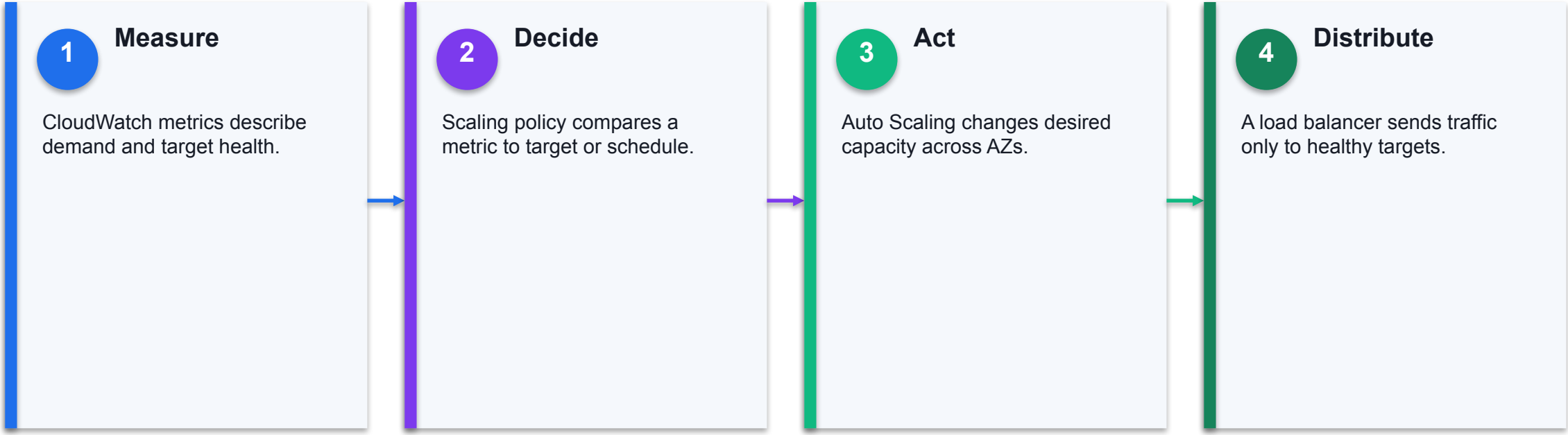
EC2 instance lifecycle and evidence · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Load balancing and Auto Scaling feedback loop



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Load balancing and Auto Scaling feedback loop · technical contract

1	INPUT / INTENT **Measure**	CloudWatch metrics describe demand and target health.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Decide**	Scaling policy compares a metric to target or schedule.	RETAIN **Approved configuration**
3	STATE CHANGE **Act**	Auto Scaling changes desired capacity across AZs.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Distribute**	A load balancer sends traffic only to healthy targets.	RETAIN **Audit, metric or test record**

CONTROL RISK Scaling on CPU when the bottleneck is queue depth can add cost without reducing latency.

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Load balancing and Auto Scaling feedback loop · decision and control

1 Target tracking

USE WHEN

Maintain a business metric target

TRADE-OFF

Needs representative metric

2 Step scaling

USE WHEN

Respond differently by breach size

TRADE-OFF

More thresholds to tune

3 Scheduled

USE WHEN

Known demand cycle

TRADE-OFF

Cannot react to surprises alone

FAILURE MODE

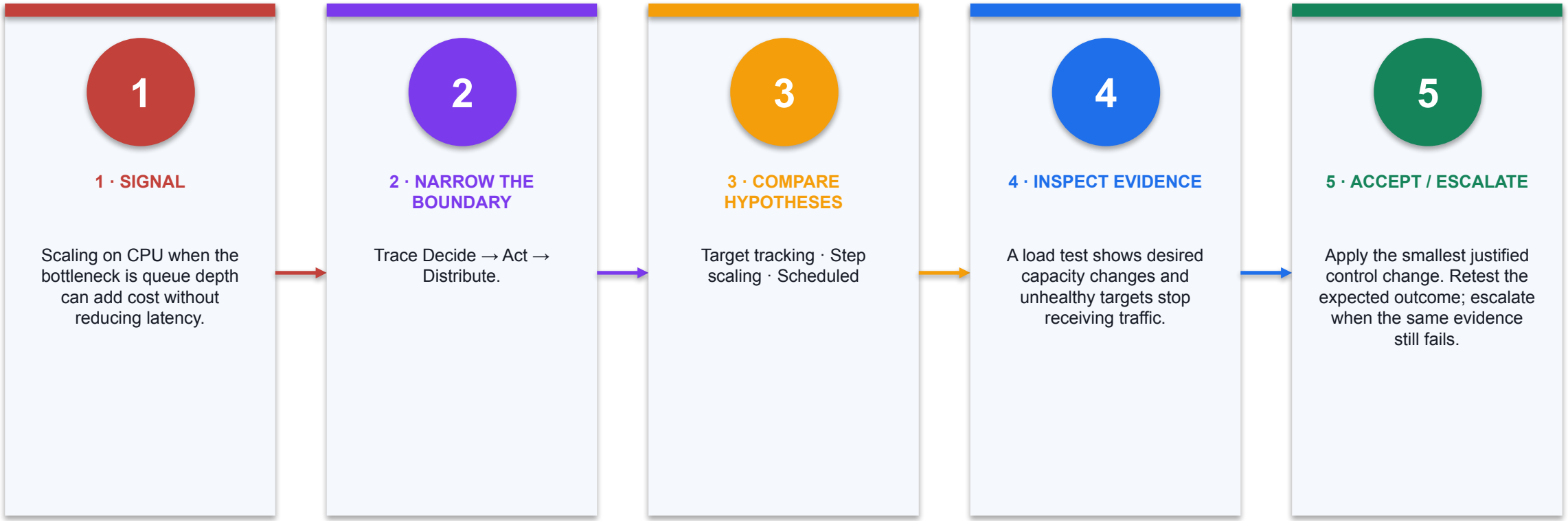
Scaling on CPU when the bottleneck is queue depth can add cost without reducing latency.

VERIFICATION EVIDENCE

A load test shows desired capacity changes and unhealthy targets stop receiving traffic.

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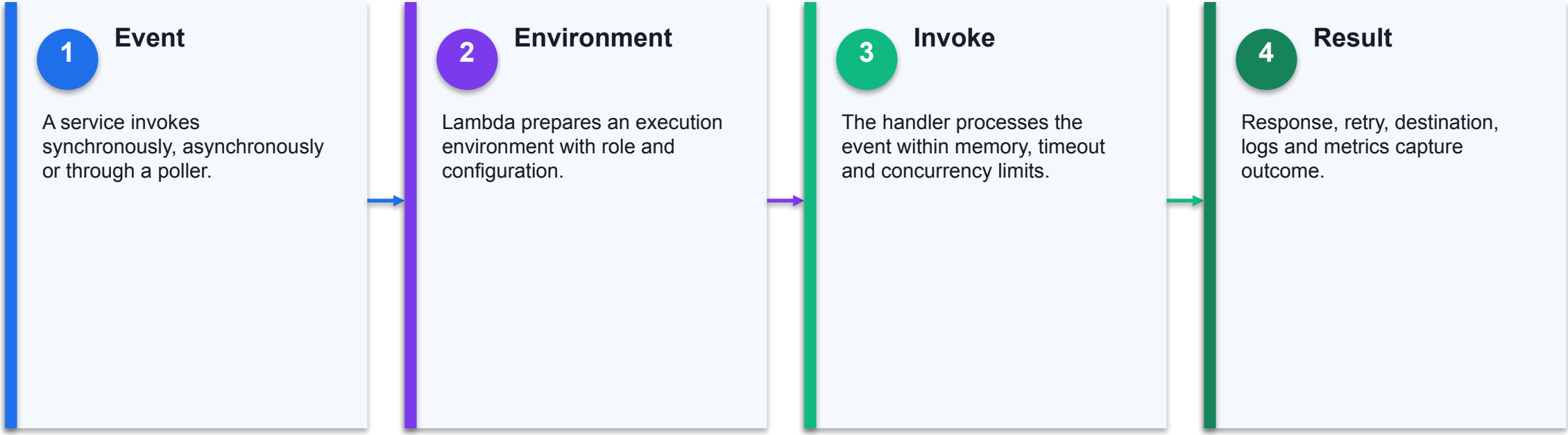
Load balancing and Auto Scaling feedback loop · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Lambda event flow and concurrency



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Lambda event flow and concurrency · technical contract

1	INPUT / INTENT **Event**	A service invokes synchronously, asynchronously or through a poller.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Environment**	Lambda prepares an execution environment with role and configuration.	RETAIN **Approved configuration**
3	STATE CHANGE **Invoke**	The handler processes the event within memory, timeout and concurrency limits.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Result**	Response, retry, destination, logs and metrics capture outcome.	RETAIN **Audit, metric or test record**

CONTROL RISK A non-idempotent handler can duplicate side effects when an event is retried.

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Lambda event flow and concurrency · decision and control

1 Synchronous

USE WHEN

Caller needs immediate response

TRADE-OFF

Caller observes function error

2 Asynchronous

USE WHEN

Event delivery with retries

TRADE-OFF

Design idempotency and failure destination

3 Stream/queue poller

USE WHEN

Ordered or buffered processing

TRADE-OFF

Batch and concurrency tuning

FAILURE MODE

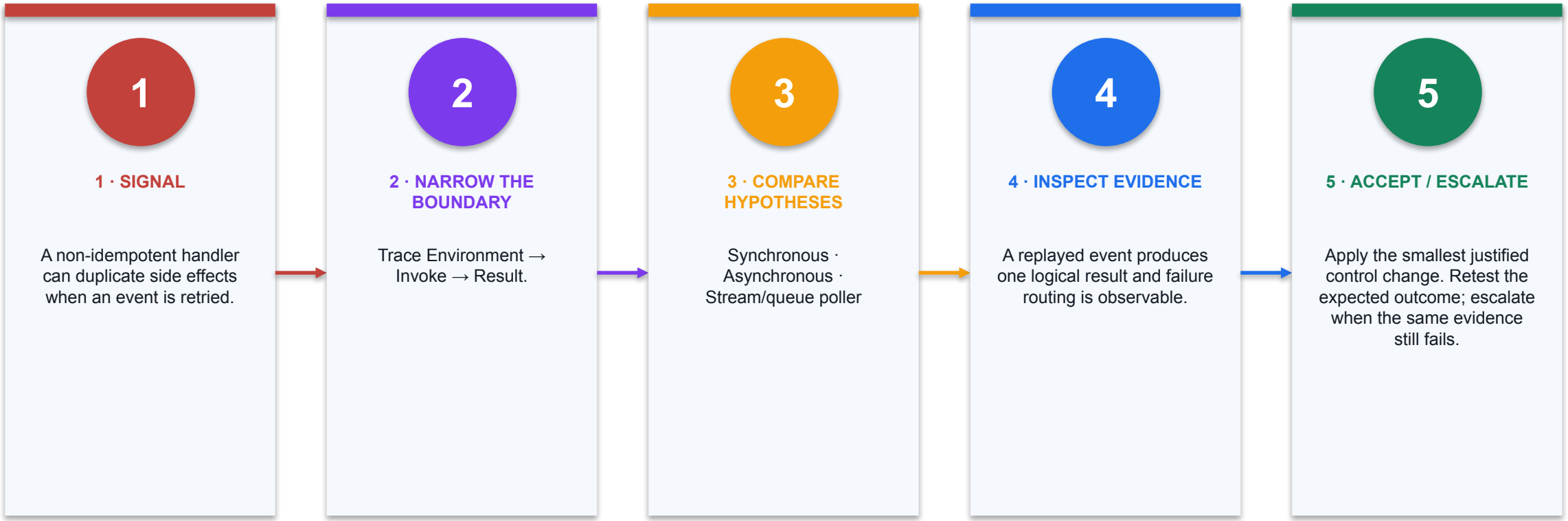
A non-idempotent handler can duplicate side effects when an event is retried.

VERIFICATION EVIDENCE

A replayed event produces one logical result and failure routing is observable.

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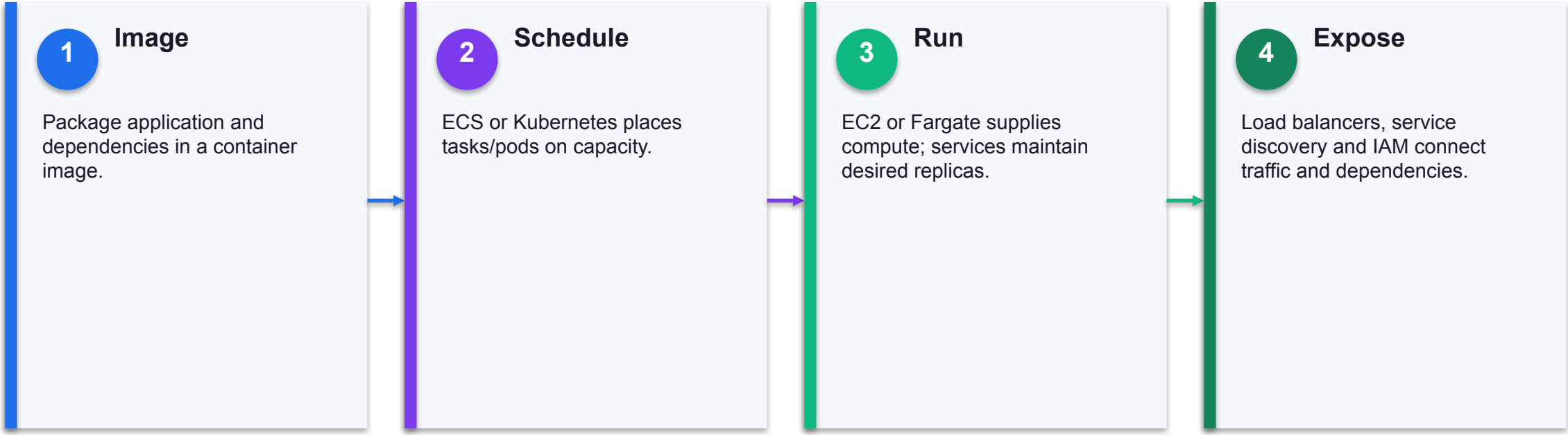
Lambda event flow and concurrency · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Containers: ECS, EKS and Fargate boundaries



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Containers: ECS, EKS and Fargate boundaries · technical contract

1	INPUT / INTENT **Image**	Package application and dependencies in a container image.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Schedule**	ECS or Kubernetes places tasks/pods on capacity.	RETAIN **Approved configuration**
3	STATE CHANGE **Run**	EC2 or Fargate supplies compute; services maintain desired replicas.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Expose**	Load balancers, service discovery and IAM connect traffic and dependencies.	RETAIN **Audit, metric or test record**

CONTROL RISK A container image can be portable while its identity, networking and storage configuration is not.

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Containers: ECS, EKS and Fargate boundaries · decision and control

1 ECS

USE WHEN

AWS-native orchestration

TRADE-OFF

Less Kubernetes portability

2 EKS

USE WHEN

Kubernetes API and ecosystem

TRADE-OFF

Cluster and Kubernetes complexity

3 Fargate

USE WHEN

Serverless task/pod capacity

TRADE-OFF

Per-task constraints and pricing

FAILURE MODE

A container image can be portable while its identity, networking and storage configuration is not.

VERIFICATION EVIDENCE

A deployment test proves image pull, task identity, health checks and service routing.

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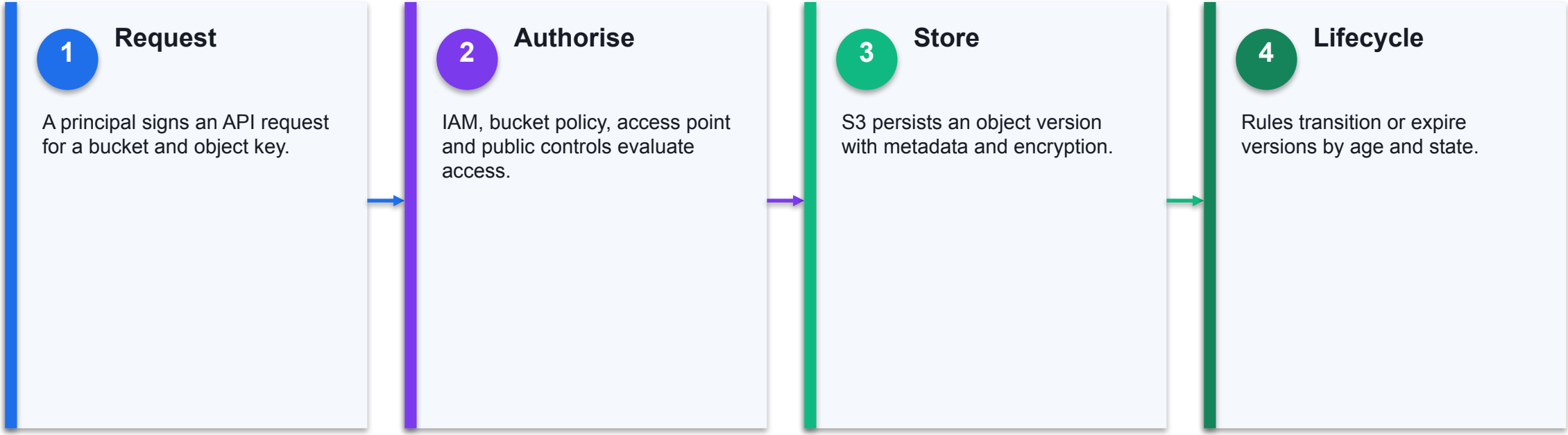
Containers: ECS, EKS and Fargate boundaries · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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S3 object lifecycle and access path



MECHANISM

Read left to right: input, control, state change, observable evidence.

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S3 object lifecycle and access path · technical contract

1	INPUT / INTENT **Request**	A principal signs an API request for a bucket and object key.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Authorise**	IAM, bucket policy, access point and public controls evaluate access.	RETAIN **Approved configuration**
3	STATE CHANGE **Store**	S3 persists an object version with metadata and encryption.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Lifecycle**	Rules transition or expire versions by age and state.	RETAIN **Audit, metric or test record**

CONTROL RISK A lifecycle rule that expires current versions can remove data earlier than business policy allows.

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S3 object lifecycle and access path · decision and control

1 Standard

USE WHEN

Frequent access

TRADE-OFF

Highest storage rate among general classes

2 Intelligent-Tiering

USE WHEN

Unknown/changing access

TRADE-OFF

Monitoring/automation fee

3 Glacier classes

USE WHEN

Archive access

TRADE-OFF

Retrieval time and charges

FAILURE MODE

A lifecycle rule that expires current versions can remove data earlier than business policy allows.

VERIFICATION EVIDENCE

A versioned test object transitions or expires exactly as the rule preview specifies.

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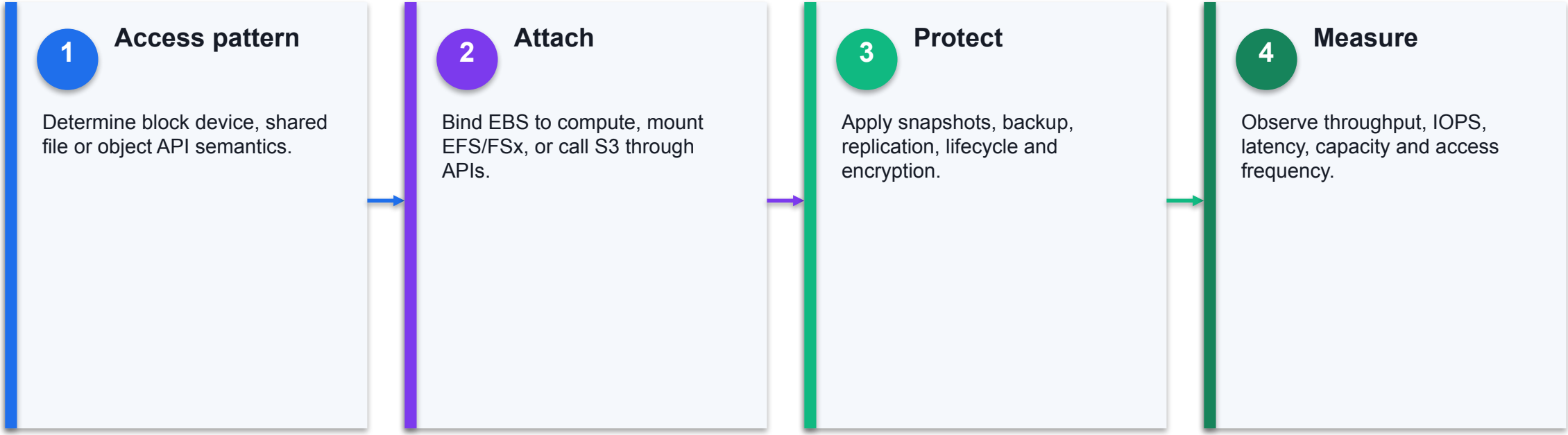
S3 object lifecycle and access path · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Block, file and object storage contracts



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Block, file and object storage contracts · technical contract

1	INPUT / INTENT **Access pattern**	Determine block device, shared file or object API semantics.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Attach**	Bind EBS to compute, mount EFS/FSx, or call S3 through APIs.	RETAIN **Approved configuration**
3	STATE CHANGE **Protect**	Apply snapshots, backup, replication, lifecycle and encryption.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Measure**	Observe throughput, IOPS, latency, capacity and access frequency.	RETAIN **Audit, metric or test record**

CONTROL RISK Choosing storage by capacity alone ignores latency, protocol and sharing requirements.

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Block, file and object storage contracts · decision and control

1 EBS

USE WHEN

Persistent block for EC2

TRADE-OFF

AZ-scoped volume attachment

2 EFS/FSx

USE WHEN

Shared managed file system

TRADE-OFF

Protocol and throughput choices

3 S3

USE WHEN

Durable object data

TRADE-OFF

Not a mounted block device contract

FAILURE MODE

Choosing storage by capacity alone ignores latency, protocol and sharing requirements.

VERIFICATION EVIDENCE

The workload performs the required read/write pattern and recovery test within targets.

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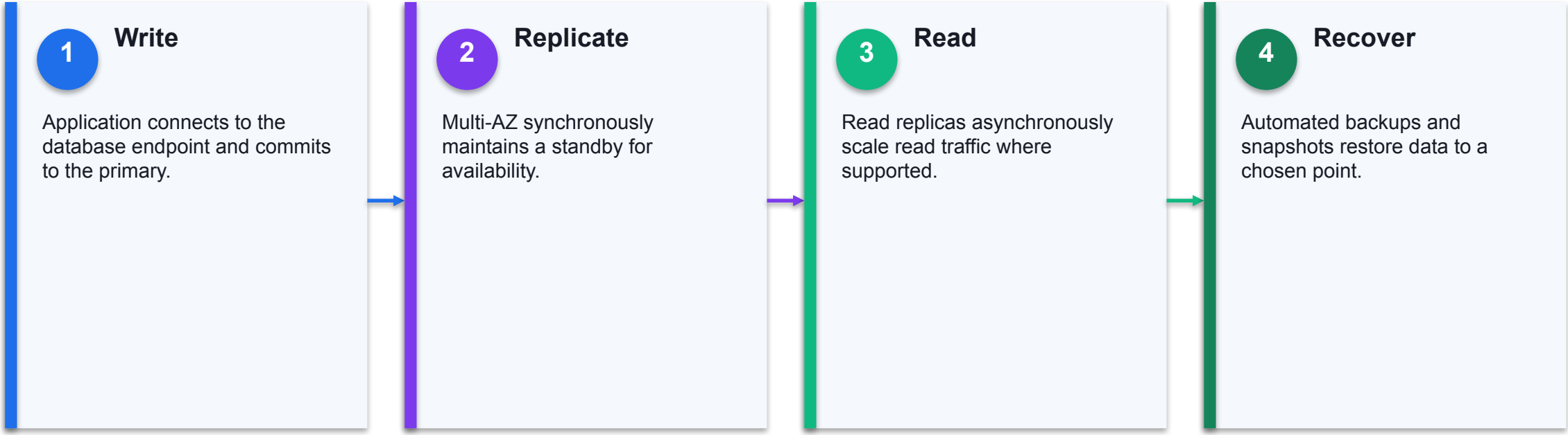
Block, file and object storage contracts · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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RDS availability, replicas and backups



MECHANISM

Read left to right: input, control, state change, observable evidence.

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RDS availability, replicas and backups · technical contract

1	INPUT / INTENT **Write**	Application connects to the database endpoint and commits to the primary.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Replicate**	Multi-AZ synchronously maintains a standby for availability.	RETAIN **Approved configuration**
3	STATE CHANGE **Read**	Read replicas asynchronously scale read traffic where supported.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Recover**	Automated backups and snapshots restore data to a chosen point.	RETAIN **Audit, metric or test record**

CONTROL RISK A read replica is not a drop-in replacement for Multi-AZ high availability.

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RDS availability, replicas and backups · decision and control

1 Multi-AZ

USE WHEN

Availability and automatic failover

TRADE-OFF

Not read scaling by itself

2 Read replica

USE WHEN

Read scale or regional copy

TRADE-OFF

Replication lag and promotion design

3 Snapshot/PITR

USE WHEN

Data recovery

TRADE-OFF

Recovery time and retained storage

FAILURE MODE

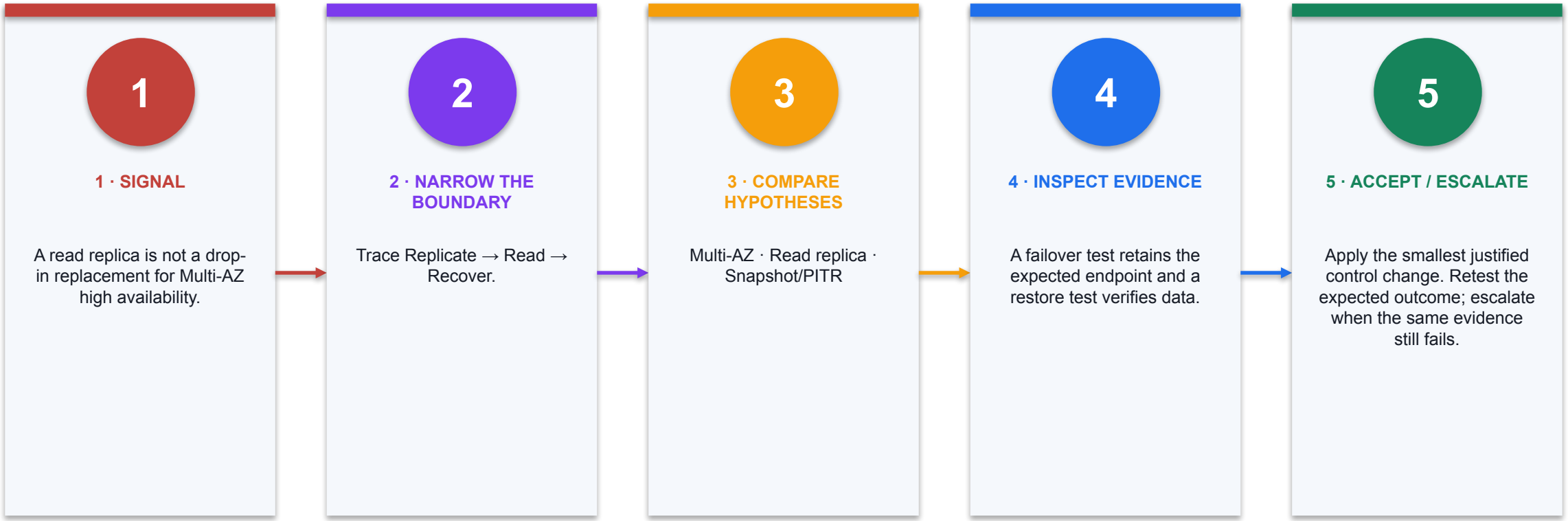
A read replica is not a drop-in replacement for Multi-AZ high availability.

VERIFICATION EVIDENCE

A failover test retains the expected endpoint and a restore test verifies data.

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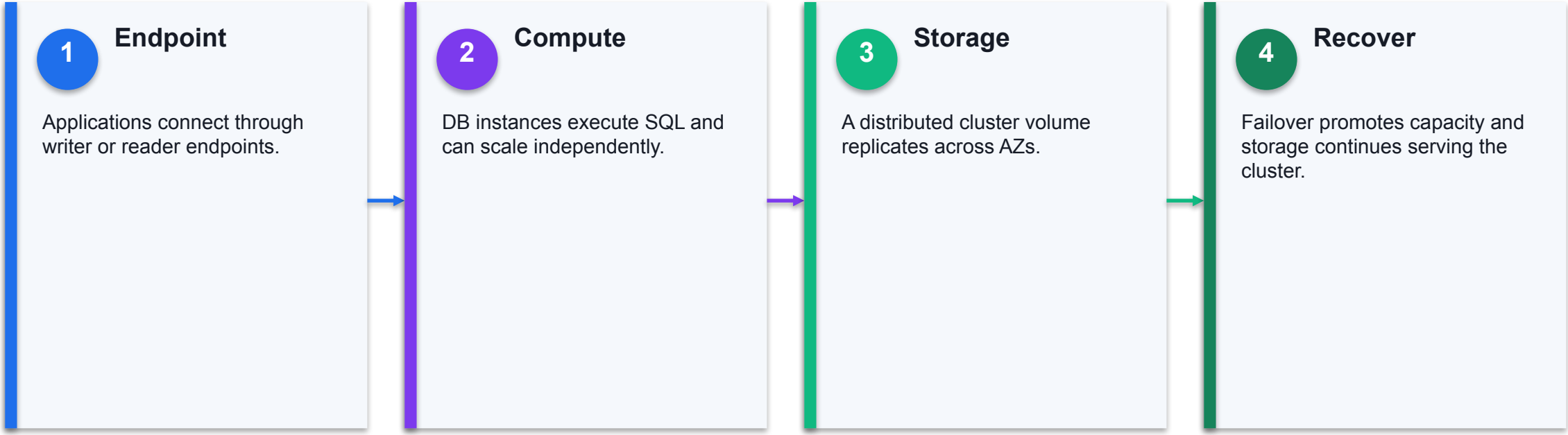
RDS availability, replicas and backups · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Aurora separates compute from distributed storage



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Aurora separates compute from distributed storage · technical contract

1	INPUT / INTENT **Endpoint**	Applications connect through writer or reader endpoints.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Compute**	DB instances execute SQL and can scale independently.	RETAIN **Approved configuration**
3	STATE CHANGE **Storage**	A distributed cluster volume replicates across AZs.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Recover**	Failover promotes capacity and storage continues serving the cluster.	RETAIN **Audit, metric or test record**

CONTROL RISK Adding readers improves read capacity but does not fix inefficient write queries.

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Aurora separates compute from distributed storage · decision and control

1 Provisioned Aurora

USE WHEN

Predictable database load

TRADE-OFF

Choose instance capacity

2 Aurora Serverless v2

USE WHEN

Variable or intermittent demand

TRADE-OFF

Capacity range and cost tuning

3 Global Database

USE WHEN

Cross-Region reads/recovery

TRADE-OFF

Replication and failover complexity

FAILURE MODE

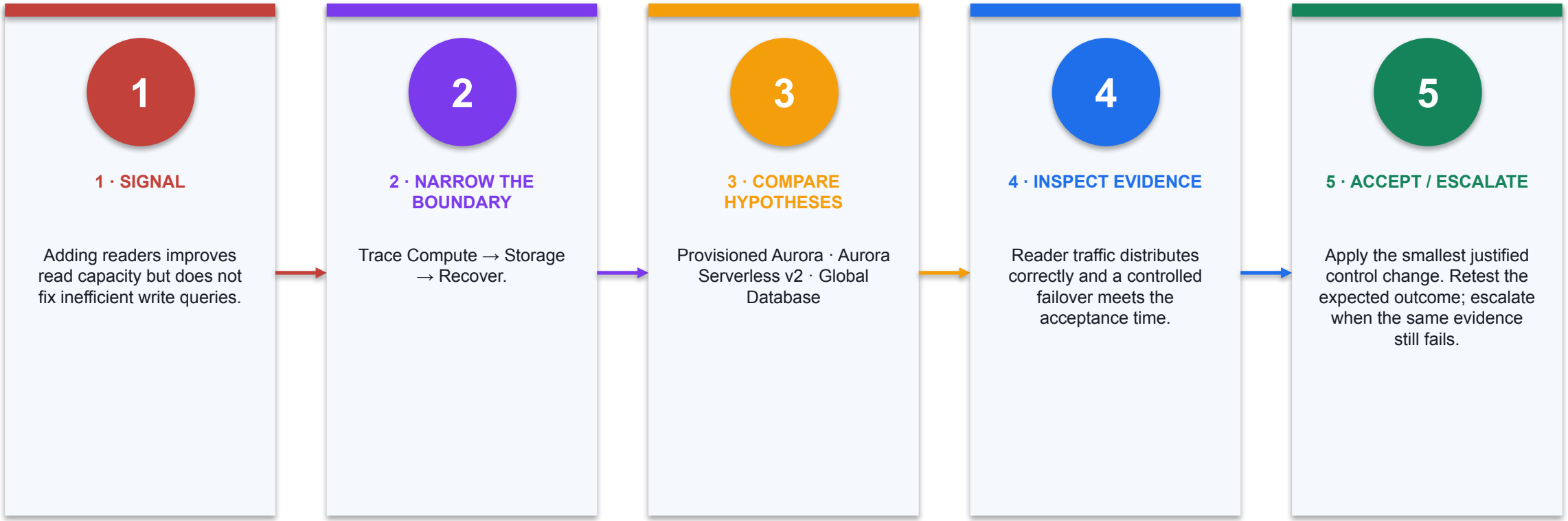
Adding readers improves read capacity but does not fix inefficient write queries.

VERIFICATION EVIDENCE

Reader traffic distributes correctly and a controlled failover meets the acceptance time.

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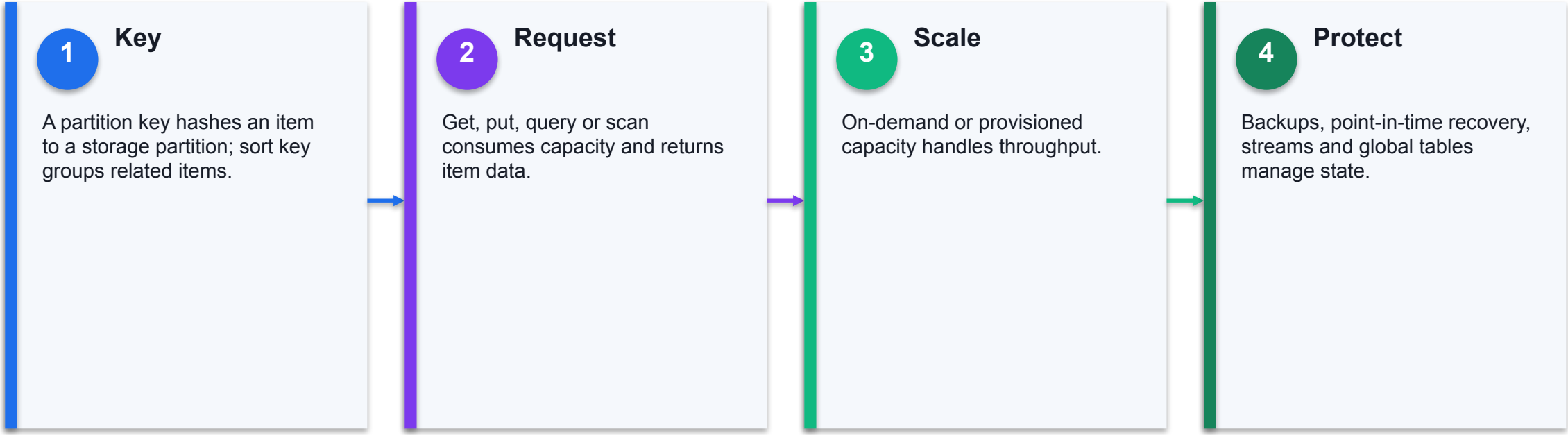
Aurora separates compute from distributed storage · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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DynamoDB partition and consistency model



MECHANISM

Read left to right: input, control, state change, observable evidence.

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DynamoDB partition and consistency model · technical contract

1	INPUT / INTENT **Key**	A partition key hashes an item to a storage partition; sort key groups related items.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Request**	Get, put, query or scan consumes capacity and returns item data.	RETAIN **Approved configuration**
3	STATE CHANGE **Scale**	On-demand or provisioned capacity handles throughput.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Protect**	Backups, point-in-time recovery, streams and global tables manage state.	RETAIN **Audit, metric or test record**

CONTROL RISK A low-cardinality partition key can create a hot partition despite high table capacity.

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DynamoDB partition and consistency model · decision and control

1 Query

USE WHEN

Known partition key and optional sort condition

TRADE-OFF

Efficient targeted access

2 Scan

USE WHEN

Read every item/filter later

TRADE-OFF

High cost and latency at scale

3 Global table

USE WHEN

Multi-Region active-active

TRADE-OFF

Conflict and replication considerations

FAILURE MODE

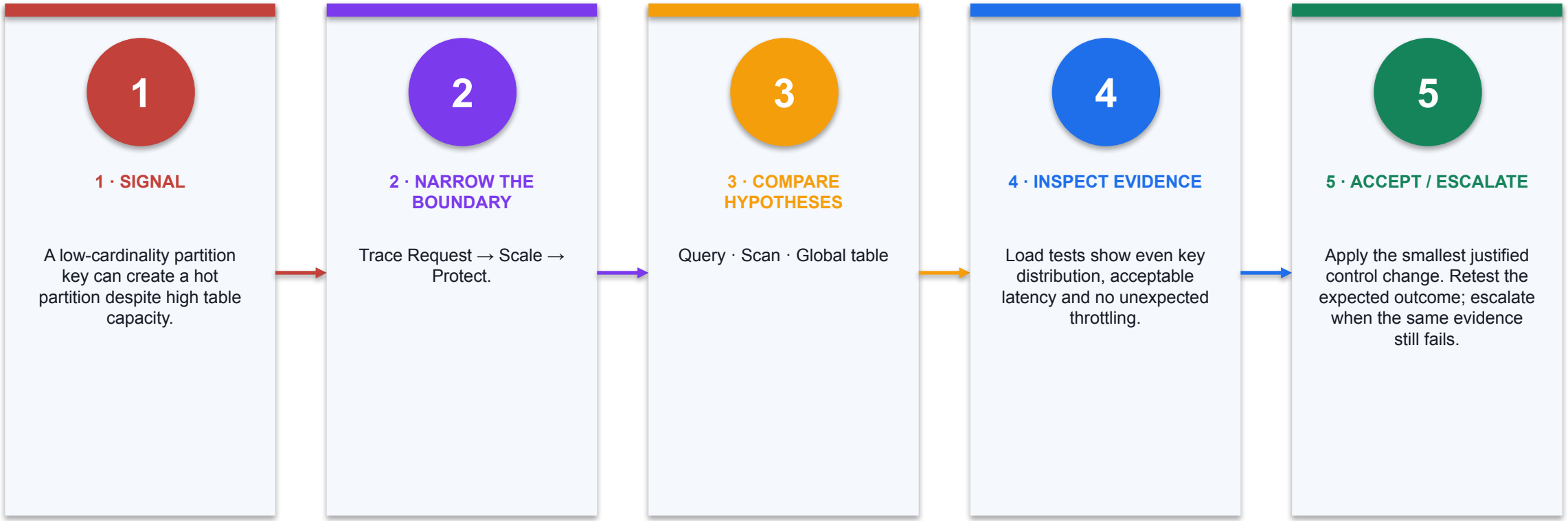
A low-cardinality partition key can create a hot partition despite high table capacity.

VERIFICATION EVIDENCE

Load tests show even key distribution, acceptable latency and no unexpected throttling.

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DynamoDB partition and consistency model · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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VPC packet path is a sequence of decisions



ROUTE EVIDENCE

Destination CIDR → target

FLOW EVIDENCE

ACCEPT / REJECT + 5-tuple

PATH EVIDENCE

Reachability result + failing component

A successful request requires a valid forward path and a valid return path across every routing and filtering boundary.

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VPC packet path is a sequence of decisions · technical contract

1	INPUT / INTENT **Source**	An ENI sends a packet from a subnet with security-group context.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Route**	The subnet route table selects local, gateway, NAT, transit or other target.	RETAIN **Approved configuration**
3	STATE CHANGE **Filter**	Network ACL and destination security group evaluate the packet.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Return**	Stateful and stateless controls govern the response path.	RETAIN **Audit, metric or test record**

CONTROL RISK A NAT gateway in a private subnet without a route to an internet gateway cannot provide egress.

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VPC packet path is a sequence of decisions · decision and control

1 Internet gateway

USE WHEN

Public IPv4/IPv6 route

TRADE-OFF

Resource still needs public addressing and rules

2 NAT gateway

USE WHEN

Private IPv4 egress

TRADE-OFF

No unsolicited inbound path

3 VPC endpoint

USE WHEN

Private service access

TRADE-OFF

Service and endpoint policy scope

FAILURE MODE

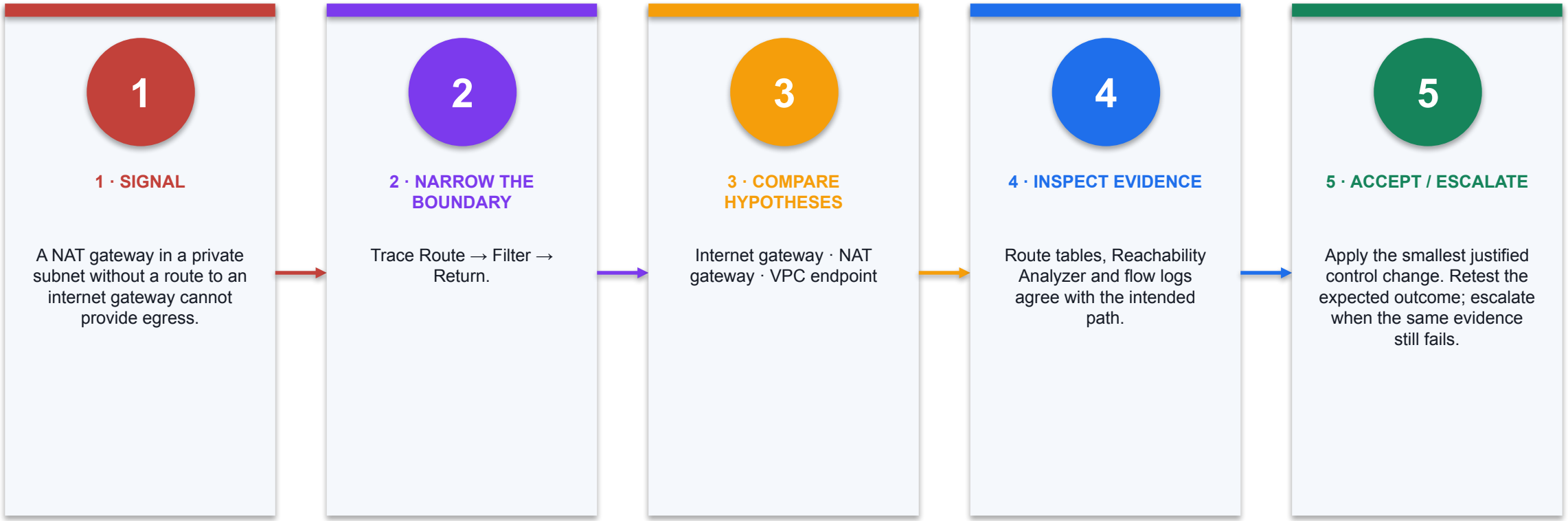
A NAT gateway in a private subnet without a route to an internet gateway cannot provide egress.

VERIFICATION EVIDENCE

Route tables, Reachability Analyzer and flow logs agree with the intended path.

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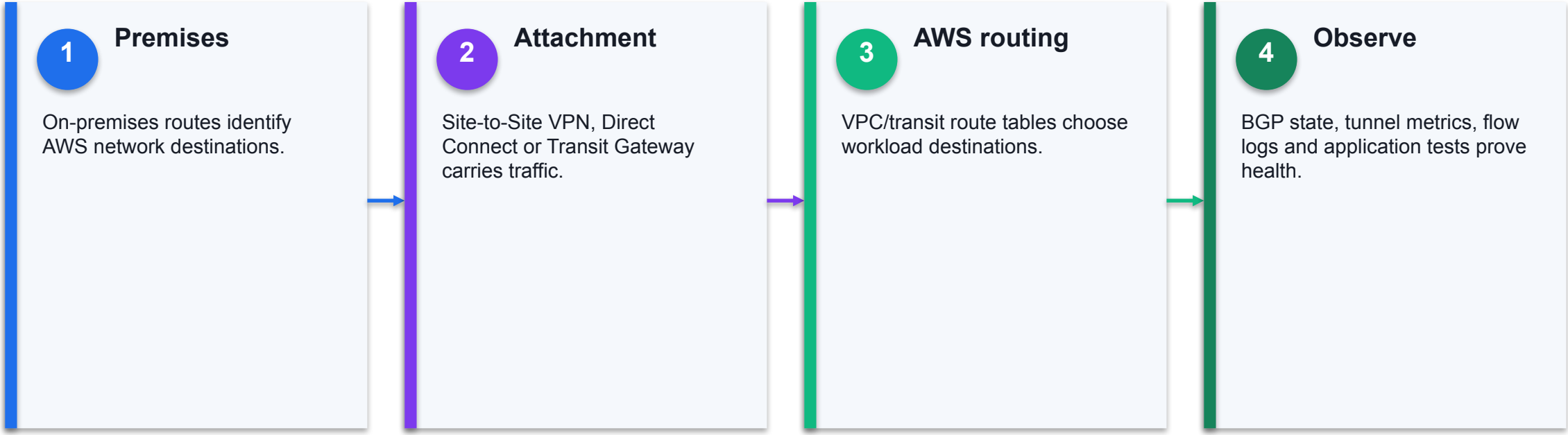
VPC packet path is a sequence of decisions · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Hybrid connectivity: encryption, bandwidth and path



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Hybrid connectivity: encryption, bandwidth and path · technical contract

1	INPUT / INTENT **Premises**	On-premises routes identify AWS network destinations.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Attachment**	Site-to-Site VPN, Direct Connect or Transit Gateway carries traffic.	RETAIN **Approved configuration**
3	STATE CHANGE **AWS routing**	VPC/transit route tables choose workload destinations.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Observe**	BGP state, tunnel metrics, flow logs and application tests prove health.	RETAIN **Audit, metric or test record**

CONTROL RISK Direct Connect is private connectivity but does not encrypt traffic by default.

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Hybrid connectivity: encryption, bandwidth and path · decision and control

1 Site-to-Site VPN

USE WHEN

Fast encrypted setup over internet

TRADE-OFF

Variable internet performance

2 Direct Connect

USE WHEN

Dedicated predictable link

TRADE-OFF

Lead time and separate encryption need

3 Transit Gateway

USE WHEN

Hub for many networks

TRADE-OFF

Routing design and attachment cost

FAILURE MODE

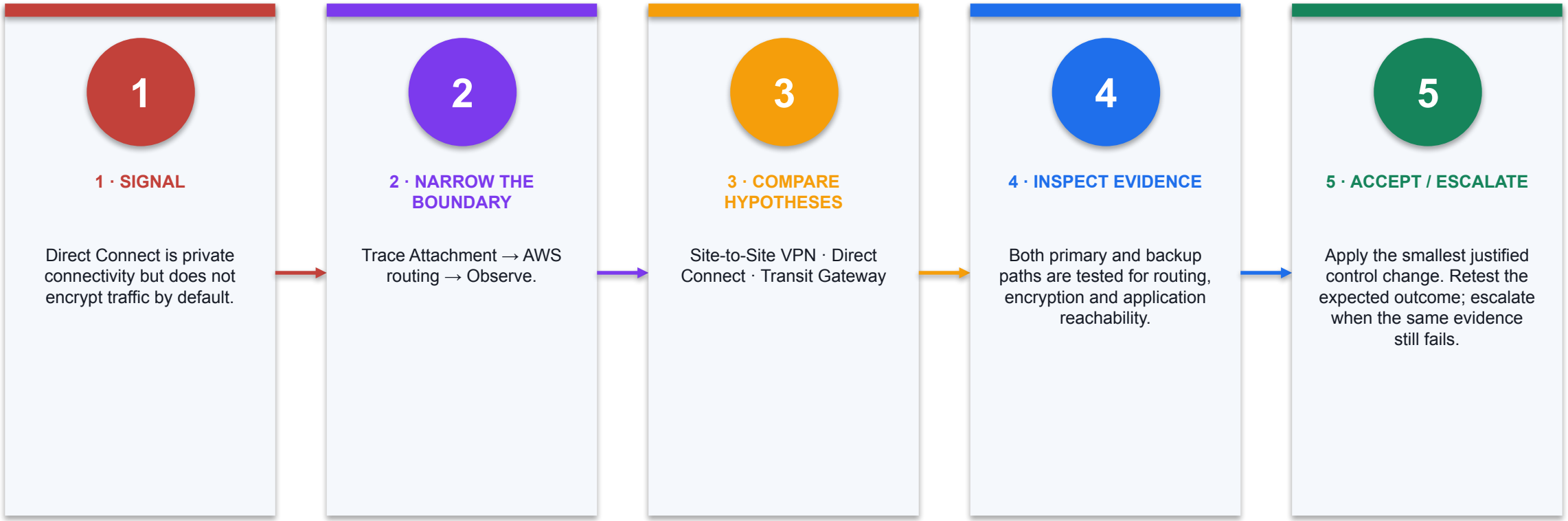
Direct Connect is private connectivity but does not encrypt traffic by default.

VERIFICATION EVIDENCE

Both primary and backup paths are tested for routing, encryption and application reachability.

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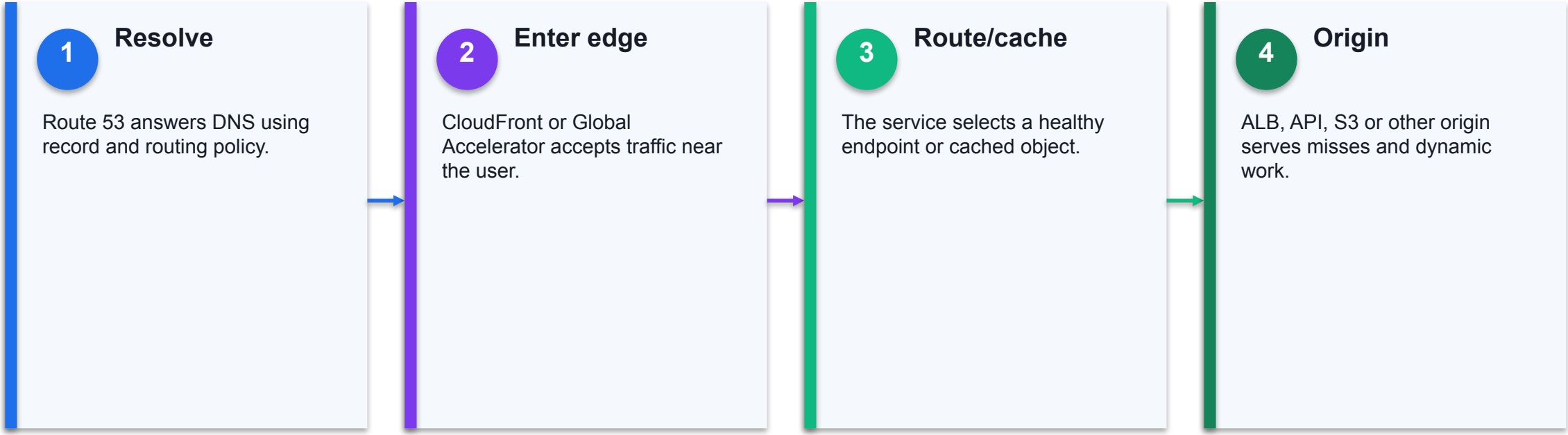
Hybrid connectivity: encryption, bandwidth and path · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Global delivery: DNS, CDN and accelerator



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Global delivery: DNS, CDN and accelerator · technical contract

1	INPUT / INTENT **Resolve**	Route 53 answers DNS using record and routing policy.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Enter edge**	CloudFront or Global Accelerator accepts traffic near the user.	RETAIN **Approved configuration**
3	STATE CHANGE **Route/cache**	The service selects a healthy endpoint or cached object.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Origin**	ALB, API, S3 or other origin serves misses and dynamic work.	RETAIN **Audit, metric or test record**

CONTROL RISK Caching personalised content under a shared key can expose one user's response to another.

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Global delivery: DNS, CDN and accelerator · decision and control

1 Route 53

USE WHEN

DNS routing and health checks

TRADE-OFF

DNS caching affects failover time

2 CloudFront

USE WHEN

HTTP content caching and edge protection

TRADE-OFF

Cache behaviour must match content

3 Global Accelerator

USE WHEN

Static anycast IPs for TCP/UDP

TRADE-OFF

No content cache

FAILURE MODE

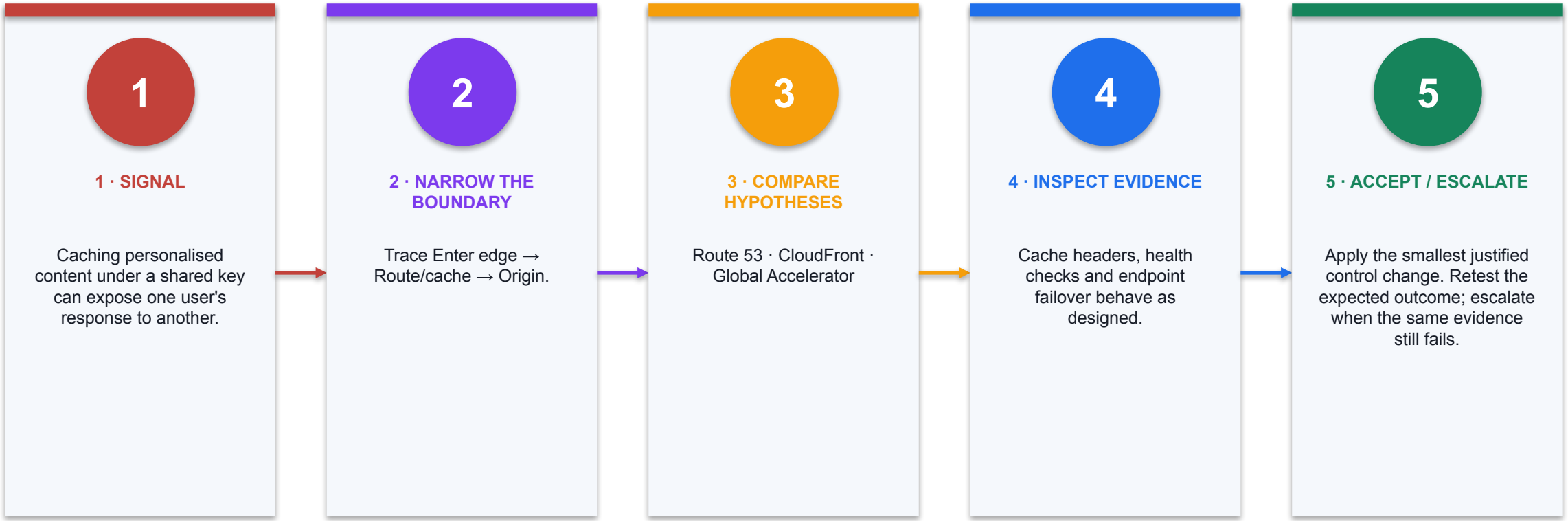
Caching personalised content under a shared key can expose one user's response to another.

VERIFICATION EVIDENCE

Cache headers, health checks and endpoint failover behave as designed.

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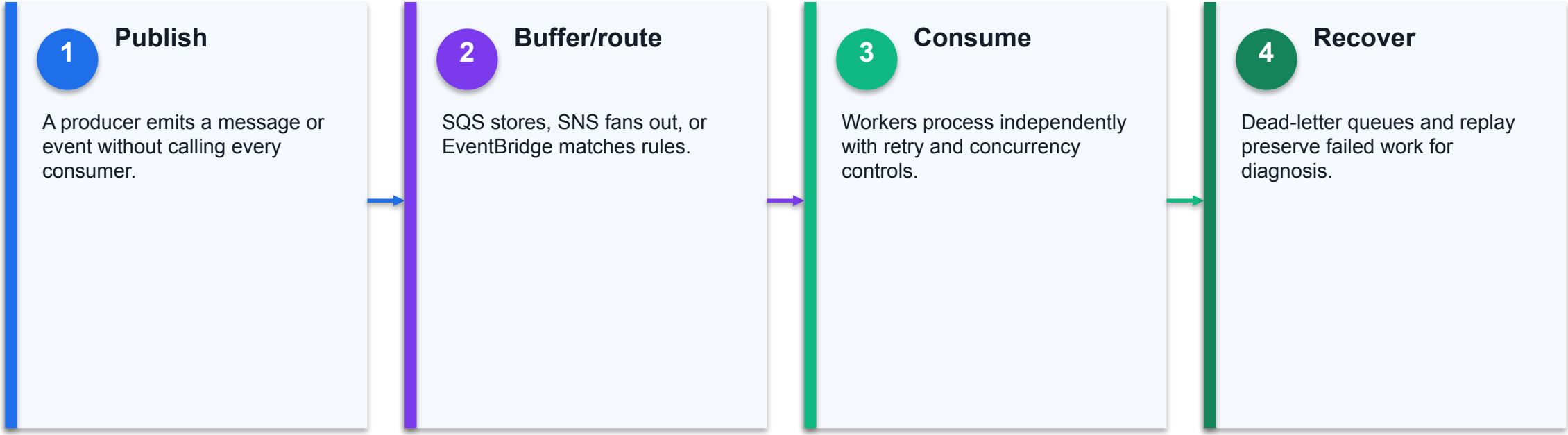
Global delivery: DNS, CDN and accelerator · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Messaging decouples rate, time and failure



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Messaging decouples rate, time and failure · technical contract

1	INPUT / INTENT **Publish**	A producer emits a message or event without calling every consumer.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Buffer/route**	SQS stores, SNS fans out, or EventBridge matches rules.	RETAIN **Approved configuration**
3	STATE CHANGE **Consume**	Workers process independently with retry and concurrency controls.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Recover**	Dead-letter queues and replay preserve failed work for diagnosis.	RETAIN **Audit, metric or test record**

CONTROL RISK Retries without idempotency can duplicate business side effects.

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Messaging decouples rate, time and failure · decision and control

1 SQS

USE WHEN

Durable work queue and back-pressure

TRADE-OFF

Consumer manages processing

2 SNS

USE WHEN

Push fan-out to subscribers

TRADE-OFF

No queue semantics by itself

3 EventBridge

USE WHEN

Rule-based event routing and SaaS integration

TRADE-OFF

Schema and rule governance

FAILURE MODE

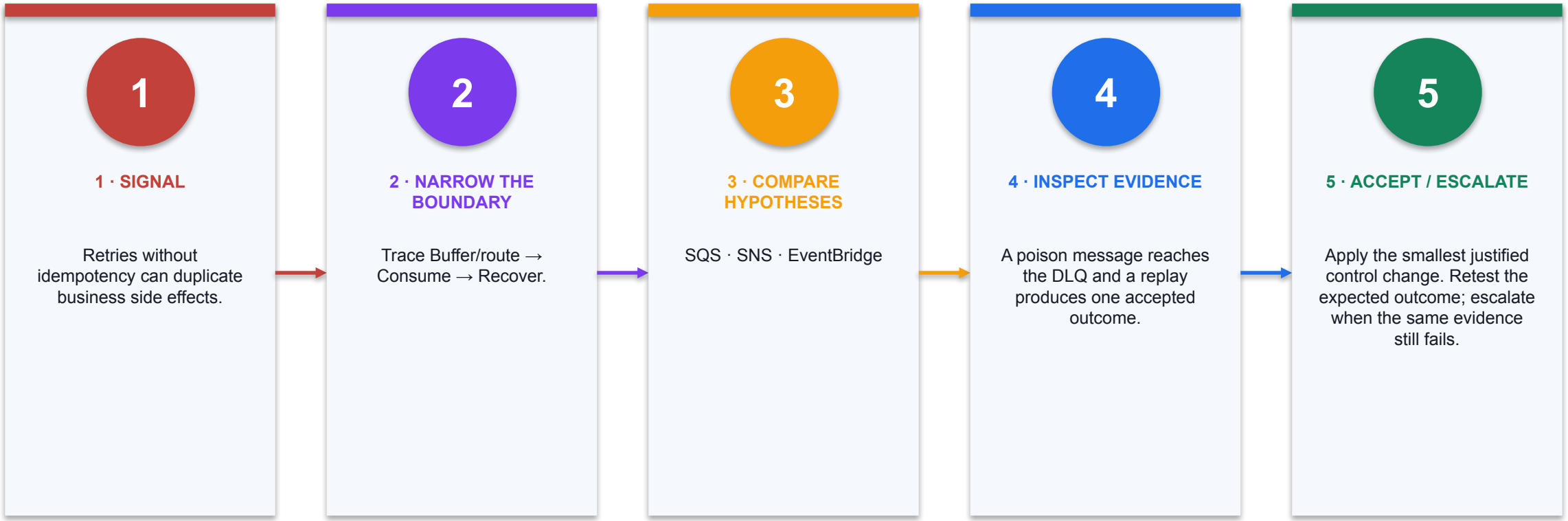
Retries without idempotency can duplicate business side effects.

VERIFICATION EVIDENCE

A poison message reaches the DLQ and a replay produces one accepted outcome.

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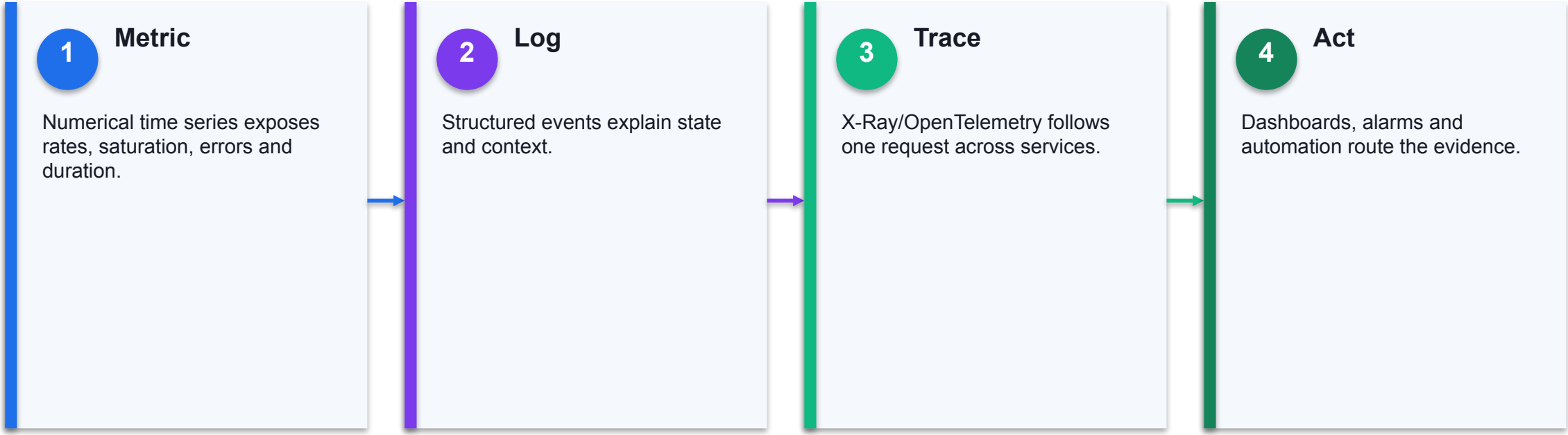
Messaging decouples rate, time and failure · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Observability: metric, log and trace evidence



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Observability: metric, log and trace evidence · technical contract

1	INPUT / INTENT **Metric**	Numerical time series exposes rates, saturation, errors and duration.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Log**	Structured events explain state and context.	RETAIN **Approved configuration**
3	STATE CHANGE **Trace**	X-Ray/OpenTelemetry follows one request across services.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Act**	Dashboards, alarms and automation route the evidence.	RETAIN **Audit, metric or test record**

CONTROL RISK A healthy average can hide a small but severe tail-latency or error population.

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Observability: metric, log and trace evidence · decision and control

1 CloudWatch metric

USE WHEN

Trend and threshold

TRADE-OFF

Low detail without dimensions

2 CloudWatch Logs

USE WHEN

Event-level diagnosis

TRADE-OFF

Needs structure and retention

3 X-Ray trace

USE WHEN

Distributed latency/error path

TRADE-OFF

Sampling and instrumentation required

FAILURE MODE

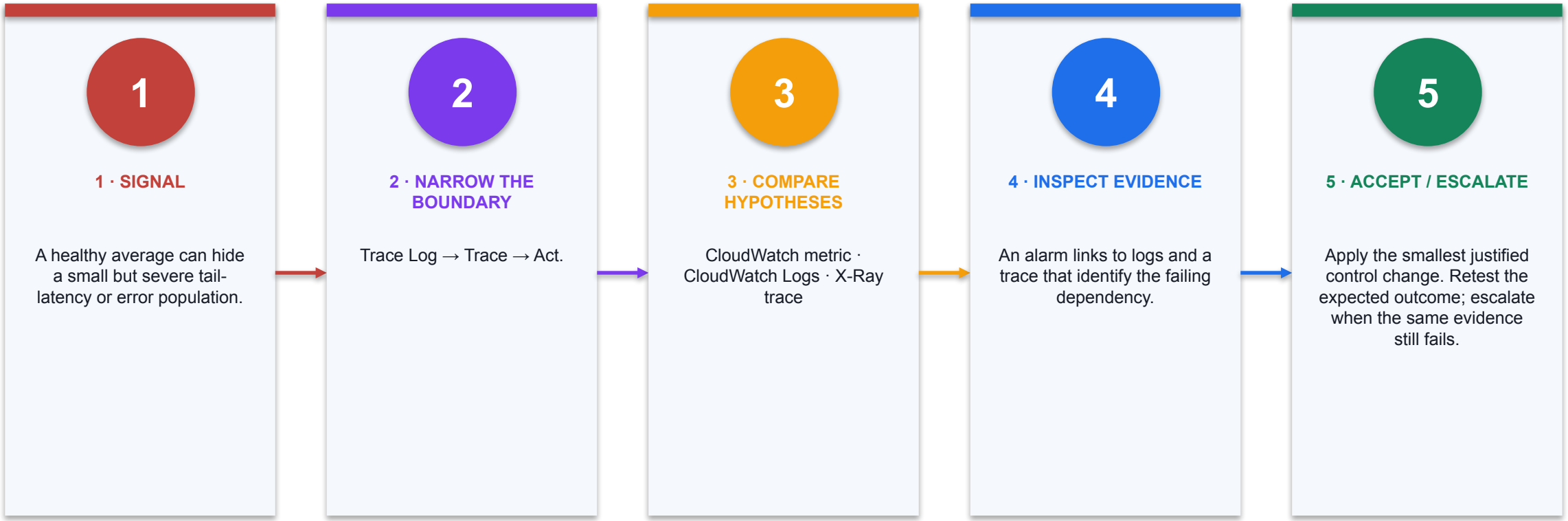
A healthy average can hide a small but severe tail-latency or error population.

VERIFICATION EVIDENCE

An alarm links to logs and a trace that identify the failing dependency.

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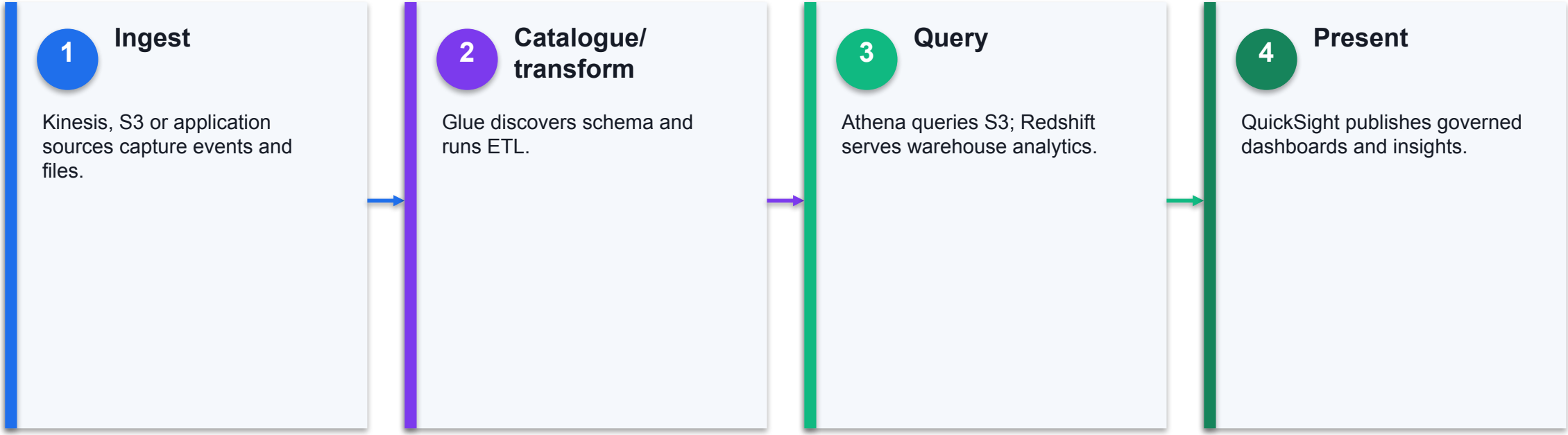
Observability: metric, log and trace evidence · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Analytics services form a data path



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Analytics services form a data path · technical contract

1	INPUT / INTENT **Ingest**	Kinesis, S3 or application sources capture events and files.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Catalogue/transform**	Glue discovers schema and runs ETL.	RETAIN **Approved configuration**
3	STATE CHANGE **Query**	Athena queries S3; Redshift serves warehouse analytics.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Present**	QuickSight publishes governed dashboards and insights.	RETAIN **Audit, metric or test record**

CONTROL RISK Querying unpartitioned text files in Athena can multiply scan cost and latency.

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Analytics services form a data path · decision and control

1 Athena

USE WHEN

Ad hoc SQL on S3

TRADE-OFF

Pay by scanned data; partition and format matter

2 Redshift

USE WHEN

Repeated warehouse analytics

TRADE-OFF

Model, load and capacity design

3 Kinesis

USE WHEN

Streaming ingestion

TRADE-OFF

Shard/throughput and consumer design

FAILURE MODE

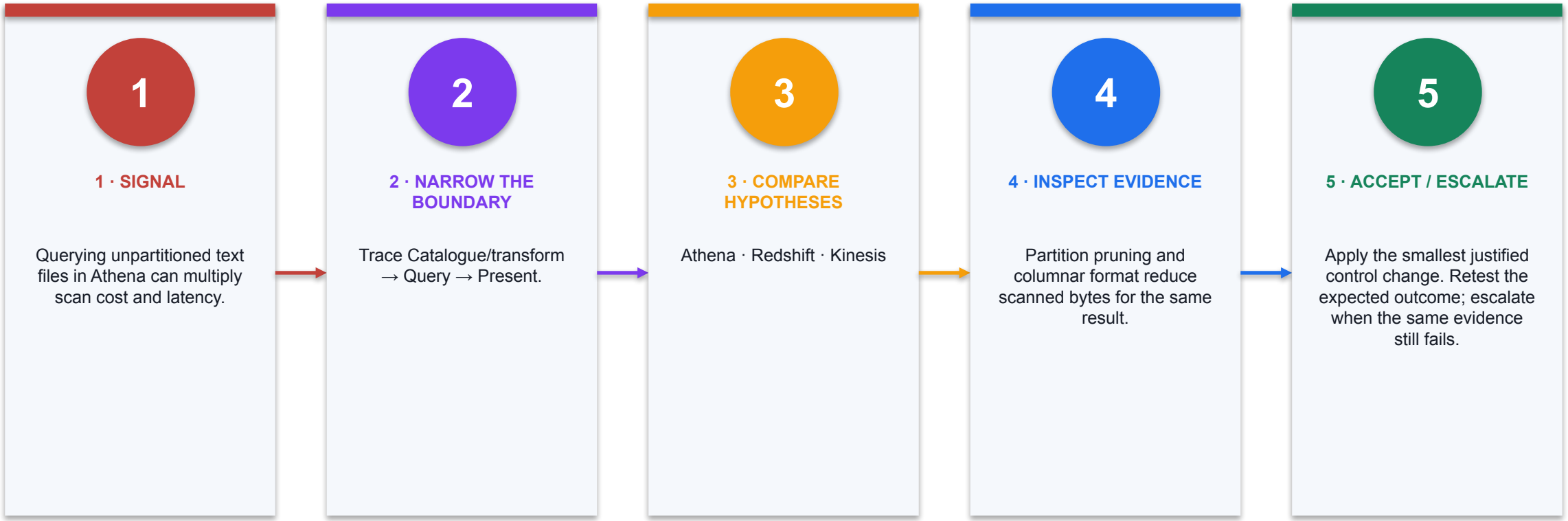
Querying unpartitioned text files in Athena can multiply scan cost and latency.

VERIFICATION EVIDENCE

Partition pruning and columnar format reduce scanned bytes for the same result.

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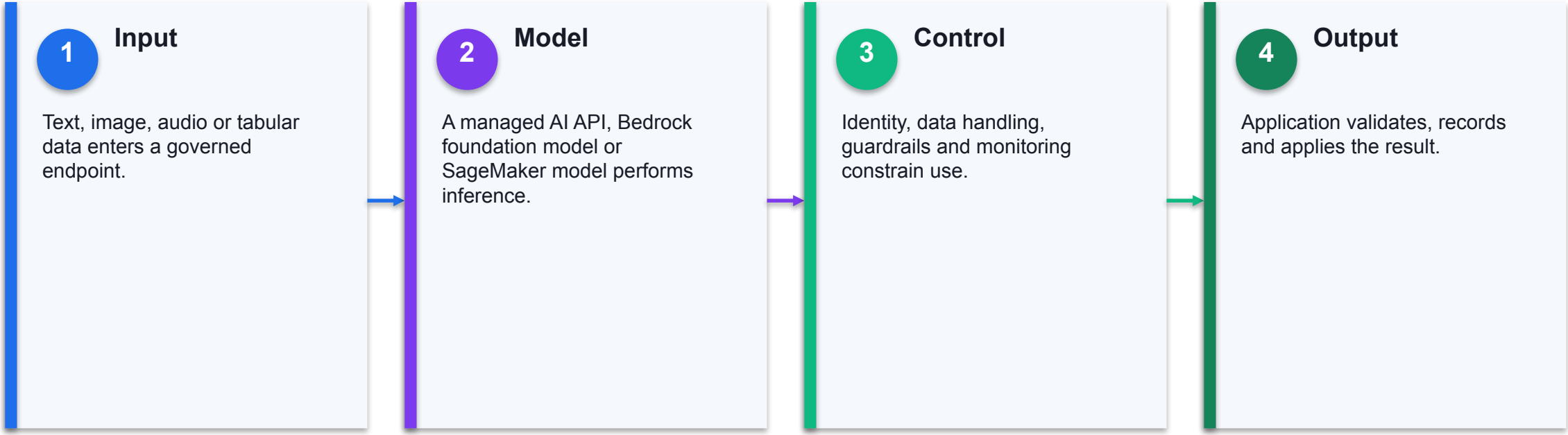
Analytics services form a data path · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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AI services: prebuilt API, foundation model or custom ML



MECHANISM

Read left to right: input, control, state change, observable evidence.

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AI services: prebuilt API, foundation model or custom ML · technical contract

1	INPUT / INTENT **Input**	Text, image, audio or tabular data enters a governed endpoint.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Model**	A managed AI API, Bedrock foundation model or SageMaker model performs inference.	RETAIN **Approved configuration**
3	STATE CHANGE **Control**	Identity, data handling, guardrails and monitoring constrain use.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Output**	Application validates, records and applies the result.	RETAIN **Audit, metric or test record**

CONTROL RISK Sending sensitive data to a model without classification and access controls creates privacy risk.

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AI services: prebuilt API, foundation model or custom ML · decision and control

1 AI service API

USE WHEN

Common vision/language tasks

TRADE-OFF

Fixed capability contract

2 Amazon Bedrock

USE WHEN

Generative AI with foundation models

TRADE-OFF

Prompt, safety and cost governance

3 SageMaker AI

USE WHEN

Build/train/deploy custom ML

TRADE-OFF

Data science and lifecycle complexity

FAILURE MODE

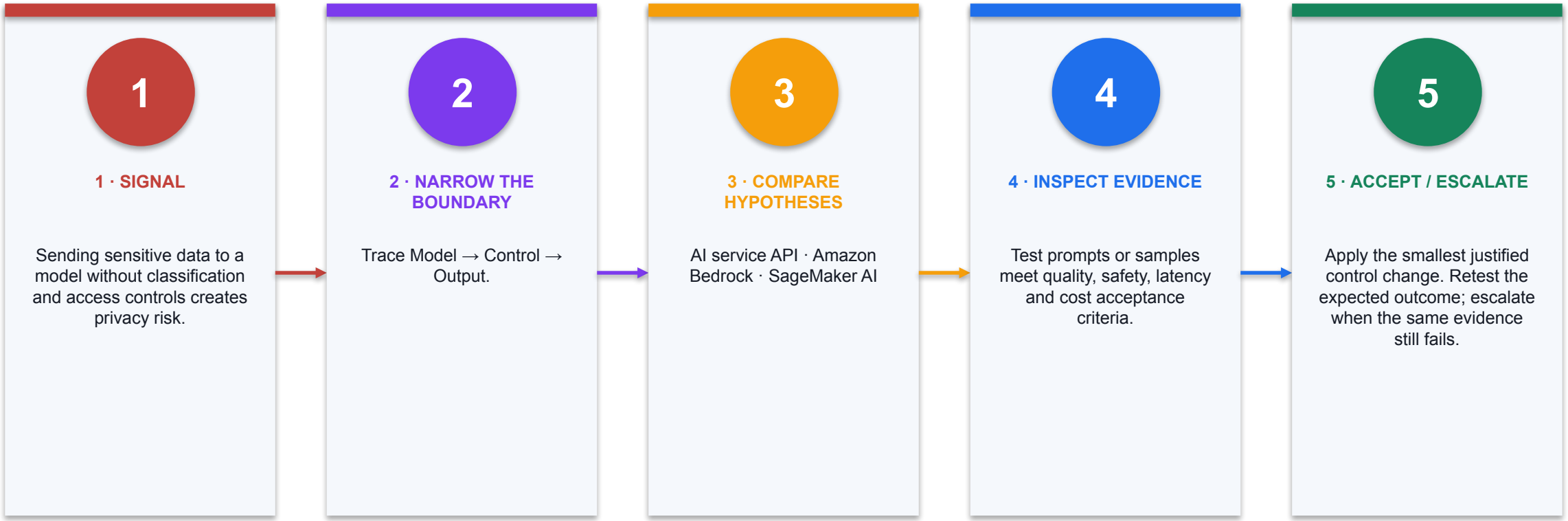
Sending sensitive data to a model without classification and access controls creates privacy risk.

VERIFICATION EVIDENCE

Test prompts or samples meet quality, safety, latency and cost acceptance criteria.

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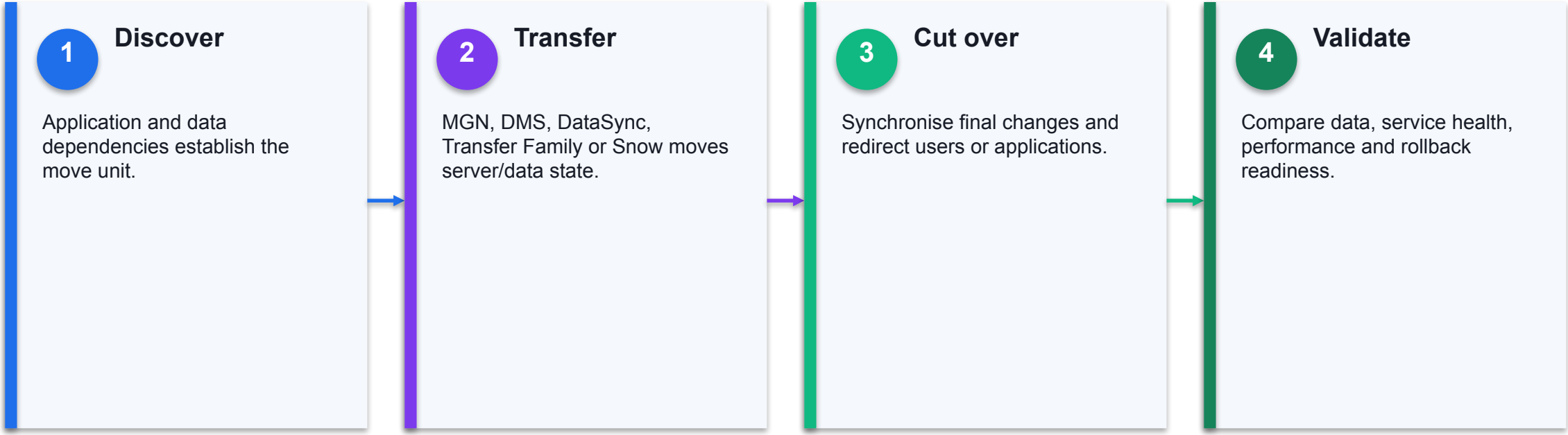
AI services: prebuilt API, foundation model or custom ML · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Migration services move different things



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Migration services move different things · technical contract

1	INPUT / INTENT **Discover**	Application and data dependencies establish the move unit.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Transfer**	MGN, DMS, DataSync, Transfer Family or Snow moves server/data state.	RETAIN **Approved configuration**
3	STATE CHANGE **Cut over**	Synchronise final changes and redirect users or applications.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Validate**	Compare data, service health, performance and rollback readiness.	RETAIN **Audit, metric or test record**

CONTROL RISK A successful byte transfer does not prove application consistency or business readiness.

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Migration services move different things · decision and control

1 Application Migration Service

USE WHEN

Lift servers to EC2

TRADE-OFF

Post-launch optimisation remains

2 Database Migration Service

USE WHEN

Continuous database replication

TRADE-OFF

Schema conversion may be separate

3 DataSync/Snow

USE WHEN

Online/offline bulk data move

TRADE-OFF

Bandwidth, device and cutover planning

FAILURE MODE

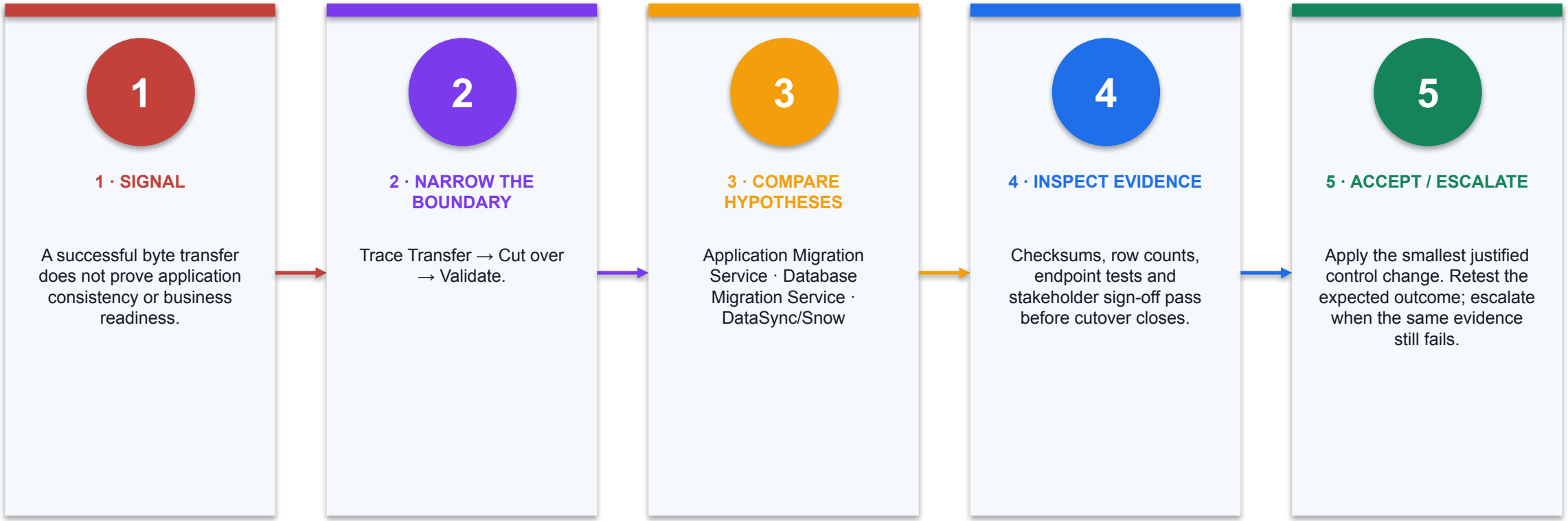
A successful byte transfer does not prove application consistency or business readiness.

VERIFICATION EVIDENCE

Checksums, row counts, endpoint tests and stakeholder sign-off pass before cutover closes.

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Migration services move different things · failure-to-evidence case



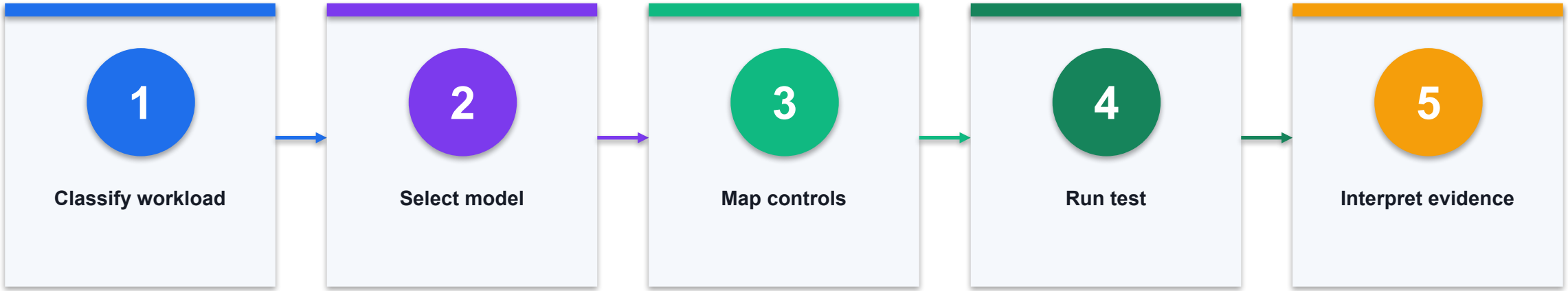
Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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DOMAIN 3 · LAB 3 · 55 MIN

Compute Selection and Health Test

A small web endpoint has a steady baseline, occasional bursts and a requirement for repeatable health evidence.



YOU'LL PRODUCE **EC2/Lambda/Fargate decision matrix, architecture sketch, health-test plan and optional sandbox evidence.**

Tools: Amazon EC2, AWS Lambda, Amazon ECS/Fargate, Elastic Load Balancing, Amazon CloudWatch · Detailed procedure: Learner Guide, Lab 3

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Compute Selection and Health Test · verification gate

1

PURPOSE

LO1 / K3 / A2 - select a compute model and test infrastructure plus application health.

2

ARTIFACT SET

EC2/Lambda/Fargate decision matrix, architecture sketch, health-test plan and optional sandbox evidence.

3

ACCEPTANCE TEST

The selected compute model follows workload attributes, both health layers have acceptance criteria, and any created resource is terminated.

EVIDENCE BOUNDARY **Amazon EC2, AWS Lambda, Amazon ECS/Fargate, Elastic Load Balancing, Amazon CloudWatch**

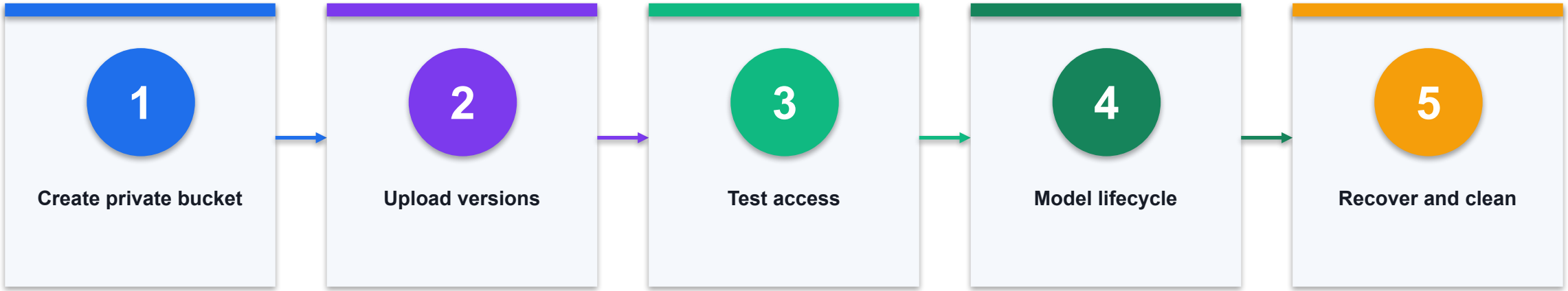
If an acceptance condition fails, retain the timestamped evidence and escalate with the tested boundary and observed result.

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DOMAIN 3 · LAB 4 · 55 MIN

S3 Protection, Versioning and Lifecycle

A team needs a private evidence bucket with recoverable versions and a documented retention path.



YOU'LL PRODUCE **Protected bucket configuration, two object versions, lifecycle rule analysis and cleanup record.**

Tools: Amazon S3, IAM Access Analyzer for S3, AWS KMS (conceptual where not authorised) · Detailed procedure: Learner Guide, Lab 4

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S3 Protection, Versioning and Lifecycle · verification gate

1

PURPOSE

LO1 / K1 / A2 - implement and verify object-storage protection and lifecycle behaviour.

2

ARTIFACT SET

Protected bucket configuration, two object versions, lifecycle rule analysis and cleanup record.

3

ACCEPTANCE TEST

Public access is blocked, two versions are visible, the prior content is recoverable, lifecycle timing is correct and cleanup is complete.

EVIDENCE BOUNDARY **Amazon S3, IAM Access Analyzer for S3, AWS KMS (conceptual where not authorised)**

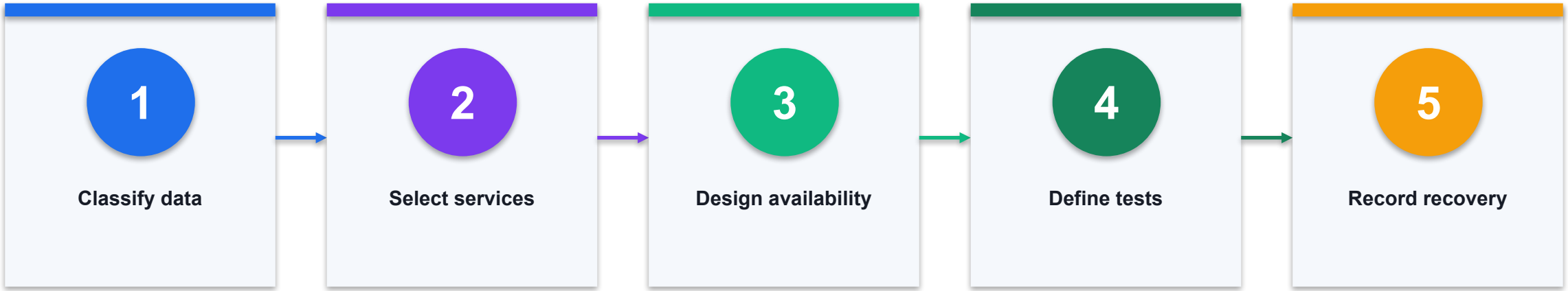
If an acceptance condition fails, retain the timestamped evidence and escalate with the tested boundary and observed result.

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DOMAIN 3 · LAB 5 · 50 MIN

Database Availability and Service Selection

A retail application needs relational transactions with AZ failover plus a high-scale session store.



YOU'LL PRODUCE **RDS Multi-AZ test plan, replica/backup comparison and DynamoDB key design.**

Tools: Amazon RDS, Amazon Aurora, Amazon DynamoDB, AWS Backup (review) · Detailed procedure: Learner Guide, Lab 5

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Database Availability and Service Selection · verification gate

1

PURPOSE

LO2 / K5 / A2 - distinguish relational availability, read scale, recovery and NoSQL access patterns.

2

ARTIFACT SET

RDS Multi-AZ test plan, replica/backup comparison and DynamoDB key design.

3

ACCEPTANCE TEST

Multi-AZ is not confused with read scaling, restore is independently tested, and the DynamoDB key supports targeted Query operations.

EVIDENCE BOUNDARY **Amazon RDS, Amazon Aurora, Amazon DynamoDB, AWS Backup (review)**

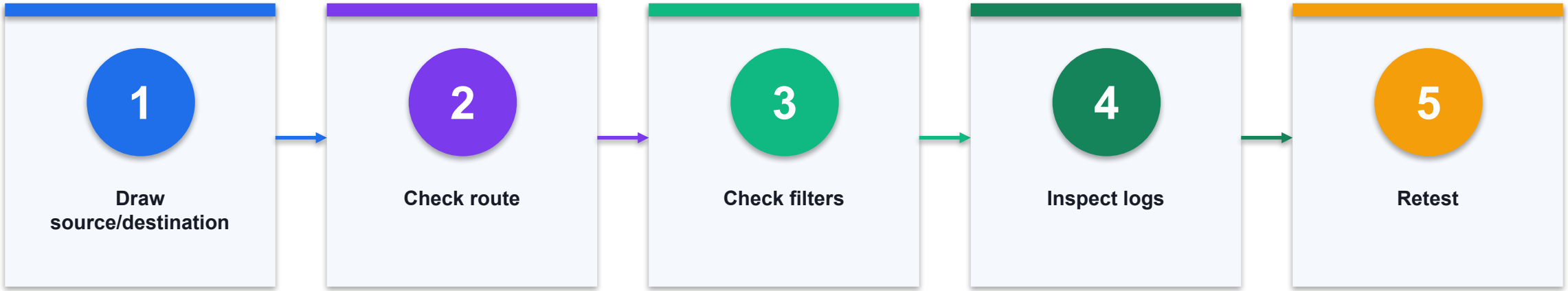
If an acceptance condition fails, retain the timestamped evidence and escalate with the tested boundary and observed result.

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DOMAIN 3 · LAB 6 · 55 MIN

VPC Traffic Path and Connectivity Diagnosis

A private application can reach its database but cannot download updates or receive traffic from the public load balancer.



YOU'LL PRODUCE **Annotated VPC path, route/security/NACL decision table, diagnosis evidence and corrected design.**

Tools: Amazon VPC, Security Groups, Network ACLs, NAT Gateway, Internet Gateway, VPC Flow Logs · Detailed procedure: Learner Guide, Lab 6

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VPC Traffic Path and Connectivity Diagnosis · verification gate

1

PURPOSE

LO2 / K2 / K4 / A3-A4 - trace and resolve a blocked cloud network path.

2

ARTIFACT SET

Annotated VPC path, route/security/NACL decision table, diagnosis evidence and corrected design.

3

ACCEPTANCE TEST

The diagram shows both directions, the root cause is supported by route/filter/log evidence, and the narrow correction passes the retest.

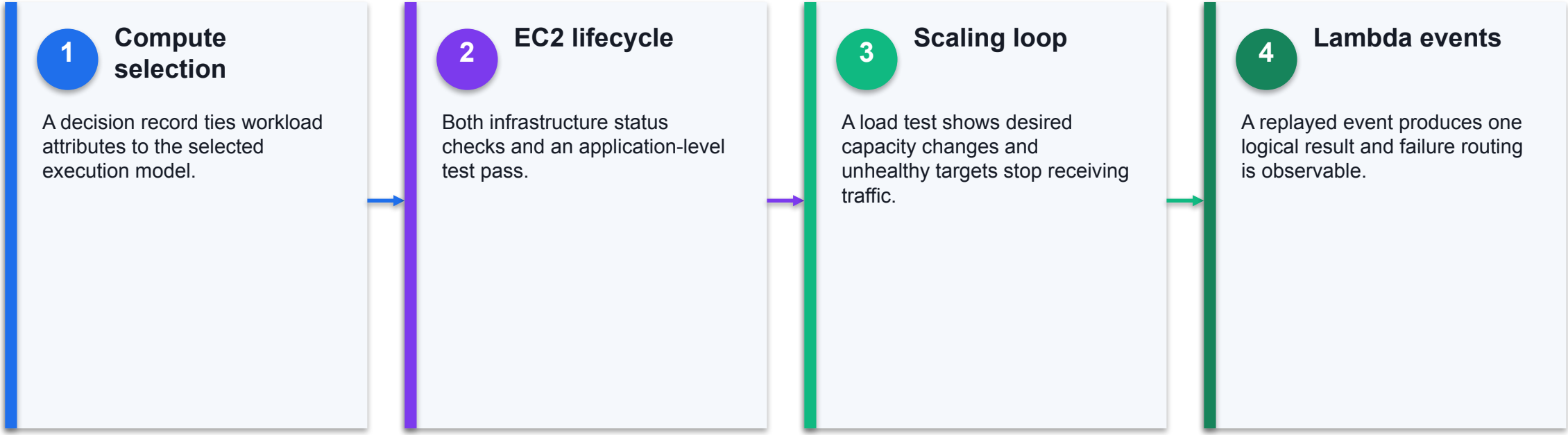
EVIDENCE BOUNDARY **Amazon VPC, Security Groups, Network ACLs, NAT Gateway, Internet Gateway, VPC Flow Logs**

If an acceptance condition fails, retain the timestamped evidence and escalate with the tested boundary and observed result.

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RECAP · VERIFY BEFORE MOVING ON

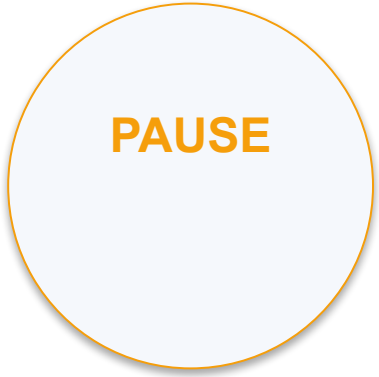
Domain 3 evidence recap



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Lunch Break

60 minutes

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04

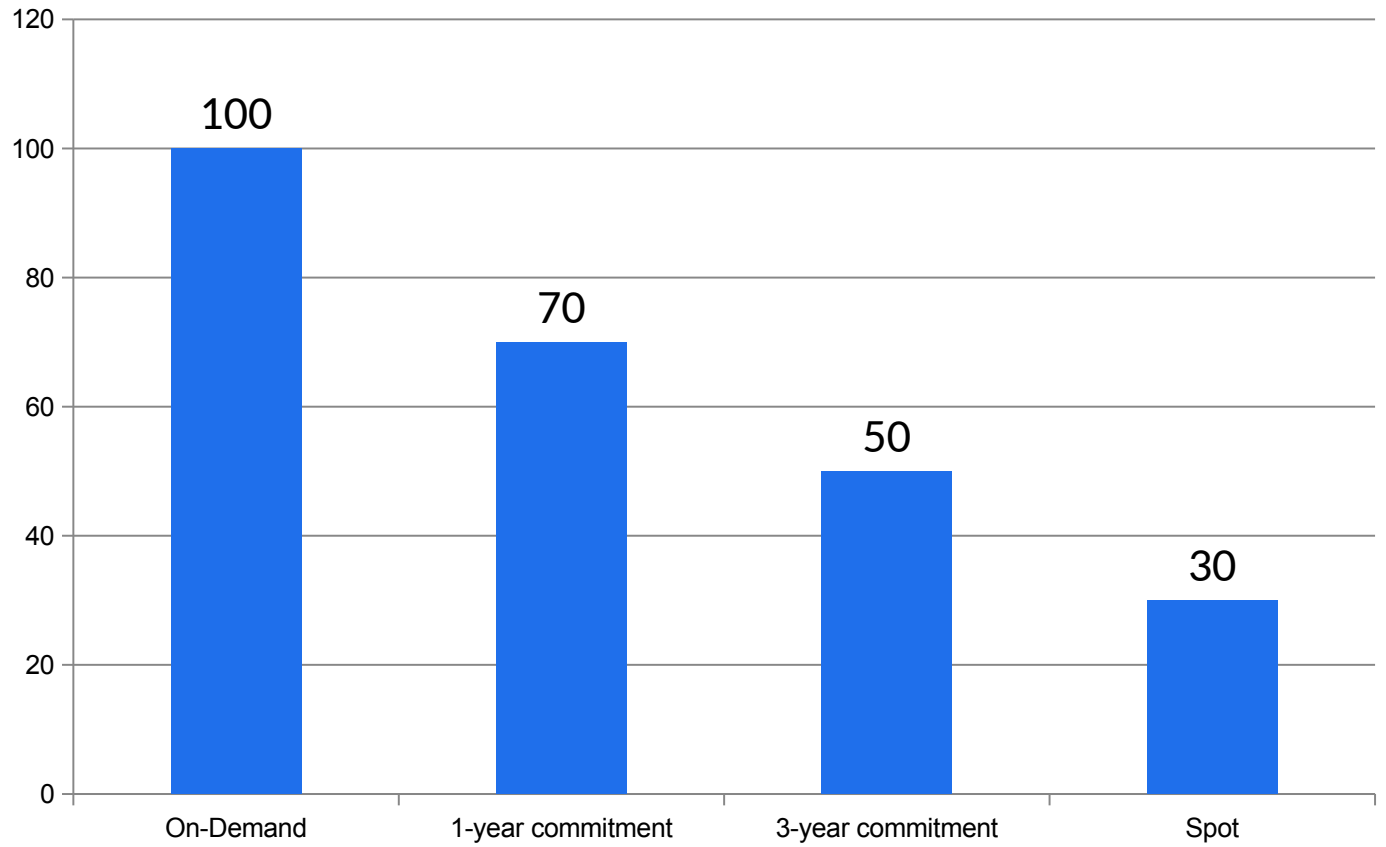
DOMAIN 4 · EXAM WEIGHT 12%

Billing, Pricing, and Support

Pricing models · cost tools · purchasing · support · governance

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Illustrative compute purchase price index



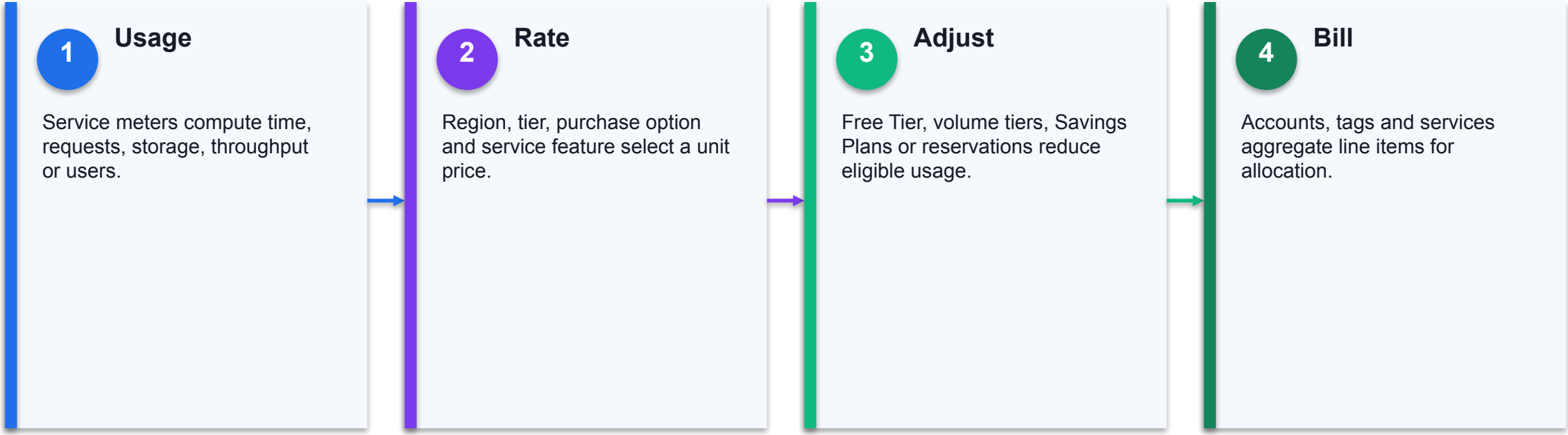
WHAT THE DATA SHOWS

Discounts trade flexibility for commitment or interruption tolerance. Values are illustrative teaching indices, not quotes.

Editable PowerPoint chart

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AWS pricing is a measured-resource equation



MECHANISM

Read left to right: input, control, state change, observable evidence.

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AWS pricing is a measured-resource equation · technical contract

1	INPUT / INTENT **Usage**	Service meters compute time, requests, storage, throughput or users.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Rate**	Region, tier, purchase option and service feature select a unit price.	RETAIN **Approved configuration**
3	STATE CHANGE **Adjust**	Free Tier, volume tiers, Savings Plans or reservations reduce eligible usage.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Bill**	Accounts, tags and services aggregate line items for allocation.	RETAIN **Audit, metric or test record**

CONTROL RISK Assuming every service is charged per hour causes inaccurate estimates and budgets.

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AWS pricing is a measured-resource equation · decision and control

1 Pay as you go

USE WHEN

Variable or uncertain demand

TRADE-OFF

Unit price can be higher

2 Save when committed

USE WHEN

Stable baseline usage

TRADE-OFF

Commitment risk

3 Pay less with scale

USE WHEN

High aggregate use

TRADE-OFF

Tier rules differ by service

FAILURE MODE

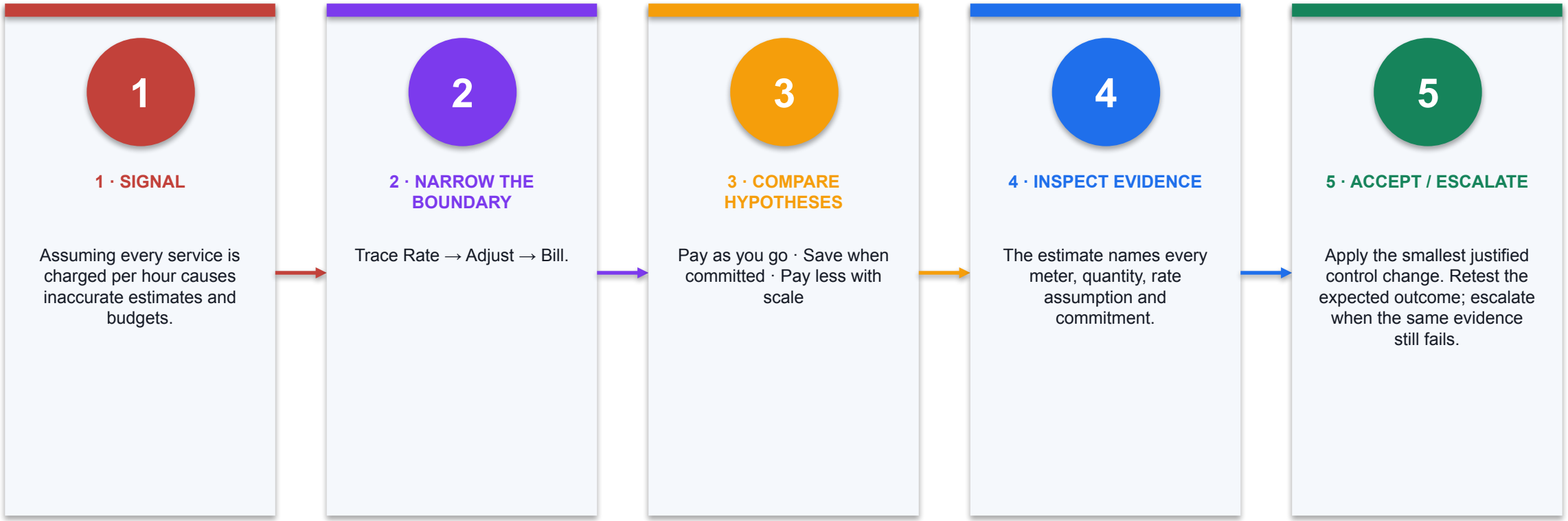
Assuming every service is charged per hour causes inaccurate estimates and budgets.

VERIFICATION EVIDENCE

The estimate names every meter, quantity, rate assumption and commitment.

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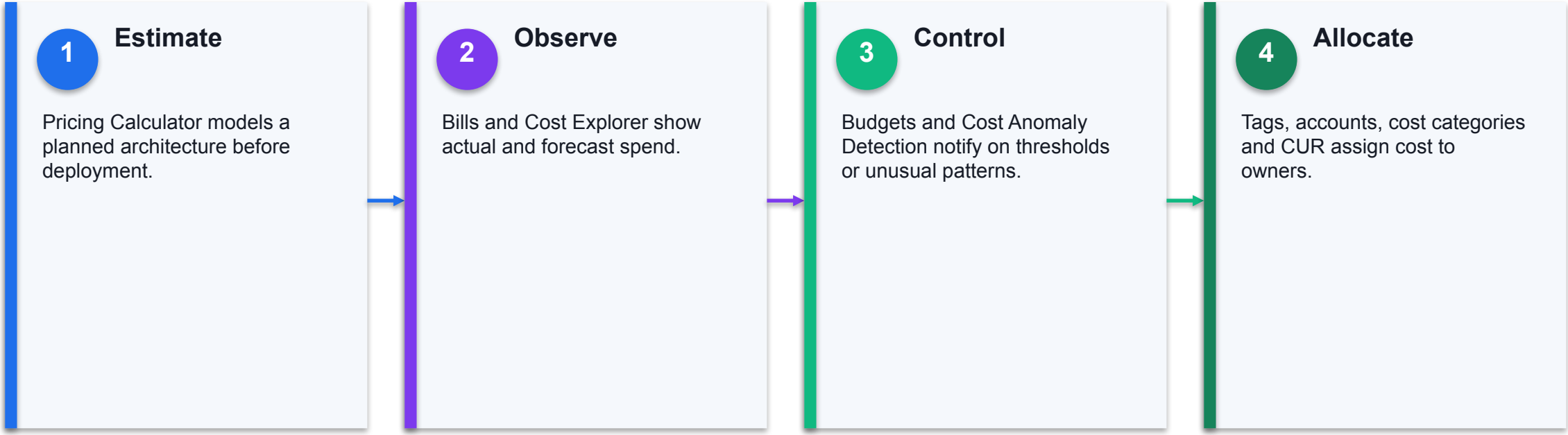
AWS pricing is a measured-resource equation · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Cost tools answer different questions



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Cost tools answer different questions · technical contract

1	INPUT / INTENT **Estimate**	Pricing Calculator models a planned architecture before deployment.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Observe**	Bills and Cost Explorer show actual and forecast spend.	RETAIN **Approved configuration**
3	STATE CHANGE **Control**	Budgets and Cost Anomaly Detection notify on thresholds or unusual patterns.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Allocate**	Tags, accounts, cost categories and CUR assign cost to owners.	RETAIN **Audit, metric or test record**

CONTROL RISK A budget does not prevent spend unless an approved budget action is configured.

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Cost tools answer different questions · decision and control

1 Pricing Calculator

USE WHEN

What might this design cost?

TRADE-OFF

Model quality depends on assumptions

2 Cost Explorer

USE WHEN

What did we spend and forecast?

TRADE-OFF

Data latency and granularity

3 Budgets

USE WHEN

When should we notify or act?

TRADE-OFF

Alert is not automatic optimisation

FAILURE MODE

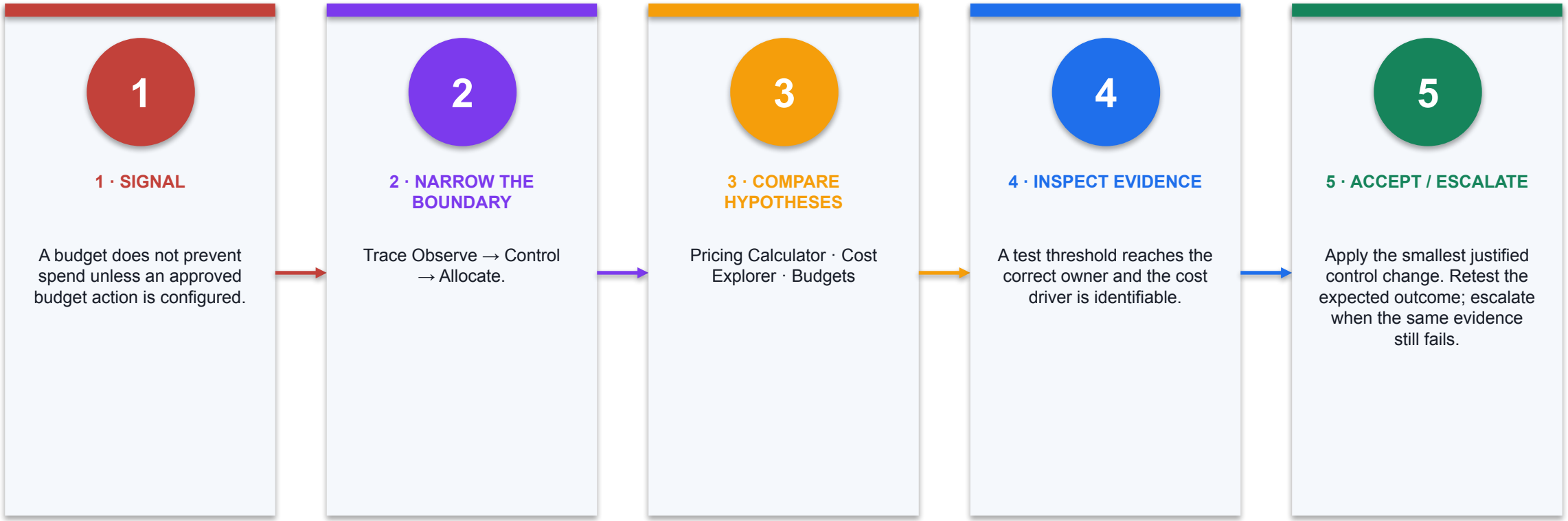
A budget does not prevent spend unless an approved budget action is configured.

VERIFICATION EVIDENCE

A test threshold reaches the correct owner and the cost driver is identifiable.

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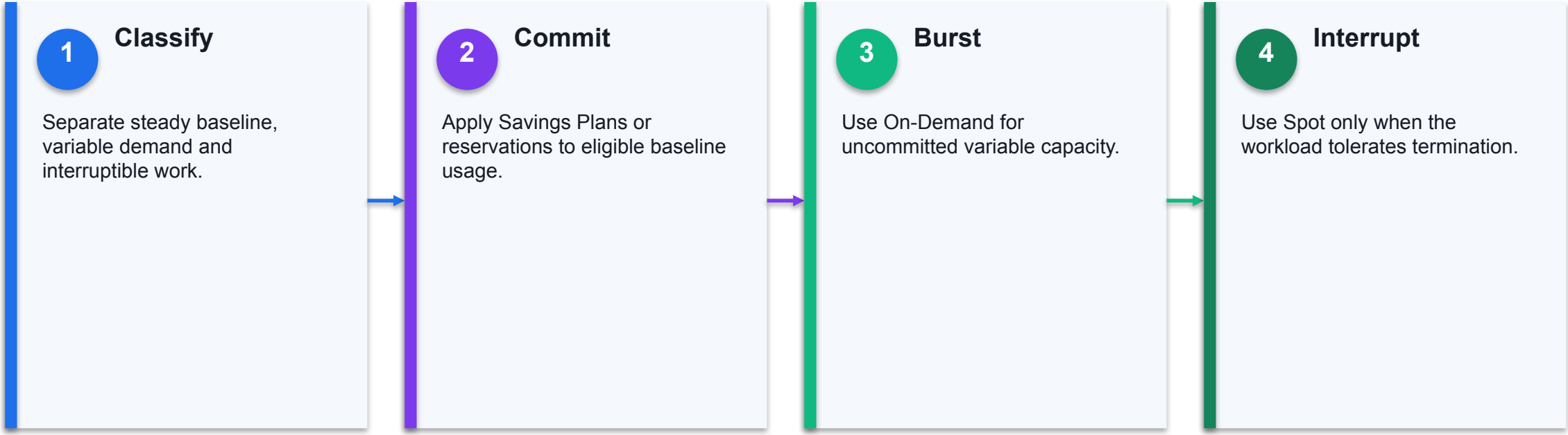
Cost tools answer different questions · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Compute purchase options match interruption and commitment



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Compute purchase options match interruption and commitment · technical contract

1	INPUT / INTENT **Classify**	Separate steady baseline, variable demand and interruptible work.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Commit**	Apply Savings Plans or reservations to eligible baseline usage.	RETAIN **Approved configuration**
3	STATE CHANGE **Burst**	Use On-Demand for uncommitted variable capacity.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Interrupt**	Use Spot only when the workload tolerates termination.	RETAIN **Audit, metric or test record**

CONTROL RISK Running a stateful single-instance database on Spot without recovery design risks data loss and outage.

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Compute purchase options match interruption and commitment · decision and control

1 On-Demand

USE WHEN

Short or unpredictable workloads

TRADE-OFF

Highest flexibility

2 Savings Plans/RI

USE WHEN

Predictable baseline

TRADE-OFF

Commitment and coverage management

3 Spot

USE WHEN

Fault-tolerant flexible jobs

TRADE-OFF

Can be interrupted

FAILURE MODE

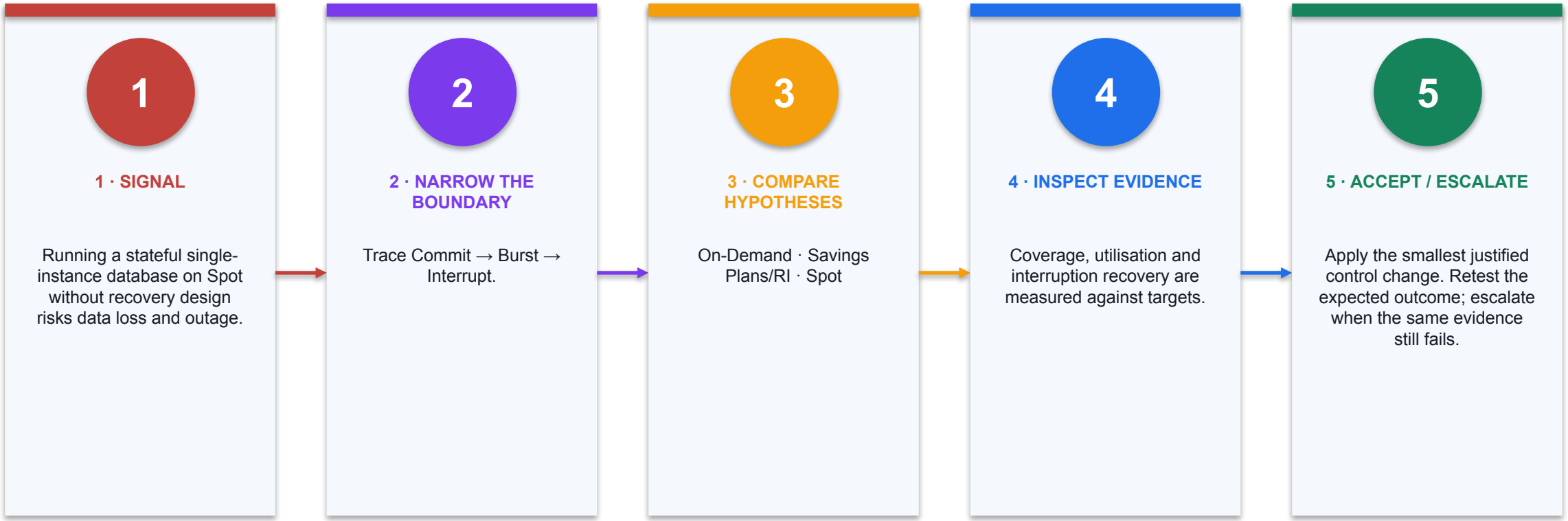
Running a stateful single-instance database on Spot without recovery design risks data loss and outage.

VERIFICATION EVIDENCE

Coverage, utilisation and interruption recovery are measured against targets.

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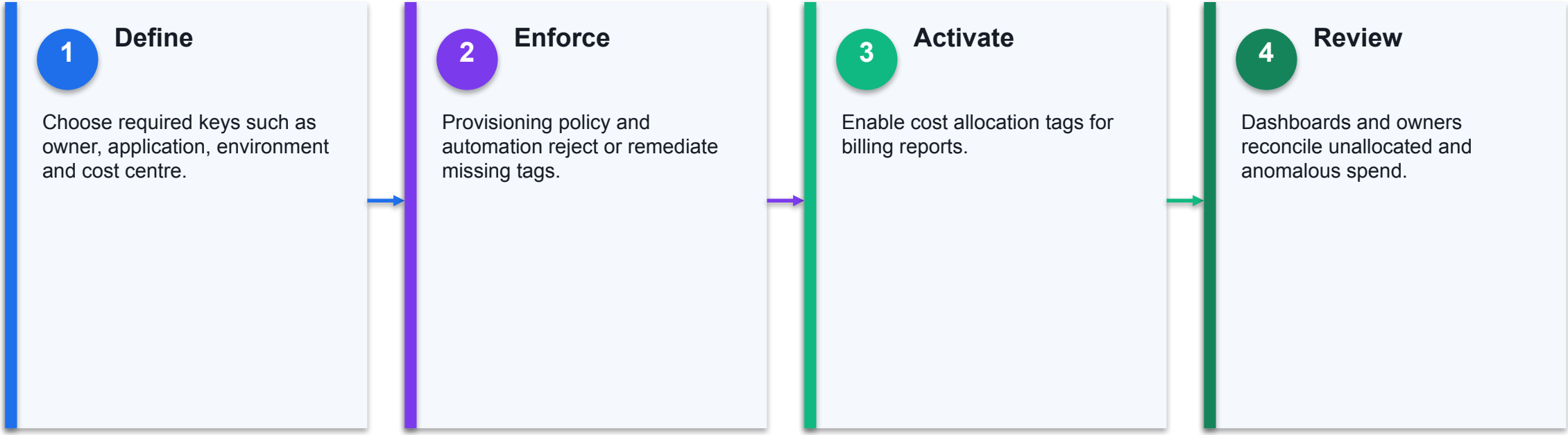
Compute purchase options match interruption and commitment · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Tagging turns spend into accountable units



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Tagging turns spend into accountable units · technical contract

1	INPUT / INTENT **Define**	Choose required keys such as owner, application, environment and cost centre.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Enforce**	Provisioning policy and automation reject or remediate missing tags.	RETAIN **Approved configuration**
3	STATE CHANGE **Activate**	Enable cost allocation tags for billing reports.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Review**	Dashboards and owners reconcile unallocated and anomalous spend.	RETAIN **Audit, metric or test record**

CONTROL RISK A tag that is present but not activated as a cost allocation tag will not appear as expected in billing analysis.

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Tagging turns spend into accountable units · decision and control

1 Account

USE WHEN

Strong cost and security boundary

TRADE-OFF

More account management

2 Tag

USE WHEN

Flexible resource attribution

TRADE-OFF

Coverage and spelling can drift

3 Cost category

USE WHEN

Business rule over billing dimensions

TRADE-OFF

Requires rule maintenance

FAILURE MODE

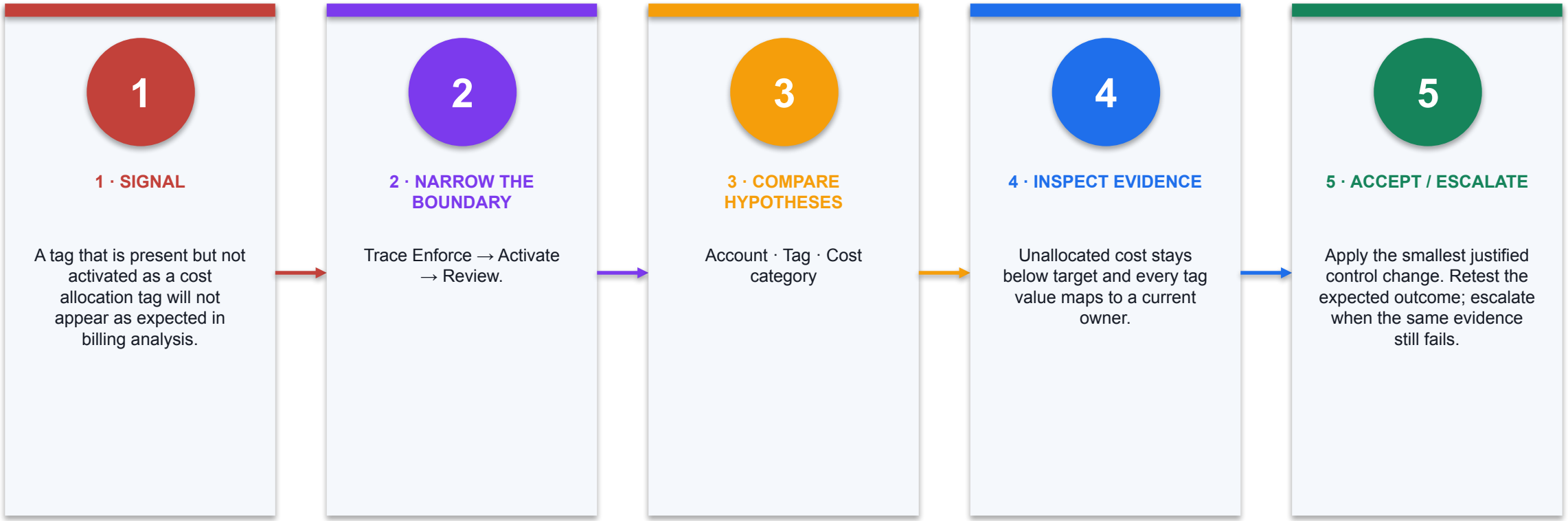
A tag that is present but not activated as a cost allocation tag will not appear as expected in billing analysis.

VERIFICATION EVIDENCE

Unallocated cost stays below target and every tag value maps to a current owner.

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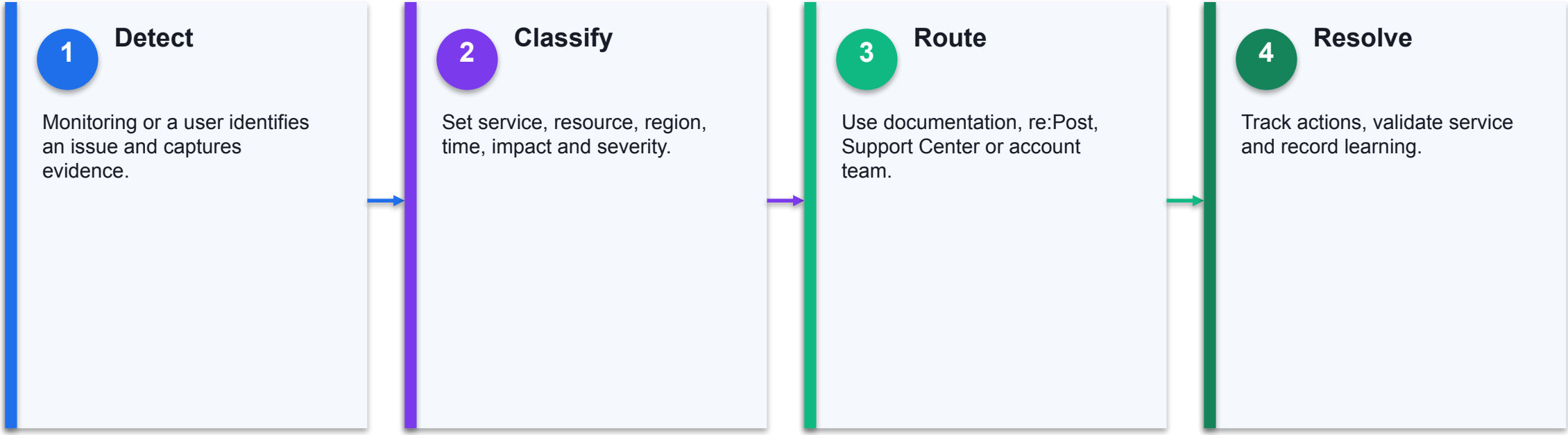
Tagging turns spend into accountable units · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Support path depends on impact and plan



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Support path depends on impact and plan · technical contract

1	INPUT / INTENT **Detect**	Monitoring or a user identifies an issue and captures evidence.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Classify**	Set service, resource, region, time, impact and severity.	RETAIN **Approved configuration**
3	STATE CHANGE **Route**	Use documentation, re:Post, Support Center or account team.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Resolve**	Track actions, validate service and record learning.	RETAIN **Audit, metric or test record**

CONTROL RISK Opening a case without resource IDs, timestamps and evidence delays diagnosis.

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Support path depends on impact and plan · decision and control

1 Basic/self-service

USE WHEN

Documentation, account and billing help

TRADE-OFF

No technical support cases

2 Business Support+

USE WHEN

Production technical support needs

TRADE-OFF

Plan cost and severity rules

3 Enterprise tiers

USE WHEN

Mission-critical proactive support

TRADE-OFF

Higher commitment and governance

FAILURE MODE

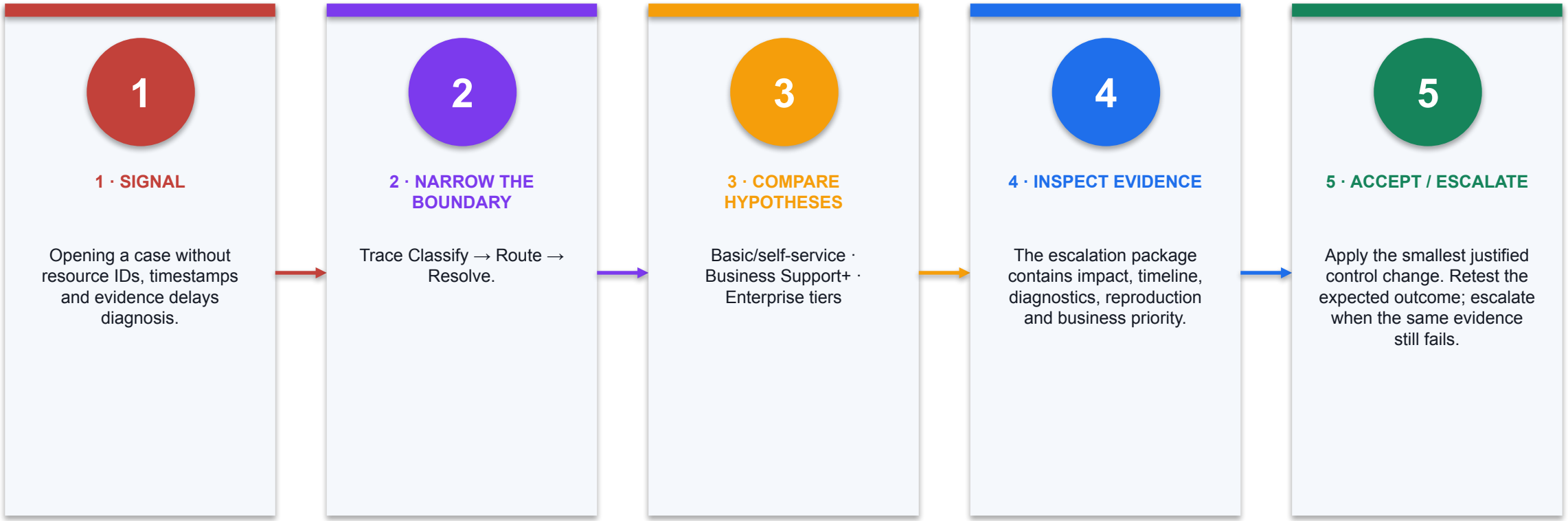
Opening a case without resource IDs, timestamps and evidence delays diagnosis.

VERIFICATION EVIDENCE

The escalation package contains impact, timeline, diagnostics, reproduction and business priority.

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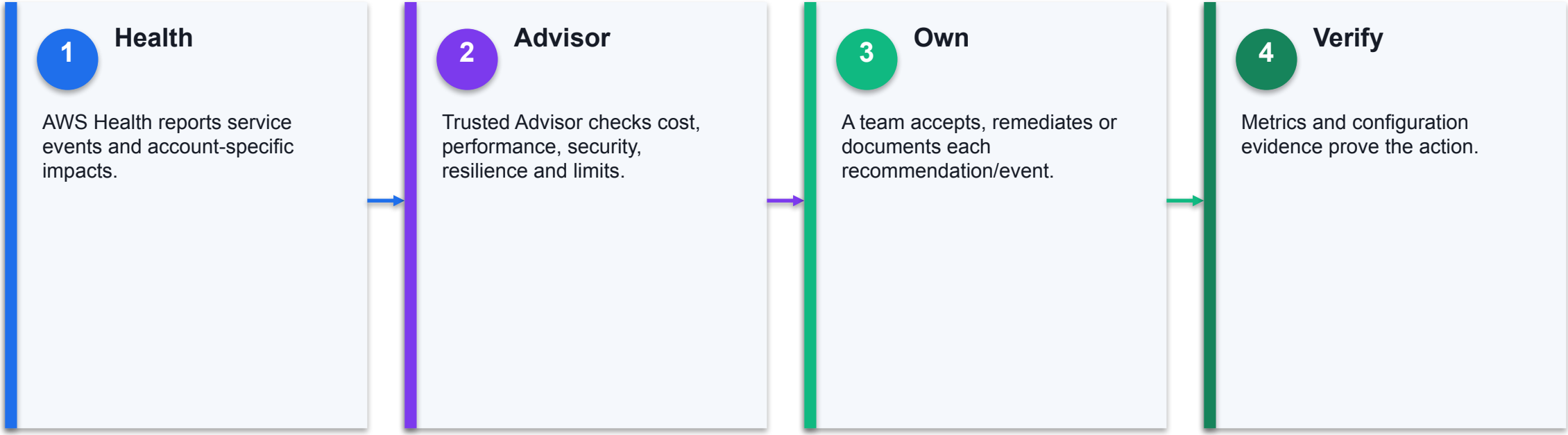
Support path depends on impact and plan · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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AWS Health and Trusted Advisor have different jobs



MECHANISM

Read left to right: input, control, state change, observable evidence.

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AWS Health and Trusted Advisor have different jobs · technical contract

1	INPUT / INTENT **Health**	AWS Health reports service events and account-specific impacts.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Advisor**	Trusted Advisor checks cost, performance, security, resilience and limits.	RETAIN **Approved configuration**
3	STATE CHANGE **Own**	A team accepts, remediates or documents each recommendation/event.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Verify**	Metrics and configuration evidence prove the action.	RETAIN **Audit, metric or test record**

CONTROL RISK A green public status page does not rule out an account-specific configuration issue.

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AWS Health and Trusted Advisor have different jobs · decision and control

1 Service Health

USE WHEN

Public regional service status

TRADE-OFF

Not account-specific diagnosis

2 AWS Health Dashboard

USE WHEN

Account events and planned changes

TRADE-OFF

Needs notification routing

3 Trusted Advisor

USE WHEN

Best-practice checks

TRADE-OFF

Available checks depend on plan

FAILURE MODE

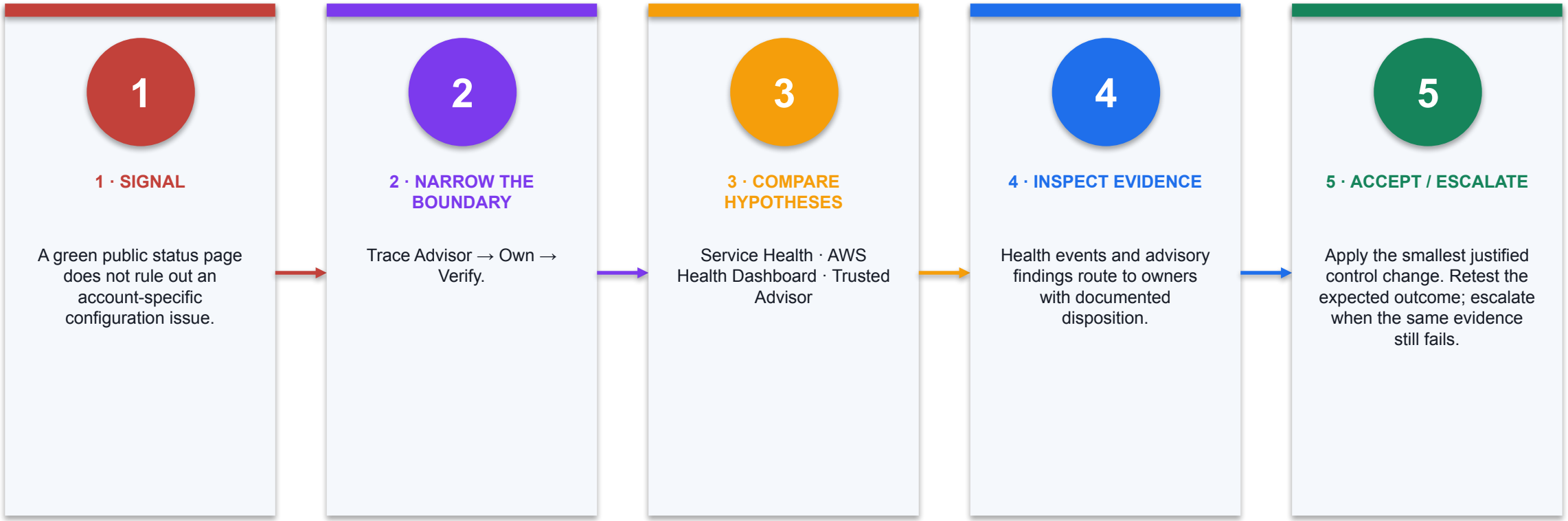
A green public status page does not rule out an account-specific configuration issue.

VERIFICATION EVIDENCE

Health events and advisory findings route to owners with documented disposition.

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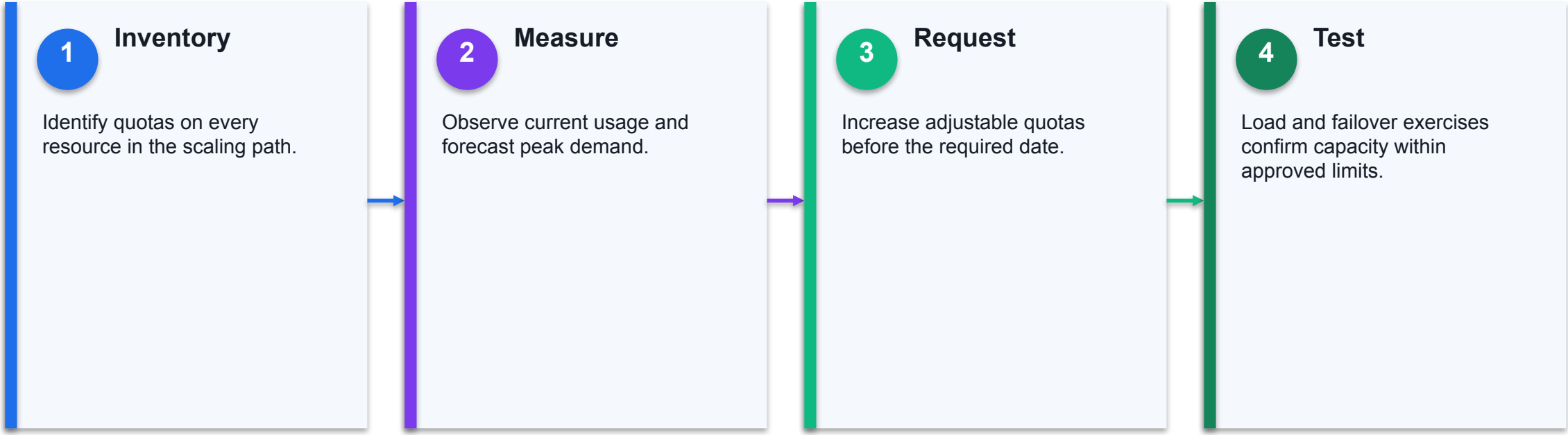
AWS Health and Trusted Advisor have different jobs · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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Service quotas are an availability control



MECHANISM

Read left to right: input, control, state change, observable evidence.

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Service quotas are an availability control · technical contract

1	INPUT / INTENT **Inventory**	Identify quotas on every resource in the scaling path.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Measure**	Observe current usage and forecast peak demand.	RETAIN **Approved configuration**
3	STATE CHANGE **Request**	Increase adjustable quotas before the required date.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Test**	Load and failover exercises confirm capacity within approved limits.	RETAIN **Audit, metric or test record**

CONTROL RISK Auto Scaling cannot add capacity after a dependent service quota is exhausted.

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Service quotas are an availability control · decision and control

1 Adjustable quota

USE WHEN

Raise through Service Quotas

TRADE-OFF

Lead time and regional scope

2 Fixed limit

USE WHEN

Redesign or shard workload

TRADE-OFF

Cannot request increase

3 Budget limit

USE WHEN

Financial guardrail

TRADE-OFF

Different from technical quota

FAILURE MODE

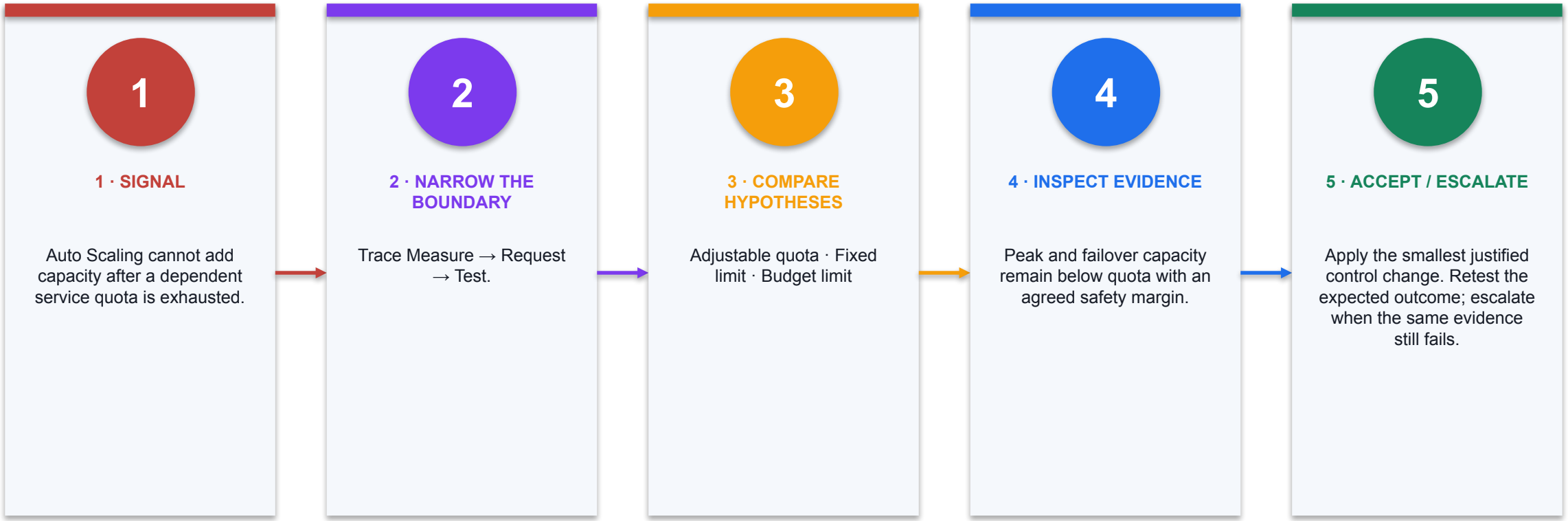
Auto Scaling cannot add capacity after a dependent service quota is exhausted.

VERIFICATION EVIDENCE

Peak and failover capacity remain below quota with an agreed safety margin.

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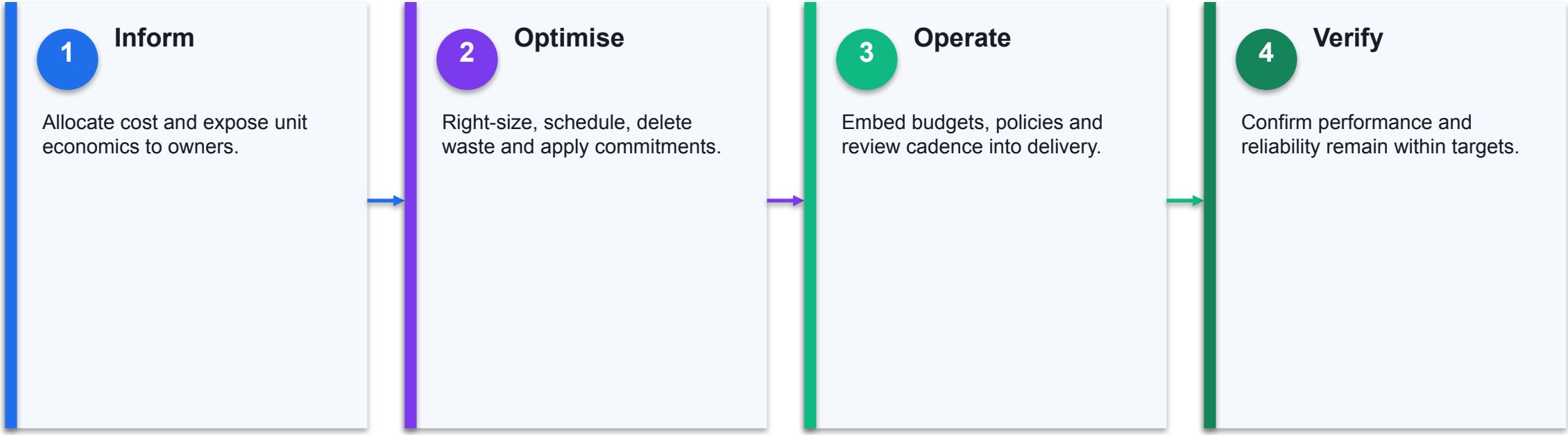
Service quotas are an availability control · failure-to-evidence case



Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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FinOps loop: visibility to verified optimisation



MECHANISM

Read left to right: input, control, state change, observable evidence.

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FinOps loop: visibility to verified optimisation · technical contract

1	INPUT / INTENT **Inform**	Allocate cost and expose unit economics to owners.	RETAIN **Declared requirement**
2	CONTROL BOUNDARY **Optimise**	Right-size, schedule, delete waste and apply commitments.	RETAIN **Approved configuration**
3	STATE CHANGE **Operate**	Embed budgets, policies and review cadence into delivery.	RETAIN **Runtime or resource state**
4	OBSERVABLE RESULT **Verify**	Confirm performance and reliability remain within targets.	RETAIN **Audit, metric or test record**

CONTROL RISK A cost reduction that breaches reliability or performance targets is not an optimisation.

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FinOps loop: visibility to verified optimisation · decision and control

1 Right-size

USE WHEN

Persistent underutilisation

TRADE-OFF

Validate performance after change

2 Schedule

USE WHEN

Non-production idle periods

TRADE-OFF

Avoid stopping required dependencies

3 Commit

USE WHEN

Stable verified baseline

TRADE-OFF

Do not cover temporary demand

FAILURE MODE

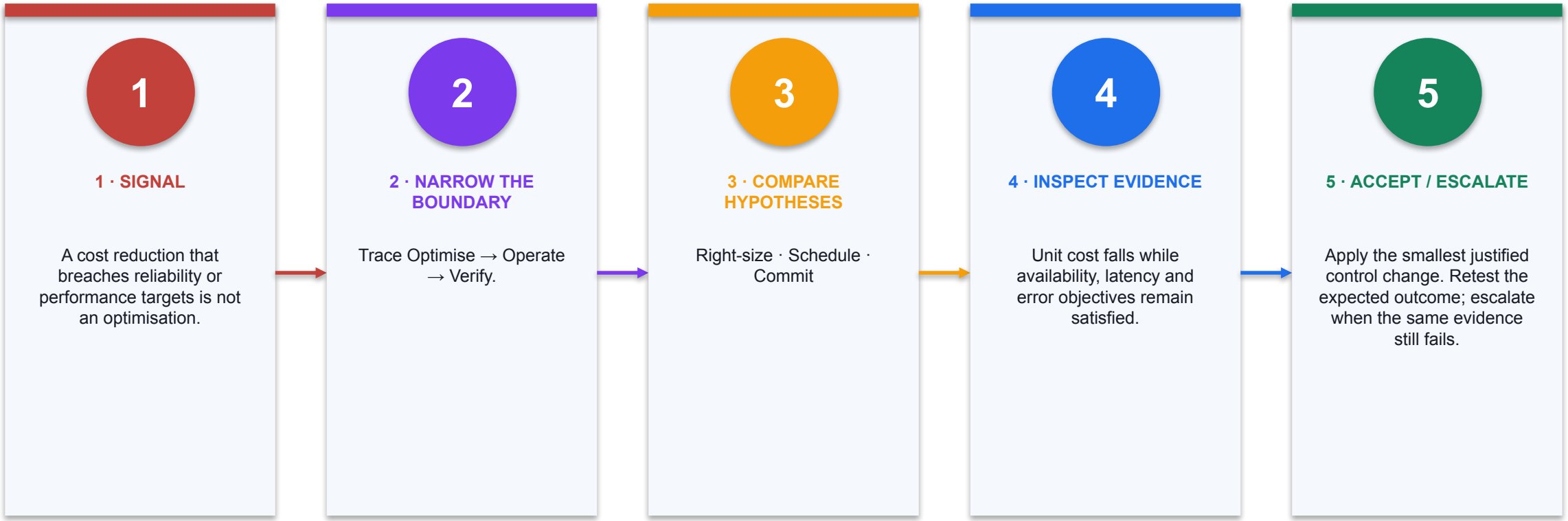
A cost reduction that breaches reliability or performance targets is not an optimisation.

VERIFICATION EVIDENCE

Unit cost falls while availability, latency and error objectives remain satisfied.

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FinOps loop: visibility to verified optimisation · failure-to-evidence case



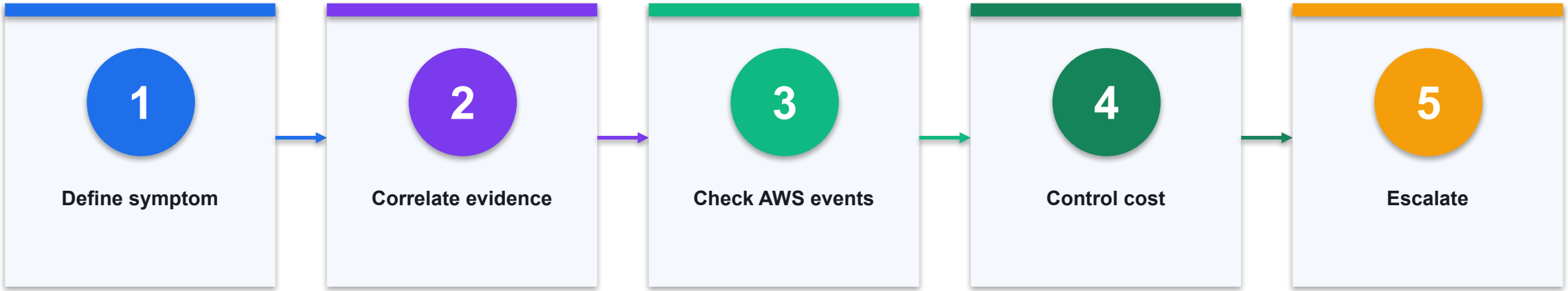
Diagnostic rule: do not change architecture, policy or capacity until the evidence identifies the failing boundary.

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DOMAIN 4 · LAB 7 · 50 MIN

Observability, Budget and Support Escalation

Users report intermittent errors while cloud spend is rising; the operations team needs evidence before escalation.



YOU'LL PRODUCE **Metric/log/trace evidence map, budget/anomaly alert, support case draft and escalation decision.**

Tools: Amazon CloudWatch, AWS X-Ray, AWS CloudTrail, AWS Health, AWS Budgets, AWS Support Center · Detailed procedure: Learner Guide, Lab 7

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Observability, Budget and Support Escalation · verification gate

1

PURPOSE

LO3 / K2 / A3 / A6 - diagnose an issue, control cost and prepare an escalation package.

2

ARTIFACT SET

Metric/log/trace evidence map, budget/anomaly alert, support case draft and escalation decision.

3

ACCEPTANCE TEST

Evidence distinguishes likely layers, cost driver and alert owner are visible, and the support draft is complete without credentials.

EVIDENCE BOUNDARY **Amazon CloudWatch, AWS X-Ray, AWS CloudTrail, AWS Health, AWS Budgets, AWS Support Center**

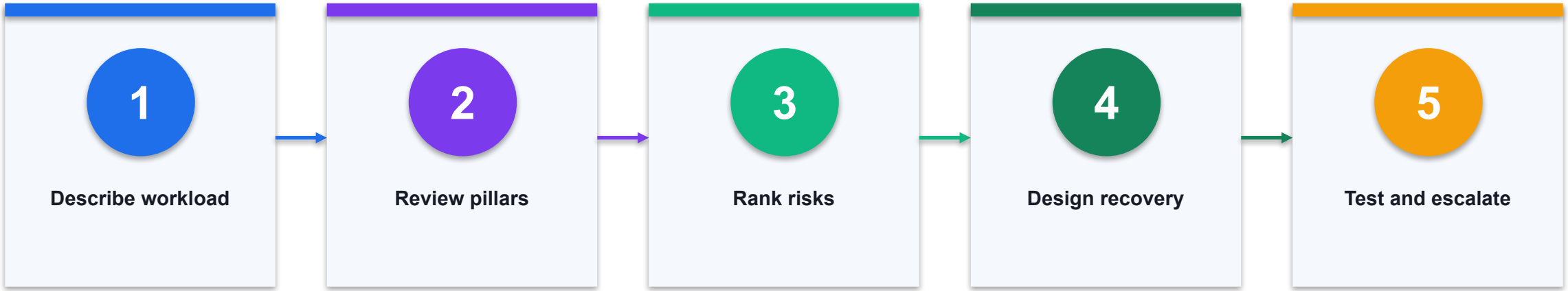
If an acceptance condition fails, retain the timestamped evidence and escalate with the tested boundary and observed result.

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DOMAIN 4 · LAB 8 · 50 MIN

Well-Architected Review and Recovery Test

A two-tier retail service must remain secure, recoverable and cost-aware as demand grows.



YOU'LL PRODUCE **Six-pillar risk register, prioritised improvement plan, RPO/RTO test and escalation path.**

Tools: AWS Well-Architected Tool or Framework, AWS Backup, AWS Resilience Hub (awareness), AWS Support · Detailed procedure: Learner Guide, Lab 8

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Well-Architected Review and Recovery Test · verification gate

1

PURPOSE

LO3 / A5-A6 - evaluate a workload, prioritise risk and define measurable recovery evidence.

2

ARTIFACT SET

Six-pillar risk register, prioritised improvement plan, RPO/RTO test and escalation path.

3

ACCEPTANCE TEST

All six pillars have evidence, risks are prioritised, the recovery test has measurable RPO/RTO criteria and unresolved risk has an owner/escalation path.

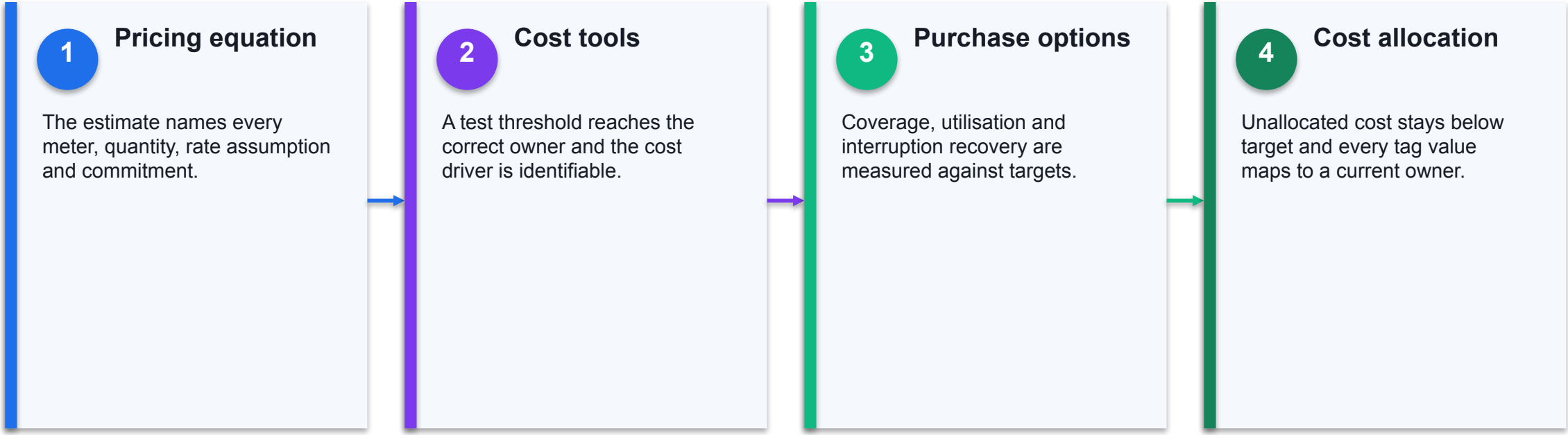
EVIDENCE BOUNDARY **AWS Well-Architected Tool or Framework, AWS Backup, AWS Resilience Hub (awareness), AWS Support**

If an acceptance condition fails, retain the timestamped evidence and escalate with the tested boundary and observed result.

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RECAP · VERIFY BEFORE MOVING ON

Domain 4 evidence recap



MECHANISM

Read left to right: input, control, state change, observable evidence.

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FINAL RECAP

Course evidence recap



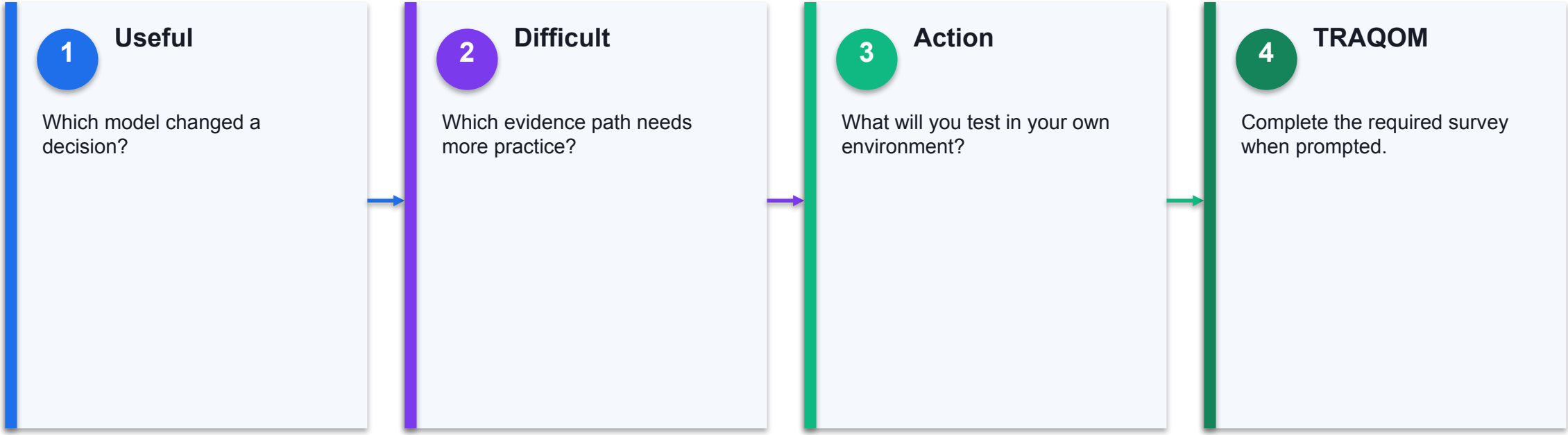
MECHANISM

Read left to right: input, control, state change, observable evidence.

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COURSE CLOSE

Course feedback



MECHANISM

Read left to right: input, control, state change, observable evidence.

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LEARNING PATH

Recommended follow-on courses



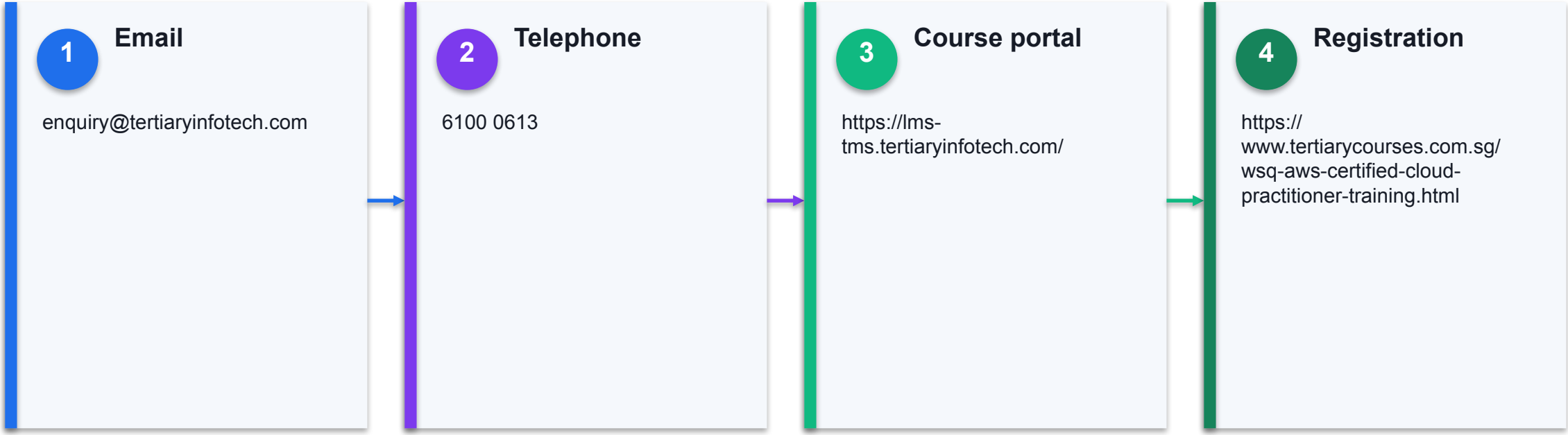
MECHANISM

Read left to right: input, control, state change, observable evidence.

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COURSE SUPPORT

Support after class



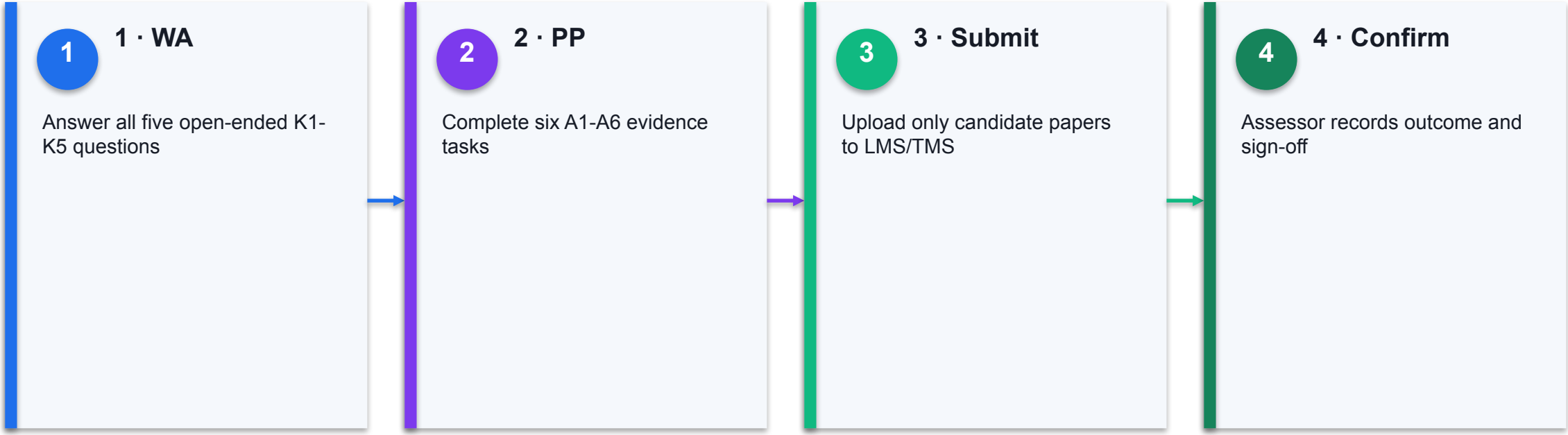
MECHANISM

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FINAL

Final assessment



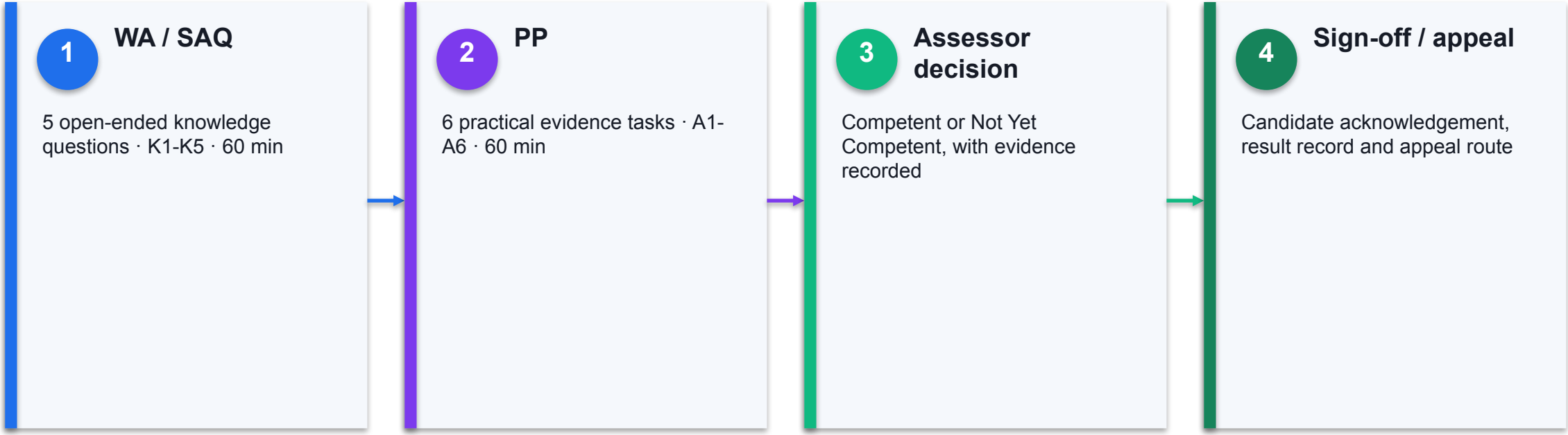
MECHANISM

Read left to right: input, control, state change, observable evidence.

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FINAL · ASSESSMENT FLOW

Assessment and sign-off flow



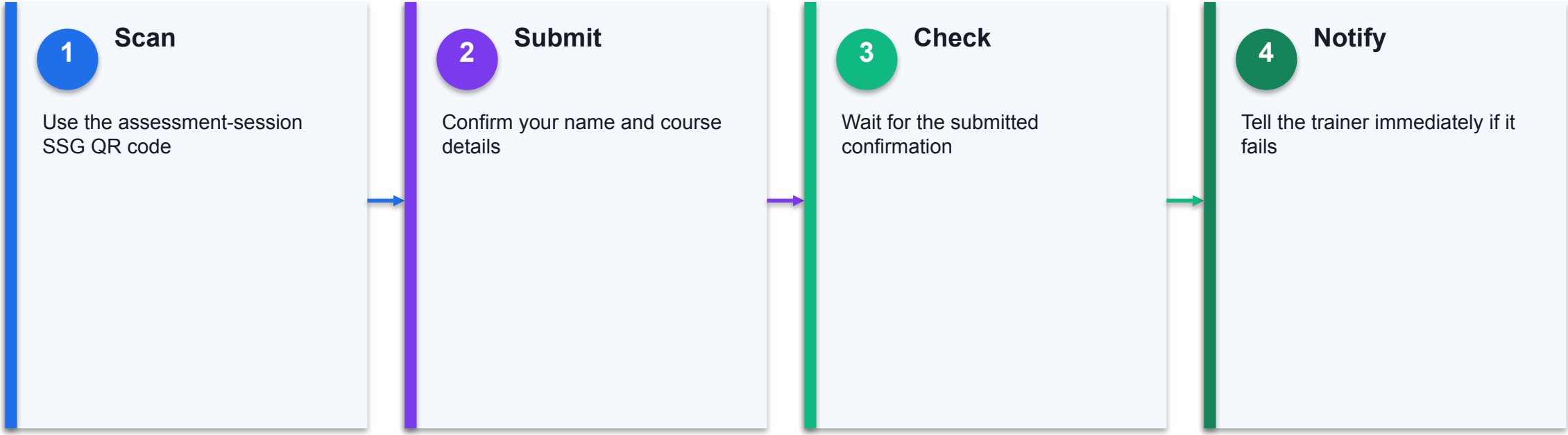
MECHANISM

Read left to right: input, control, state change, observable evidence.

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FINAL · MANDATORY

Digital Attendance (Assessment)



MECHANISM

Read left to right: input, control, state change, observable evidence.

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AWS CERTIFIED CLOUD PRACTITIONER · CLF-C02

Thank You

Explain the mechanism · choose the service · verify the evidence

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